Problems

1. Find analytical expressions for the magnitude and phase response for each $G(s)$ below. [Section: 10.1]

a. $G(s) = \dfrac{1}{s(s+2)(s+4)}$

b. $G(s) = \dfrac{(s+5)}{(s+2)(s+4)}$

c. $G(s) = \dfrac{(s+3)(s+5)}{s(s+2)(s+4)}$

2. For each function in Problem 1, make a plot of the log-magnitude and the phase, using log-frequency in rad/s as the ordinate. Do not use asymptotic approximations. [Section: 10.1]

3. For each function in Problem 1, make a polar plot of the frequency response. [Section: 10.1]

4. For each function in Problem 1, sketch the Bode asymptotic magnitude and asymptotic phase plots. Compare your results with your answers to Problem 1. [Section: 10.2]

5. Sketch the Nyquist diagram for each of the systems in Figure P10.1. [Section: 10.4]

6. Draw the polar plot from the separate magnitude and phase curves shown in Figure P10.2. [Section: 10.1]

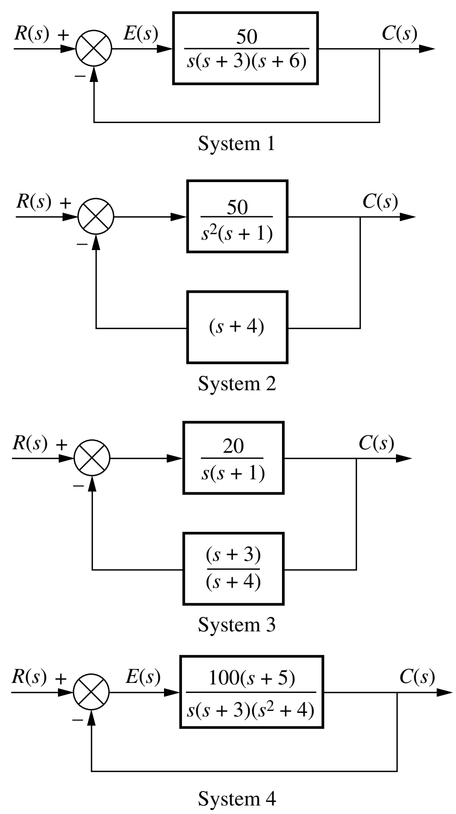

FIGURE P10.1

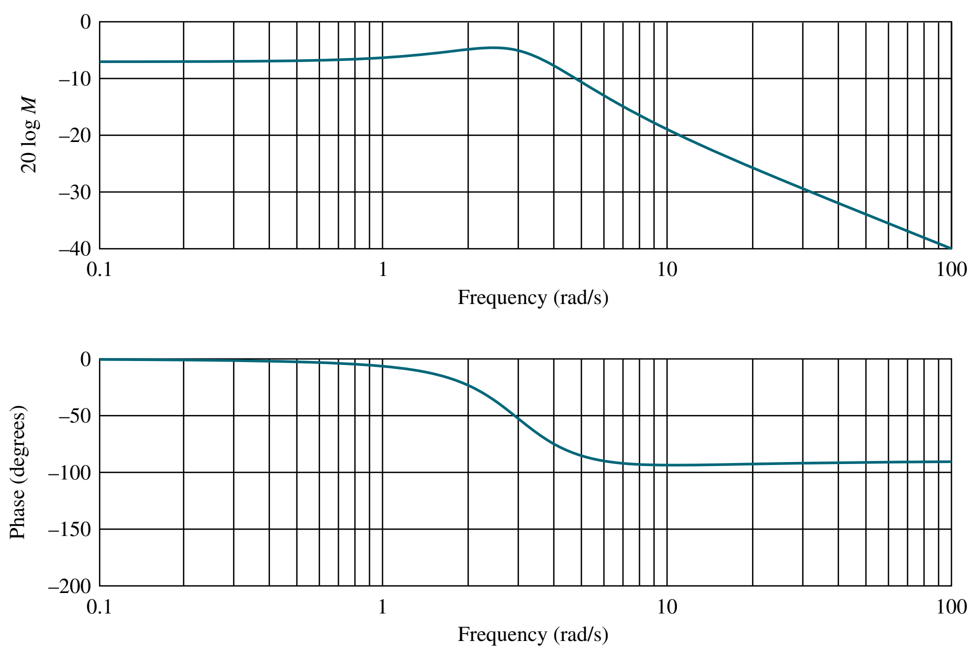

FIGURE P10.2

7. Draw the separate magnitude and phase curves from the polar plot shown in Figure P10.3. [Section: 10.1]

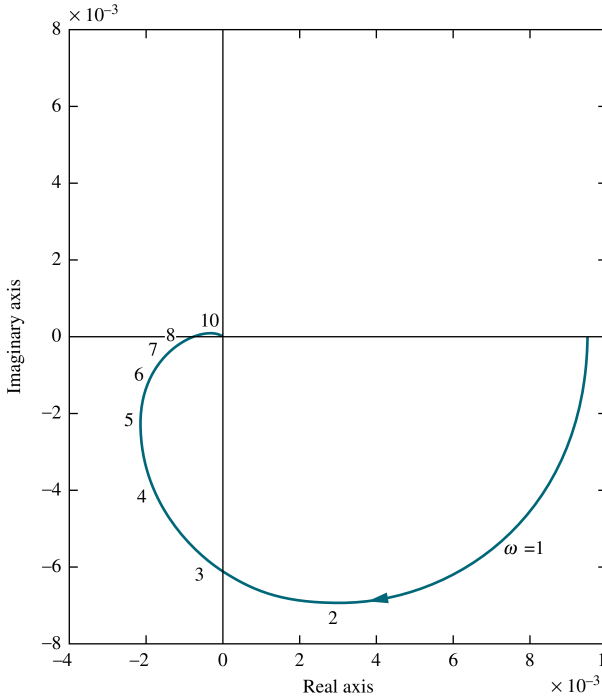

FIGURE P10.3

8. Write a program in MATLAB that will do the following:

MATLAB **ML**

a. Plot the Nyquist diagram of a system

b. Display the real-axis crossing value and frequency

Apply your program to a unity feedback system with

$$G(s) = \frac{K(s+5)}{(s^2+6s+100)(s^2+4s+25)}$$

9. Using the Nyquist criterion, find out whether each system of Problem 5 is stable. [Section: 10.3]

10. Using the Nyquist criterion, find the range of K for stability for each of the systems in Figure P10.4. [Section: 10.3]

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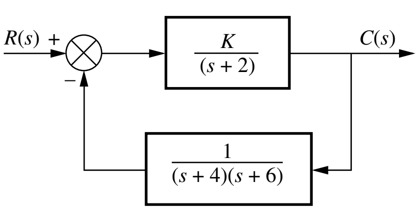
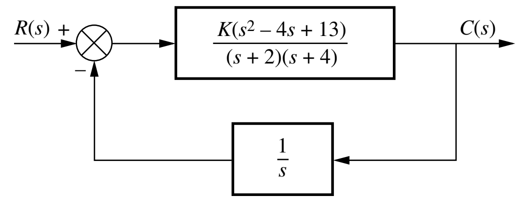
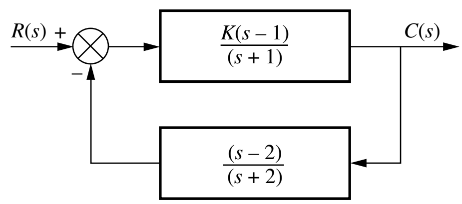


System 1

FIGURE P10.4 (*figure continues*)



System 2



System 3

FIGURE P10.4 (*Continued*)

11. For each system of Problem 10, find the gain margin and phase margin if the value of K in each part of Problem 10 is [Section: 10.6]

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a. $K = 1000$

b. $K = 100$

c. $K = 0.1$

12. Write a program in MATLAB that will do the following:

MATLAB **ML**

a. Allow a value of gain, K, to be entered from the keyboard

b. Display the Bode plots of a system for the entered value of K

c. Calculate and display the gain and phase margin for the entered value of K

Test your program on a unity feedback system with $G(s) = K/[s(s+3)(s+12)]$.

13. Use MATLAB's LTI Viewer to find the gain margin, phase margin, zero dB frequency, and 180° frequency for a unity feedback system with

Gui Tool **GUIT**

$$G(s) = \frac{8000}{(s+6)(s+20)(s+35)}$$

Use the following methods:

a. The Nyquist diagram

b. Bode plots

14. Derive Eq. (10.54), the closed-loop bandwidth in terms of ζ and ω_n of a two-pole system. [Section: 10.8]

15. For each closed-loop system with the following performance characteristics, find the closed-loop bandwidth: [Section: 10.8]

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a. $\zeta = 0.2,\ T_s = 3$ seconds

b. $\zeta = 0.2,\ T_p = 3$ seconds

c. $T_s = 4$ seconds, $T_p = 2$ seconds

d. $\zeta = 0.3,\ T_r = 4$ seconds

16. Consider the unity feedback system of Figure 10.10. For each $G(s)$ that follows, use the M and N circles to make a plot of the closed-loop frequency response: [Section: 10.9]

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a. $G(s) = \dfrac{10}{s(s+1)(s+2)}$

b. $G(s) = \dfrac{1000}{(s+3)(s+4)(s+5)(s+6)}$

c. $G(s) = \dfrac{50(s+3)}{s(s+2)(s+4)}$

17. Repeat Problem 16, using the Nichols chart in place of the M and N circles. [Section: 10.9]

18. Using the results of Problem 16, estimate the percent overshoot that can be expected in the step response for each system shown. [Section: 10.10]

19. Use the results of Problem 17 to estimate the percent overshoot if the gain term in the numerator of the forward path of each part of the problem is respectively changed as follows: [Section: 10.10]

a. From 10 to 30

b. From 1000 to 2500

c. From 50 to 75

20. Write a program in MATLAB that will do the following:

MATLAB ML

a. Allow a value of gain, K, to be entered from the keyboard

b. Display the closed-loop magnitude and phase frequency response plots of a unity feedback system with an open-loop transfer function, $KG(s)$

c. Calculate and display the peak magnitude, frequency of the peak magnitude, and bandwidth for the closed-loop frequency response and the entered value of K

Test your program on the system of Figure P10.5 for $K = 40$.

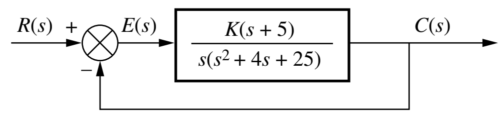

FIGURE P10.5

21. Use MATLAB's LTI Viewer with the Nichols plot to find the gain margin, phase margin, zero dB frequency, and 180° frequency for a unity feedback system with the forward-path transfer function

Gui Tool GUIT

$$G(s) = \frac{5(s+6)}{s(s^2+4s+15)}$$

22. Write a program in MATLAB that will do the following:

MATLAB ML

a. Make a Nichols plot of an open-loop transfer function

b. Allow the user to read the Nichols plot display and enter the value of M_p

c. Make closed-loop magnitude and phase plots

d. Display the expected values of percent overshoot, settling time, and peak time

e. Plot the closed-loop step response

Test your program on a unity feedback system with the forward-path transfer function

$$G(s) = \frac{5(s+6)}{s(s^2+4s+15)}$$

and explain any discrepancies.

23. Using Bode plots, estimate the transient response of the systems in Figure P10.6. [Section: 10.10]

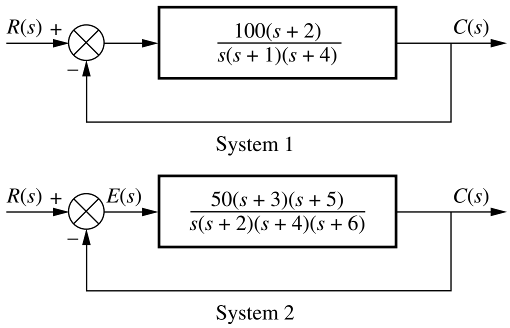

FIGURE P10.6

24. For the system of Figure P10.5, do the following: [Section: 10.10]

a. Plot the Bode magnitude and phase plots.

b. Assuming a second-order approximation, estimate the transient response of the system if $K = 40$.

c. Use MATLAB or any other program to check your assumptions by simulating the step response of the system.

MATLAB ML

25. The Bode plots for a plant, $G(s)$, used in a unity feedback system are shown in Figure P10.7. Do the following:

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a. Find the gain margin, phase margin, zero dB frequency, 180° frequency, and the closed-loop bandwidth.

b. Use your results in Part **a** to estimate the damping ratio, percent overshoot, settling time, and peak time.

26. Write a program in MATLAB that will use an open-loop transfer function, $G(s)$, to do the following:

MATLAB ML

a. Make a Bode plot

b. Use frequency response methods to estimate the percent overshoot, settling time, and peak time

c. Plot the closed-loop step response

Test your program by comparing the results to those obtained for the systems of Problem 23.

27. The open-loop frequency response shown in Figure P10.8 was experimentally obtained from a unity feedback system. Estimate the percent overshoot and steady-state error of the closed-loop system. [Sections: 10.10, 10.11]

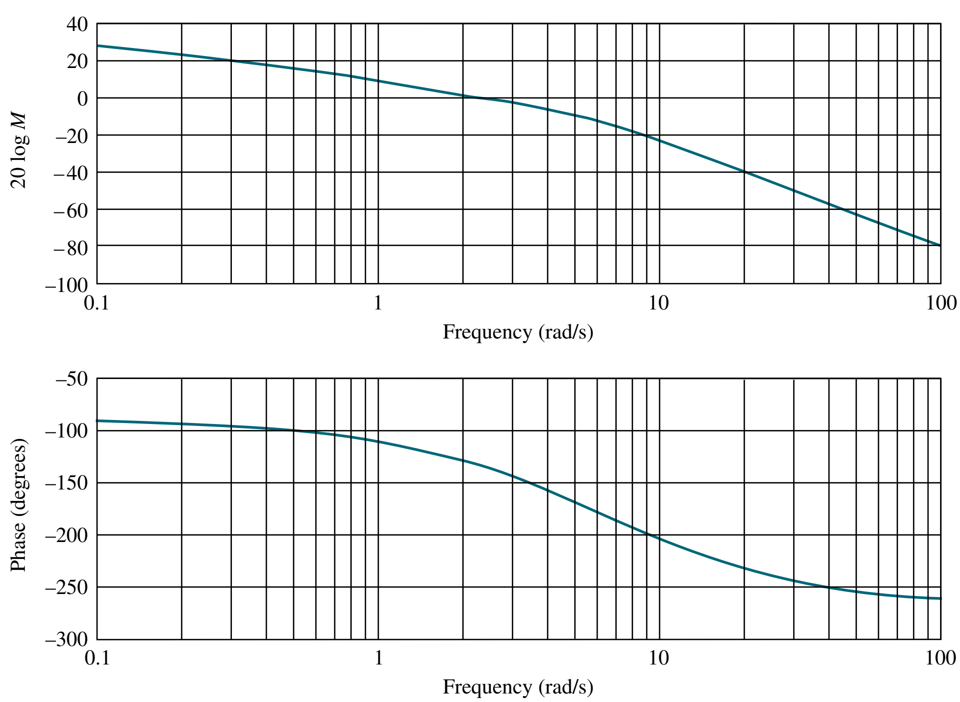

FIGURE P10.7

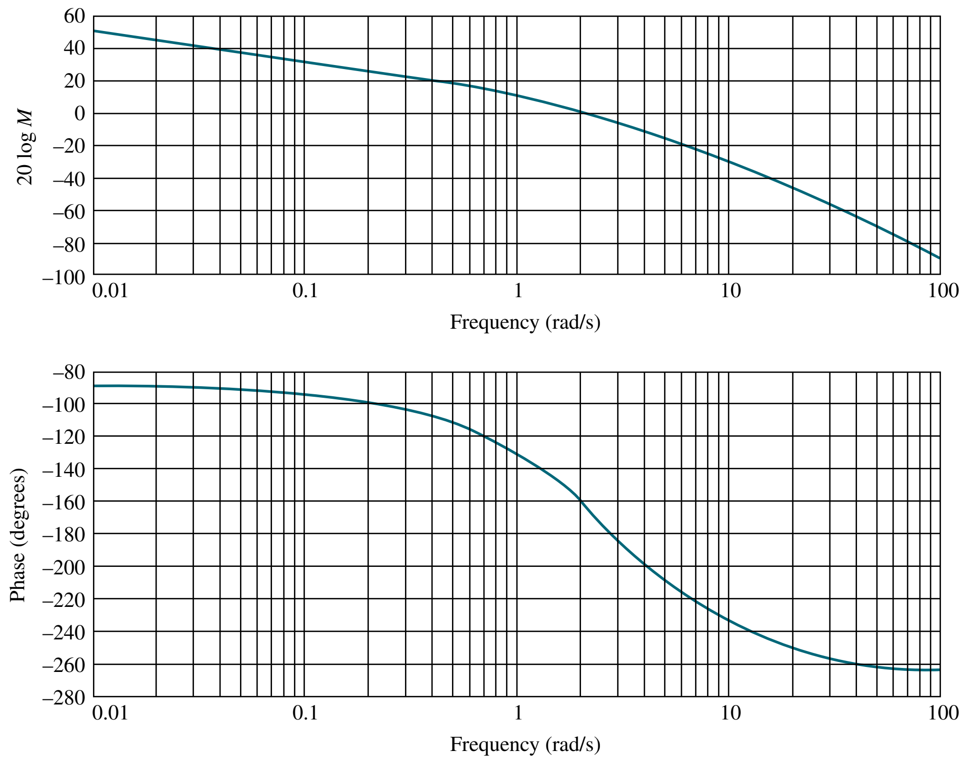

FIGURE P10.8

28. Consider the system in Figure P10.9. [Section: 10.12]

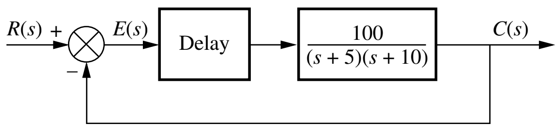

FIGURE P10.9

a. Find the phase margin if the system is stable for time delays of 0, 0.1, 0.2, 0.5, and 1 second.

b. Find the gain margin if the system is stable for each of the time delays given in Part **a**.

c. For what time delays mentioned in Part **a** is the system stable?

d. For each time delay that makes the system unstable, how much reduction in gain is required for the system to be stable?

29. Given a unity feedback system with the forward-path transfer function

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$$G(s) = \frac{K}{(s+1)(s+3)(s+6)}$$

and a delay of 0.5 second, find the range of gain, K, to yield stability. Use Bode plots and frequency response techniques. [Section: 10.12]

30. Given a unity feedback system with the forward-path transfer function

$$G(s) = \frac{K}{s(s+3)(s+12)}$$

and a delay of 0.5 second, make a second-order approximation and estimate the percent overshoot if $K = 40$. Use Bode plots and frequency response techniques. [Section: 10.12]

31. `Use the MATLAB function pade(T,n) to model the delay in Problem 30. Obtain the unit step response and evaluate your second-order approximation in Problem 30.`

MATLAB
ML

32. For the Bode plots shown in Figure P10.10, determine the transfer function by hand or via MATLAB. [Section: 10.13]

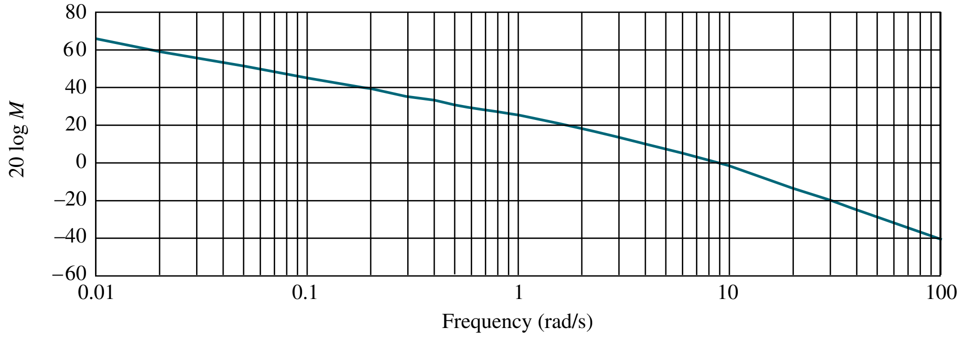

FIGURE P10.10

33. Repeat Problem 32 for the Bode plots shown in Figure P10.11. [Section: 10.13]

34. An overhead crane consists of a horizontally moving trolley of mass m_T dragging a load of mass m_L, which dangles from its bottom surface at the end of a rope of fixed length, L. The position of the trolley is controlled in the feedback configuration shown in Figure 10.20. Here,

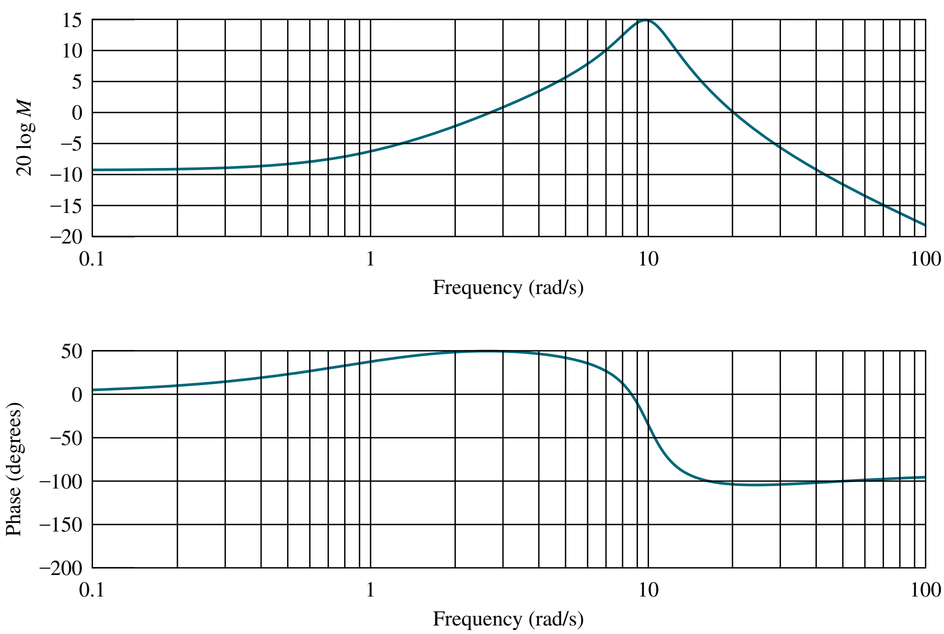

FIGURE P10.11

$G(s) = KP(s)$, $H = 1$, and

$$P(s) = \frac{X_T(s)}{F_T(s)} = \frac{1}{m_T} \frac{s^2 + \omega_0^2}{s^2(s^2 + a\omega_0^2)}$$

The input is $f_T(t)$, the input force applied to the trolley. The output is $x_T(t)$, the trolley displacement. Also, $\omega_0 = \sqrt{\frac{g}{L}}$ and $a = (m_L + m_T)/m_T$ (*Marttinen, 1990*) Make a qualitative Bode plot of the system assuming $a > 1$.

35. A room's temperature can be controlled by varying the radiator power. In a specific room, the transfer function from indoor radiator power, $\dot{Q}$, to room temperature, T in °C is (*Thomas, 2005*)

$$P(s) = \frac{T(s)}{\dot{Q}(s)} = \frac{(1 \times 10^{-6})s^2 + (1.314 \times 10^{-9})s + (2.66 \times 10^{-13})}{s^3 + 0.00163\, s^2 + (5.272 \times 10^{-7})s + (3.538 \times 10^{-11})}$$

The system is controlled in the closed-loop configuration shown in Figure 10.20 with $G(s) = KP(s)$, $H = 1$.

a. Draw the corresponding Nyquist diagram for $K = 1$.

b. Obtain the gain and phase margins.

c. Find the range of K for the closed-loop stability. Compare your result with that of Problem 61, Chapter 6.

36. The open-loop dynamics from dc voltage armature to angular position of a robotic manipulator joint is given by $P(s) = \frac{48500}{s^2 + 2.89s}$ (*Low, 2005*).

a. Draw by hand a Bode plot using asymptotic approximations for magnitude and phase.

b. `Use MATLAB to plot the exact Bode plot and compare with your sketch from Part` **a.** MATLAB ML

37. Problem 49, Chapter 8 discusses a magnetic levitation system with a plant transfer function $P(s) = -\frac{1300}{s^2 - 860^2}$ (*Galvão, 2003*). Assume that the plant is in cascade with an $M(s)$ and that the system will be controlled by the loop shown in Figure 10.20, where $G(s) = M(s)P(s)$ and $H = 1$. For each $M(s)$ that follows, draw the Nyquist diagram when $K = 1$, and find the range of closed-loop stability for $K > 0$.

a. $M(s) = -K$

b. $M(s) = -\frac{K(s + 200)}{s + 1000}$

c. Compare your results with those obtained in Problem 49, Chapter 8.

38. The simplified and linearized model for the transfer function of a certain bicycle from steer angle (δ) to roll angle (φ) is given by (*Åstrom, 2005*)

$$P(s) = \frac{\varphi(s)}{\delta(s)} = \frac{10(s + 25)}{s^2 + 25}$$

Assume the rider can be represented by a gain K, and that the closed-loop system is shown in Figure 10.20 with $G(s) = KP(s)$ and $H = 1$. Use the Nyquist stability criterion to find the range of K for closed-loop stability.

39. The control of the radial pickup position of a digital versatile disk (DVD) was discussed in Problem 48, Chapter 9. There, the open-loop transfer function from coil input voltage to radial pickup position was given as (*Bittanti, 2002*)

$$P(s) = \frac{0.63}{\left(1 + \frac{0.36}{305.4}s + \frac{s^2}{305.4^2}\right)\left(1 + \frac{0.04}{248.2}s + \frac{s^2}{248.2^2}\right)}$$

Assume the plant is in cascade with a controller,

$$M(s) = \frac{0.5(s + 1.63)}{s(s + 0.27)}$$

and in the closed-loop configuration shown in Figure 10.20, where $G(s) = M(s)P(s)$ and $H = 1$. Do the following:

a. Draw the open-loop frequency response in a Nichols chart.

b. Predict the system's response to a unit step input. Calculate the $\%OS$, c_{final}, and T_s.

c. `Verify the results of Part` **b** `using MATLAB simulations.` MATLAB ML

40. The Soft Arm, used to feed people with disabilities, was discussed in Problem 57 in Chapter 6. Assuming the system block diagram shown in Figure P10.12, use frequency response techniques to determine the following (*Kara, 1992*):

a. Gain margin, phase margin, zero dB frequency, and 180° frequency

b. Is the system stable? Why?

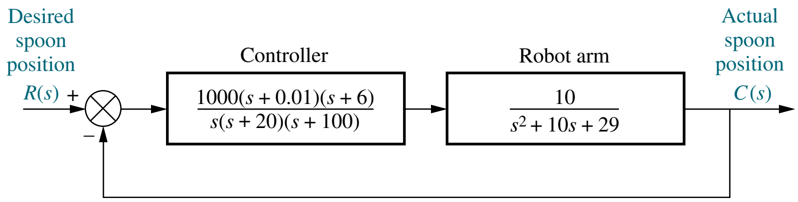

FIGURE P10.12 Soft Arm position control system block diagram

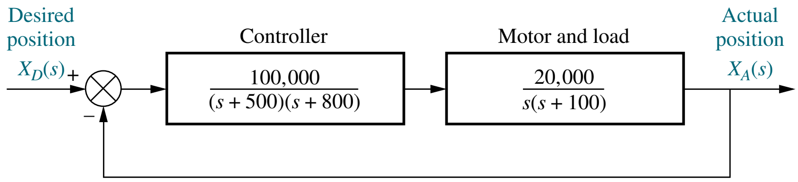

FIGURE P10.13 Floppy disk drive block diagram

41. A floppy disk drive was discussed in Problem 57 in Chapter 8. Assuming the system block diagram shown in Figure P10.13, use frequency response techniques to determine the following:

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a. Gain margin, phase margin, zero dB frequency, 180° frequency, and closed-loop bandwidth

b. Percent overshoot, settling time, and peak time

c. `Use MATLAB to simulate the closed-loop step response and compare the results to those obtained in Part` **`b.`**

MATLAB ML

42. Industrial robots, such as that shown in Figure P10.14, require accurate models for design of high performance. Many transfer function models for industrial robots assume interconnected rigid bodies with the drive-torque source modeled as a pure gain, or first-order system. Since the motions associated with the robot are connected to the drives through flexible linkages rather than rigid linkages, past modeling does not explain the resonances observed. An accurate, small-motion, linearized model has been developed that takes into consideration the flexible drive. The transfer function

$$G(s) = 999.12\frac{(s^2+8.94s+44.7^2)}{(s+20.7)(s^2+34.858s+60.1^2)}$$

relates the angular velocity of the robot base to electrical current commands (*Good*, *1985*). Make a Bode plot of the frequency response and identify the resonant frequencies.

FIGURE P10.14 Robot performing construction of computer memory units (© Michael Rosenfield/Science Faction/ © Corbis).

43. The charge-coupled device (CCD) that is used in video movie cameras to convert images into electrical signals can be used as part of an automatic focusing system in cameras. Automatic focusing can be implemented by focusing the center of the image on a charge-coupled device array through two lenses. The separation of the two images on the CCD is related to the focus. The camera senses the separation, and a computer drives the lens and

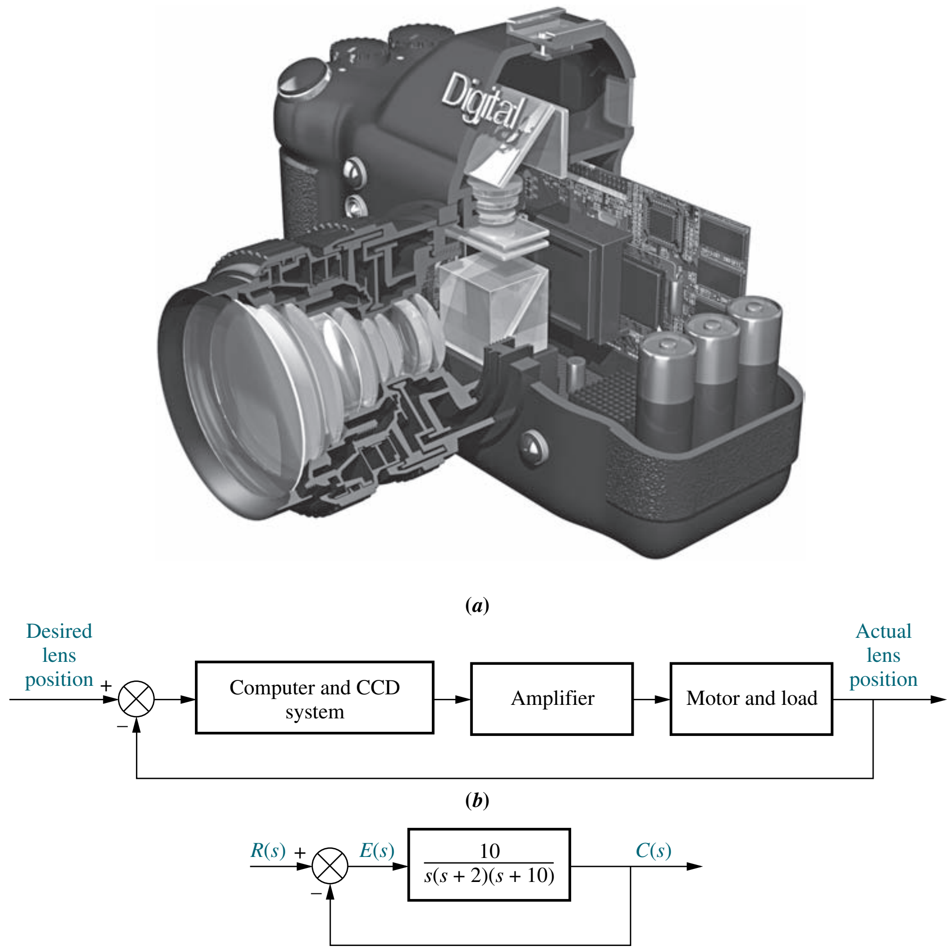

FIGURE P10.15 **a.** A cutaway view of a digital camera showing parts of the CCD automatic focusing system (© Stephen Sweet/iStockphoto); **b.** functional block diagram; **c.** block diagram

focuses the image. The automatic focus system is a position control, where the desired position of the lens is an input selected by pointing the camera at the subject. The output is the actual position of the lens. The camera in Figure P10.15(*a*) uses a CCD automatic focusing system. Figure P10.15(*b*) shows the automatic focusing feature represented as a position control system. Assuming the simplified model shown in Figure P10.15(*c*), draw the Bode plots and estimate the percent overshoot for a step input.

44. A ship's roll can be stabilized with a control system. A voltage applied to the fins' actuators creates a roll torque that is applied to the ship. The ship, in response to the roll torque, yields a roll angle. Assuming the block diagram for the roll control system shown in Figure P10.16, determine the gain and phase margins for the system.

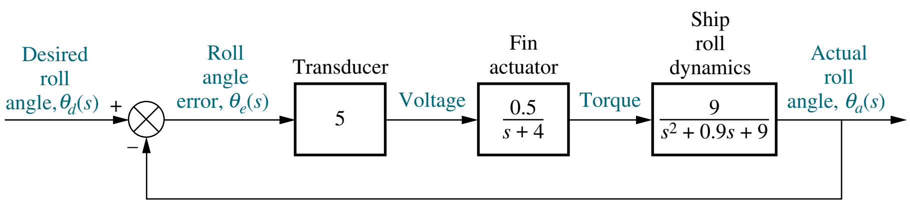

FIGURE P10.16 Block diagram of a ship's roll-stabilizing system

45. The linearized model of a particular network link working under TCP/IP and controlled using a random early detection (RED) algorithm can be described by Figure 10.20 where $G(s) = M(s)P(s)$, $H = 1$, and (*Hollot, 2001*)

$$M(s) = \frac{0.005L}{s + 0.005};\ P(s) = \frac{140625e^{-0.1s}}{(s + 2.67)(s + 10)}$$

a. Plot the Nichols chart for $L = 1$. Is the system closed-loop stable?

b. Find the range of L for closed-loop stability.

c. Use the Nichols chart to predict $\%OS$ and T_s for $L = 0.95$. Make a hand sketch of the expected unit step response.

d. Verify Part **c** with a Simulink unit step response simulation. Simulink SL

46. In the TCP/IP network link of Problem 45, let $L = 0.8$, but assume that the amount of delay is an unknown variable.

a. Plot the Nyquist diagram of the system for zero delay, and obtain the phase margin.

b. Find the maximum delay allowed for closed-loop stability.

47. Thermal flutter of the Hubble Space Telescope (HST) produces errors for the pointing control system. Thermal flutter of the solar arrays occurs when the spacecraft passes from sunlight to darkness and when the spacecraft is in daylight. In passing from daylight to darkness, an end-to-end bending oscillation of frequency f_1 rad/s is experienced. Such oscillations interfere with the pointing control system of the HST. A filter with the transfer function

$$G_f(s) = \frac{1.96(s^2 + s + 0.25)(s^2 + 1.26s + 9.87)}{(s^2 + 0.015s + 0.57)(s^2 + 0.083s + 17.2)}$$

is proposed to be placed in cascade with the PID controller to reduce the bending (*Wie, 1992*).

a. Obtain the frequency response of the filter and estimate the bending frequencies that will be reduced.

b. Explain why this filter will reduce the bending oscillations if these oscillations are thought to be disturbances at the output of the control system.

48. An experimental holographic media storage system uses a flexible photopolymer disk. During rotation, the disk tilts, making information retrieval difficult. A system that compensates for the tilt has been developed. For this, a laser beam is focused on the disk surface and disk variations are measured through reflection. A mirror is in turn adjusted to align with the disk and makes information retrieval possible. The system can be represented by a unity feedback system in which a controller with transfer function

$$G_C(s) = \frac{78.575(s + 436)^2}{(s + 132)(s + 8030)}$$

and a plant

$$P(s) = \frac{1.163 \times 10^8}{s^3 + 962.5s^2 + 5.958 \times 10^5 s + 1.16 \times 10^8}$$

form an open loop transmission $L(s) = G_c(s)P(s)$ (*Kim, 2009*).

a. Use MATLAB to obtain the system's Nyquist diagram. Find out if the system is stable. MATLAB ML

b. Find the system's phase margin.

c. Use the value of phase margin obtained in **b.** to calculate the expected system's overshoot to a step input.

d. Simulate the system's response to a unit step input and verify the %OS calculated in **c.**

49. The design of cruise control systems in heavy vehicles such as big rigs is especially challenging due to the extreme variations in payload. A typical frequency response for the transfer function from fuel mass flow to vehicle speed is shown in Figure P10.17.

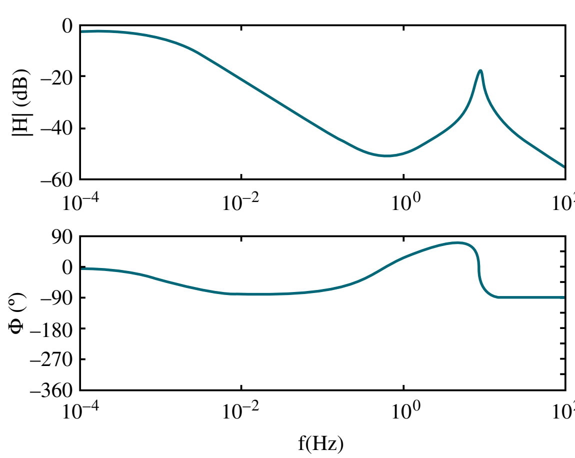

FIGURE P10.17

This response includes the dynamics of the engine, the gear box, the propulsion shaft, the differential, the drive shafts, the chassis, the payload, and tire dynamics. Assume that the system is controlled in a closed-loop, unity-feedback loop using a proportional compensator (*van der Zalm*, *2008*).

a. Make a plot of the Nyquist diagram that corresponds to the Bode plot of Figure P10.17.

b. Assuming there are no open-loop poles in the right half-plane, find out if the system is closed-loop stable when the proportional gain $K = 1$.

c. Find the range of positive K for which the system is closed-loop stable.

50. LabVIEW LV MATLAB ML

Use LabVIEW with the Control Design and Simulation Module, and MathScript RT Module and modify the CDEx Nyquist Analysis.vi to obtain the range of K for stability using the Nyquist plot for any system you enter. In addition, design a LabVIEW VI that will accept as an input the polynomial numerator and polynomial denominator of an open-loop transfer function and obtain a Nyquist plot for a value of $K = 10,000$. Your VI will also display the following as generated from the Nyquist plot: (1) gain margin, (2) phase margin, (3) zero dB frequency, and (4) 180 degrees frequency. Use the system and results of Skill-Assessment Exercise 10.6 to test your VIs.

51. LabVIEW LV MATLAB ML

Use LabVIEW with the Control Design and Simulation Module, and MathScript RT Module to build a VI that will accept an open-loop transfer function, plot the Bode diagram, and plot the closed-loop step response. Your VI will also use the CD Parametric Time Response.vi to display (1) rise time, (2) peak time, (3) settling time, (4) percent overshoot, (5) steady-state value, and (6) peak value. Use the system in Skill-Assessment Exercise 10.9 to test your VI. Compare the results obtained from your VI with those obtained in Skill-Assessment Exercise 10.9.

52. MATLAB ML

The block diagram of a cascade system used to control water level in a steam generator of a nuclear power plant (*Wang, 2009*) was presented in Figure P.6.19. In that system, the level controller, $G_{LC}(s)$, is the *master* controller and the feed-water flow controller, $G_{FC}(s)$, is the slave controller. Consider that the inner feedback loop is replaced by its equivalent transfer function, $G_{WX}(s)$.

Using numerical values in (*Wang, 2009*) and (*Bhambhani, 2008*) the transfer functions with a 1 second pure delay are:

$$G_{fw}(s) = \frac{2 \cdot e^{-\tau s}}{s(T_1 s + 1)} = \frac{2 \cdot e^{-s}}{s(25s + 1)};$$

$$G_{WX}(s) = \frac{(4s + 1)}{3(3.333s + 1)};$$

$$G_{LC}(s) = K_{P_{LC}} + K_{D_{LC}} s = 1.5(10s + 1).$$

Use MATLAB or any other program to:

a. Obtain Bode magnitude and phase plots for this system using a fifth-order Padé approximation (available in MATLAB). Note on these plots, if applicable, the gain and phase margins.

b. Plot the response of the system, $c(t)$, to a unit step input, $r(t) = u(t)$. Note on the $c(t)$ curve the rise time, T_r, the settling time, T_s, the final value of the output, and, if applicable, the percent overshoot, $\%OS$, and mid peak time, T_p.

c. Repeat the above two steps for a pure delay of 1.5 seconds.

PROGRESSIVE ANALYSIS AND DESIGN PROBLEMS

53. High-speed rail pantograph. Problem 21 in Chapter 1 discusses active control of a pantograph mechanism for high-speed rail systems. In Problem 79(a), Chapter 5, you found the block diagram for the active pantograph control system. In Chapter 8, Problem 72, you designed the gain to yield a closed-loop step response with 30% overshoot. A plot of the step response should have shown a settling time greater than 0.5 second as well as a high-frequency oscillation superimposed over the step response. In Chapter 9, Problem 55, we reduced the settling time to about 0.3 second, reduced the step response steady-state error to zero, and eliminated the high-frequency oscillations by using a notch filter (*O'Connor, 1997*). Using the equivalent forward transfer function found in Chapter 5 cascaded with the notch filter specified in Chapter 9, do the following using frequency response techniques:

a. Plot the Bode plots for a total equivalent gain of 1 and find the gain margin, phase margin, and 180° frequency.

b. Find the range of K for stability.

c. Compare your answer to Part **b** with your answer to Problem 67, Chapter 6. Explain any differences.

54. Control of HIV/AIDS. The linearized model for an HIV/AIDS patient treated with RTIs was obtained in Chapter 6 as (*Craig, 2004*);

$$P(s) = \frac{Y(s)}{U_1(s)} = \frac{-520s - 10.3844}{s^3 + 2.6817s^2 + 0.11s + 0.0126}$$

a. Consider this plant in the feedback configuration in Figure 10.20 with $G(s) = P(s)$ and $H(s) = 1$. Obtain the Nyquist diagram. Evaluate the system for closed-loop stability.

b. Consider this plant in the feedback configuration in Figure 10.20 with $G(s) = -P(s)$ and $H(s) = 1$. Obtain the Nyquist diagram. Evaluate the system for closed-loop stability. Obtain the gain and phase margins.

55. Hybrid vehicle. In Problem 8.74 we used MATLAB to plot the root locus for the speed control of an HEV rearranged as a unity-feedback system, as shown in Figure P7.34 (*Preitl, 2007*). The plant and compensator were given by

MATLAB ML

$$G(s) = \frac{K(s+0.6)}{(s+0.5858)(s+0.0163)}$$

and we found that $K = 0.78$, resulted in a critically damped system.

a. Use MATLAB or any other program to plot

i. The Bode magnitude and phase plots for that system, and

ii. The response of the system, $c(t)$, to a step input, $r(t) = 4\,u(t)$. Note on the $c(t)$ curve the rise time, T_r, and settling time, T_s, as well as the final value of the output.

b. Now add an integral gain to the controller, such that the plant and compensator transfer function becomes

$$G(s) = \frac{K_1(s+Z_c)(s+0.6)}{s(s+0.5858)(s+0.0163)}$$

where $K_1 = 0.78$ and $Z_c = \dfrac{K_2}{K_1} = 0.4$. Use MATLAB or any other program to do the following:

i. Plot the Bode magnitude and phase plots for this case.

ii. Obtain the response of the system to a step input, $r(t) = 4\,u(t)$. Plot $c(t)$ and note on it the rise time, T_r, percent overshoot, $\%OS$, peak time, T_p, and settling time, T_s.

c. Does the response obtained in **a.** or **b.** resemble a second-order overdamped, critically damped, or underdamped response? Explain.

Cyber Exploration Laboratory

Experiment 10.1

Objective To examine the relationships between open-loop frequency response and stability, open-loop frequency response and closed-loop transient response, and the effect of additional closed-loop poles and zeros upon the ability to predict closed-loop transient response

Minimum Required Software Packages MATLAB, and the Control System Toolbox

Prelab

1. Sketch the Nyquist diagram for a unity negative feedback system with a forward transfer function of $G(s) = \dfrac{K}{s(s+2)(s+10)}$. From your Nyquist plot, determine the range of gain, K, for stability.

2. Find the phase margins required for second-order closed-loop step responses with the following percent overshoots: 5%, 10%, 20%, 30%.

Lab

1. Using the SISO Design Tool, produce the following plots simultaneously for the system of Prelab 1: root locus, Nyquist diagram, and step response. Make plots for the following values of K: 50, 100, the value for marginal stability found in Prelab 1, and a value above that found for marginal stability. Use the zoom tools when required to produce an illustrative plot. Finally, change the gain by grabbing and moving the closed-loop poles along the root locus and note the changes in the Nyquist diagram and step response.
2. Using the SISO Design Tool, produce Bode plots and closed-loop step responses for a unity negative feedback system with a forward transfer function of $G(s) = \dfrac{K}{s(s+10)^2}$. Produce these plots for each value of phase margin found in Prelab 2. Adjust the gain to arrive at the desired phase margin by grabbing the Bode magnitude curve and moving it up or down. Observe the effects, if any, upon the Bode phase plot. For each case, record the value of gain and the location of the closed-loop poles.
3. Repeat Lab 2 for $G(s) = \dfrac{K}{s(s+10)}$.

Postlab

1. Make a table showing calculated and actual values for the range of gain for stability as found in Prelab 1 and Lab 1.
2. Make a table from the data obtained in Lab 2 itemizing phase margin, percent overshoot, and the location of the closed-loop poles.
3. Make a table from the data obtained in Lab 3 itemizing phase margin, percent overshoot, and the location of the closed-loop poles.
4. For each Postlab task 1 to 3, explain any discrepancies between the actual values obtained and those expected.

Experiment 10.2

Objective To use LabVIEW and Nichols charts to determine the closed-loop time response performance.

Minimum Required Software Packages LabVIEW, Control Design and Simulation Module, MathScript RT Module, and MATLAB

Prelab

1. Assume a unity-feedback system with a forward-path transfer function, $G(s) = \dfrac{100}{s(s+5)}$. Use MATLAB or any method to determine gain and phase margins. In addition, find the percent overshoot, settling time, and peak time of the closed-loop step response.
2. Design a LabVIEW VI that will create a Nichols chart. Adjust the Nichols chart's scale to estimate gain and phase margins. Then, prompt the user to enter the values of

gain and phase margin found from the Nichols chart. In response, your VI will produce the percent overshoot, settling time, and peak time of the closed-loop step response.

Lab Run your VI for the system given in the Prelab. Test your VI with other systems of your choice.

Postlab Compare the closed-loop performance calculated in the Prelab with those produced by your VI.

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T E N

Frequency Response Techniques

SOLUTION TO CASE STUDY CHALLENGE

Antenna Control: Stability Design and Transient Performance

First find the forward transfer function, G(s).

Pot:

$$K_1 = \frac{10}{\pi} = 3.18$$

Preamp:

$$K$$

Power amp:

$$G_1(s) = \frac{100}{s(s+100)}$$

Motor and load:

$$J = 0.05 + 5\,(\frac{1}{5})^2 = 0.25\ ;\ D = 0.01 + 3\,(\frac{1}{5})^2 = 0.13;\ \frac{K_t}{Ra} = \frac{1}{5}\ ;\ K_b = 1.$$

Therefore,

$$G_m(s) = \frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J}}{s(s+\frac{1}{J}(D + \frac{K_t K_b}{R_a}))} = \frac{0.8}{s(s+1.32)}\ .$$

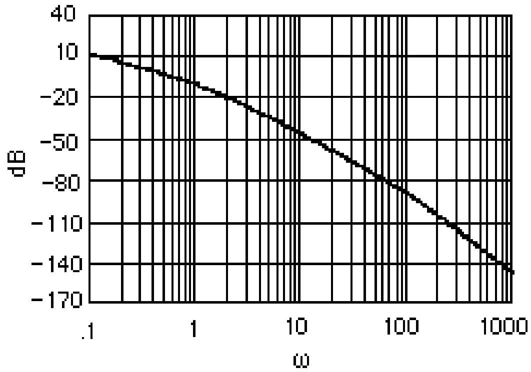
Gears:

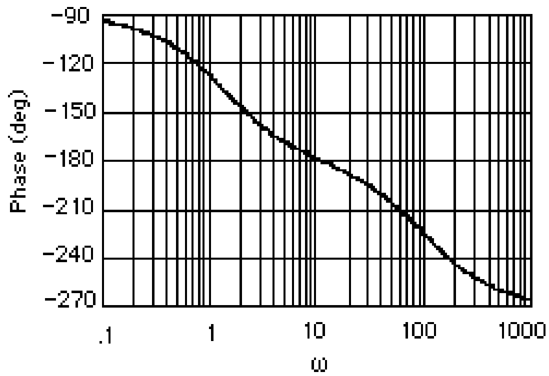
$$K_2 = \frac{50}{250} = \frac{1}{5}$$

Therefore,

$$G(s) = K_1 K G_1(s) G_m(s) K_2 = \frac{50.88K}{s(s+1.32)(s+100)}$$

Plotting the Bode plots for K = 1,





a. Phase is 180^{o} at $\omega = 11.5$ rad/s. At this frequency the gain is - 48.41 dB, or K = 263.36. Therefore, for stability, 0 < K < 263.36.

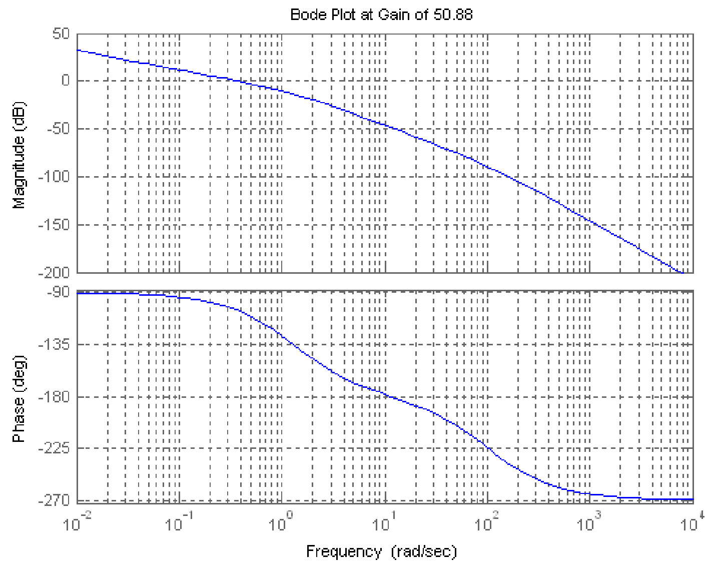
b. If K = 3, the magnitude curve will be 9.54 dB higher and go through zero dB at $\omega = 0.94$ rad/s. At this frequency, the phase response is -125.99^{o}. Thus, the phase margin is $180^{\text{o}} - 125.99^{\text{o}} = 54.01^{\text{o}}$. Using Eq. (10.73), $\zeta = 0.528$. Eq. (4.38) yields %OS = 14.18%.

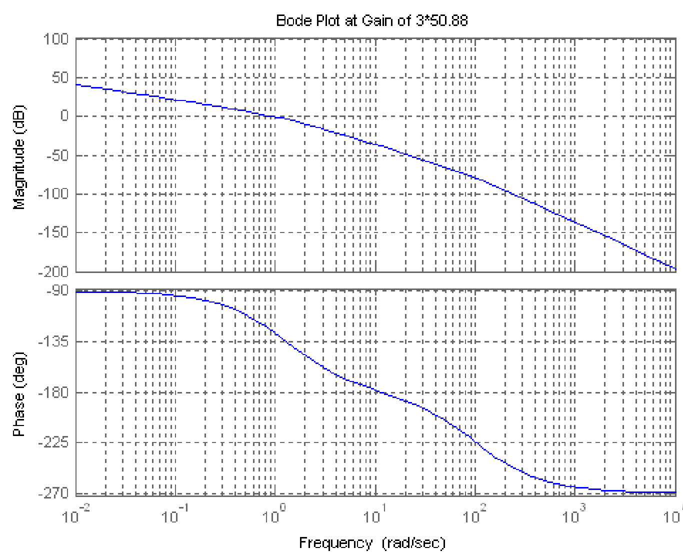
c.

Program:
```
numga=50.88;
denga=poly([0 -1.32 -100]);
'Ga(s)'
Ga=tf(numga,denga);
Gazpk=zpk(Ga)
'(a)'
bode(Ga)
title('Bode Plot at Gain of 50.88')
pause
[Gm,Pm,Wcp,Wcg]=margin(Ga);
'Gain for Stability'
Gm
pause
'(b)'
numgb=50.88*3;
dengb=denga;
'Gb(s)'
Gb=tf(numgb,dengb);
Gbzpk=zpk(Gb)
bode(Gb)
title('Bode Plot at Gain of 3*50.88')
[Gm,Pm,Wcp,Wcg]=margin(Gb);
'Phase Margin'
Pm
for z=0:.01:1
Pme=atan(2*z/(sqrt(-2*z^2+sqrt(1+4*z^4))))*(180/pi);
if Pm-Pme<=0;
break
end
end
z
percent=exp(-z*pi/sqrt(1-z^2))*100
```

Computer response:
```
ans =
```

```
Ga(s)

Zero/pole/gain:
      50.88
------------------
s (s+100) (s+1.32)

ans =

(a)

ans =

Gain for Stability

Gm =

  262.8585

ans =

(b)

ans =

Gb(s)

Zero/pole/gain:
      152.64
------------------
s (s+100) (s+1.32)

ans =

Phase Margin

Pm =

   53.9644

z =

    0.5300

percent =

   14.0366
```





ANSWERS TO REVIEW QUESTIONS

1. a. Transfer functions can be modeled easily from physical data; **b.** Steady-state error requirements can be considered easily along with the design for transient response; **c.** Settles ambiguities when sketching root locus; (d) Valuable tool for analysis and design of nonlinear systems.

2. A sinusoidal input is applied to a system. The sinusoidal output's magnitude and phase angle is measured in the steady-state. The ratio of the output magnitude divided by the input magnitude is the magnitude response at the applied frequency. The difference between the output phase angle and the input phase angle is the phase response at the applied frequency. If the magnitude and phase response are plotted over a range of different frequencies, the result would be the frequency response for the system.

3. Separate magnitude and phase curves; polar plot

4. If the transfer function of the system is G(s), let s=jω. The resulting complex number's magnitude is the magnitude response, while the resulting complex number's angle is the phase response.

5. Bode plots are asymptotic approximations to the frequency response displayed as separate magnitude and phase plots, where the magnitude and frequency are plotted in dB.

6. Negative 6 dB/octave which is the same as 20 dB/decade

7. Negative 24 dB/octave or 80 dB/decade

8. Negative 12 dB/octave or 40 dB/decade

9. Zero degrees until 0.2; a negative slope of 45^o/decade from a frequency of 0.2 until 20; a constant -90^o phase from a frequency of 20 until ∞

10. Second-order systems require a correction near the natural frequency due to the peaking of the curve for different values of damping ratio. Without the correction the accuracy is in question.

11. Each pole yields a maximum difference of 3.01 dB at the break frequency. Thus for a pole of multiplicity three, the difference would be 3x3.01 or 9.03 dB at the break frequency, - 4.

12. Z = P - N, where Z = # of closed-loop poles in the right-half plane, P = # of open-loop poles in the right-half plane, and N = # of counter-clockwise encirclements of -1 made by the mapping.

13. Whether a system is stable or not since the Nyquist criterion tells us how many rhp the system has

14. A Nyquist diagram, typically, is a mapping, through a function, of a semicircle that encloses the right half plane.

15. Part of the Nyquist diagram is a polar frequency response plot since the mapping includes the positive jω axis.

16. The contour must bypass them with a small semicircle.

17. We need only map the positive imaginary axis and then determine that the gain is less than unity when the phase angle is 180^o.

18. We need only map the positive imaginary axis and then determine that the gain is greater than unity when the phase angle is 180^o.

19. The amount of additional open-loop gain, expressed in dB and measured at 180^o of phase shift, required to make a closed-loop system unstable.

20. The phase margin is the amount of additional open-loop phase shift, Φ_M, required at unity gain to make the closed-loop system unstable.

21. Transient response can be obtained from (1) the closed-loop frequency response peak, (2) phase margin

22. a. Find T(jω)=G(jω)/[1+G(jω)H(jω)] and plot in polar form or separate magnitude and phase plots. **b.** Superimpose G(jω)H(jω) over the M and N circles and plot. **c.** Superimpose G(jω)H(jω) over the Nichols chart and plot.

23. For Type zero: K_p = low frequency gain; For Type 1: K_v = frequency value at the intersection of the initial slope with the frequency axis; For Type 2: K_a = square root of the frequency value at the intersection of the initial slope with the frequency axis.

24. No change at all

25. A straight line of negative slope, ωT, where T is the time delay

26. When the magnitude response is flat and the phase response is flat at 0^o.

SOLUTIONS TO PROBLEMS

1.

a.

$$G(s) = \frac{1}{s(s+2)(s+4)};\quad G(j\omega) = \frac{1}{-6\omega^2 + i(8\omega - \omega^3)}$$

$$M(\omega) = \frac{1}{\sqrt{(8\omega-\omega^3)^2 + (6\omega^2)^2}};\quad \Phi(\omega) = -\left(\pi + \arctan\left[\frac{8-\omega^2}{-6\omega}\right]\right)$$

b.

$$G(s) = \frac{(s+5)}{(s+2)(s+4)};\quad G(j\omega) = \frac{(\omega^2+40) - i(\omega^2+22)\omega}{\omega^4 + 20\omega^2 + 64}$$

$$M(\omega) = \frac{\sqrt{(\omega^2+40)^2 + \omega^2(\omega^2+22)^2}}{\omega^4 + 20\omega^2 + 64};\quad \Phi(\omega) = \arctan\left(\frac{-[\omega^2+22]\omega}{\omega^2+40}\right)$$

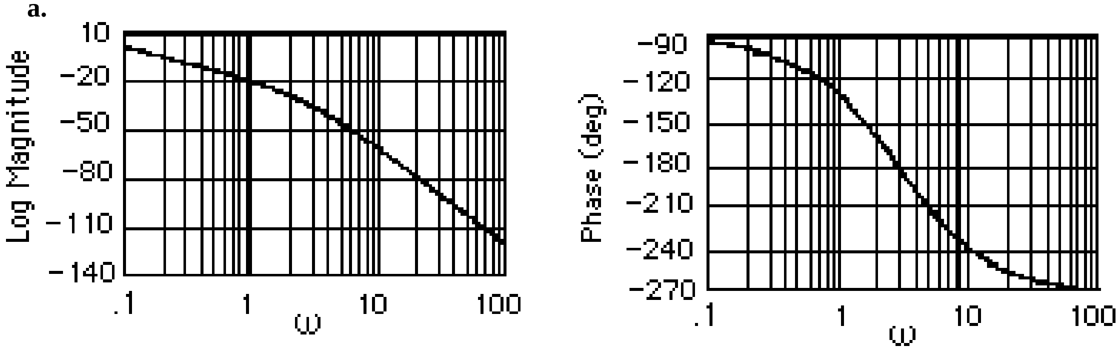
c.

$$G(s) = \frac{(s+3)(s+5)}{s(s+4)(s+2)};\quad G(j\omega) = \frac{-2\omega(\omega^2+13) - i(\omega^4 + 25\omega^2 + 120)}{\omega^5 + 20\omega^3 + 64\omega}$$

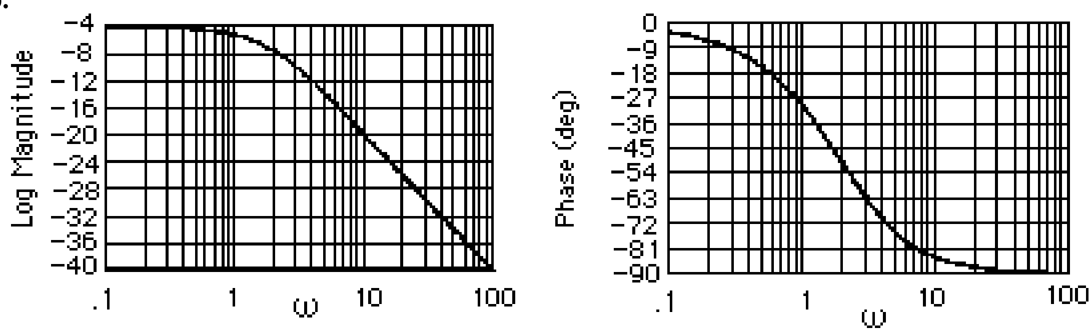
$$M(\omega) = \frac{\sqrt{(2\omega[\omega^2+13])^2 + (\omega^4+25\omega^2+120)^2}}{\omega^5 + 20\omega^3 + 64\omega};\quad \Phi(\omega) = \pi + \arctan\left(\frac{\omega^4+25\omega^2+120}{2\omega[\omega^2+13]}\right)$$

2.

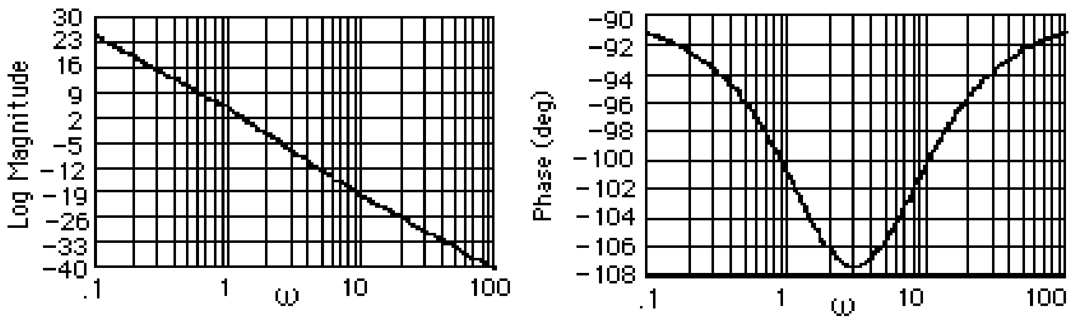
a.



b.

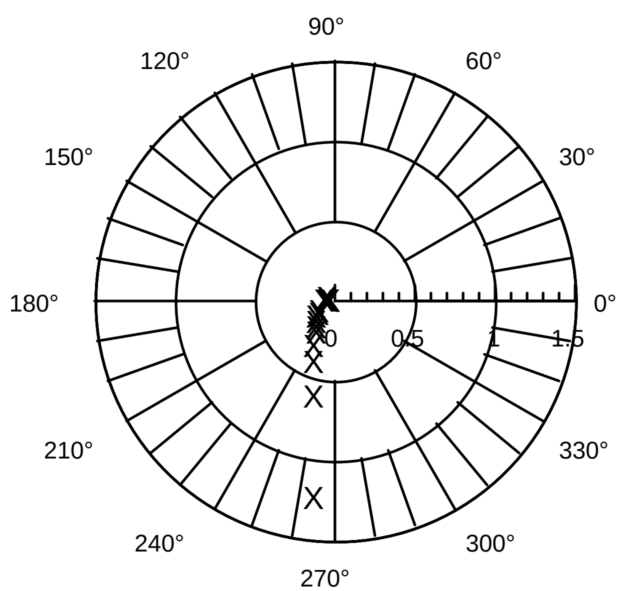

Log Magnitude
-4
-8
-12
-16
-20
-24
-28
-32
-36
-40
.1
1
ω
10
100
Phase (deg)
0
-9
-18
-27
-36
-45
-54
-63
-72
-81
-90
.1
1
ω
10
100

c.

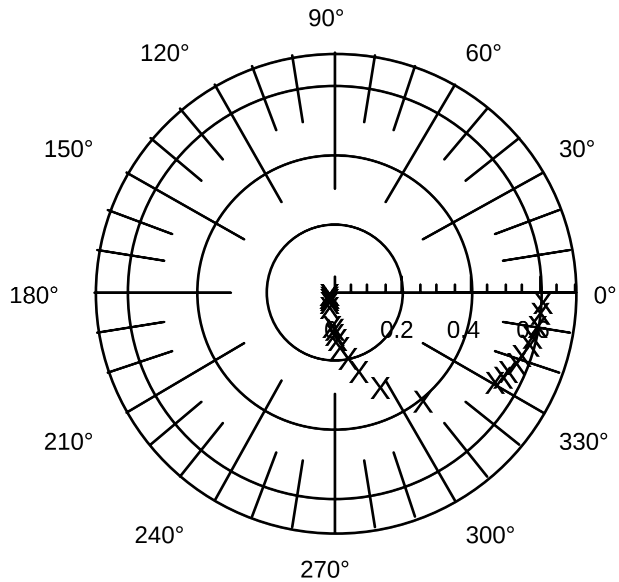

Log Magnitude
30
23
16
9
2
-5
-12
-19
-26
-33
-40
.1
1
ω
10
100
Phase (deg)
-90
-92
-94
-96
-98
-100
-102
-104
-106
-108
.1
1
ω
10
100

3.

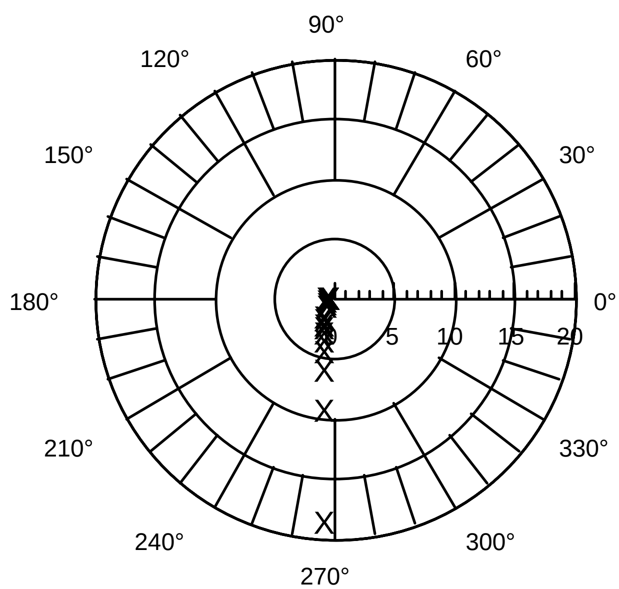
a.


90°
120°
60°
150°
30°
180°
0°
0
0.5
1
1.5
210°
330°
240°
300°
270°

b.

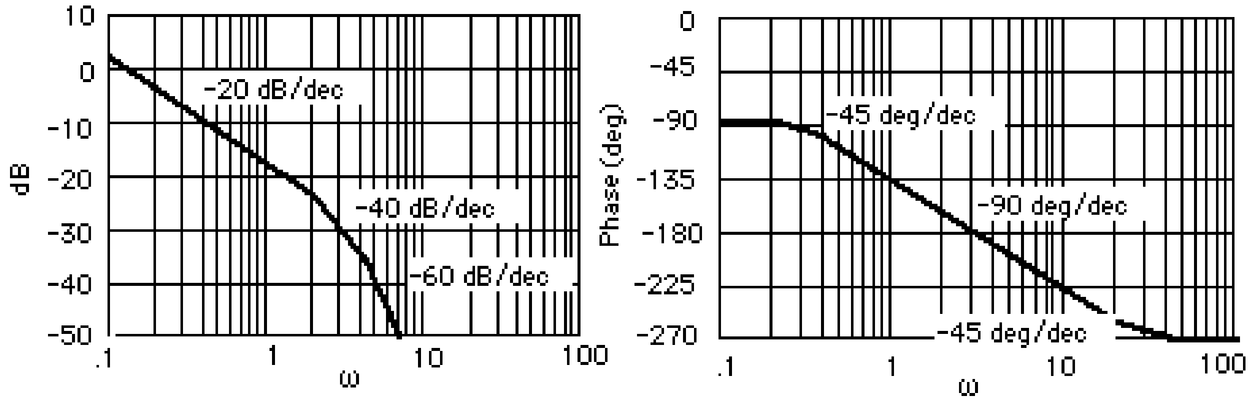

90°
120°
60°
150°
30°
180°
0°
0.2
0.4
210°
330°
240°
300°
270°

c.

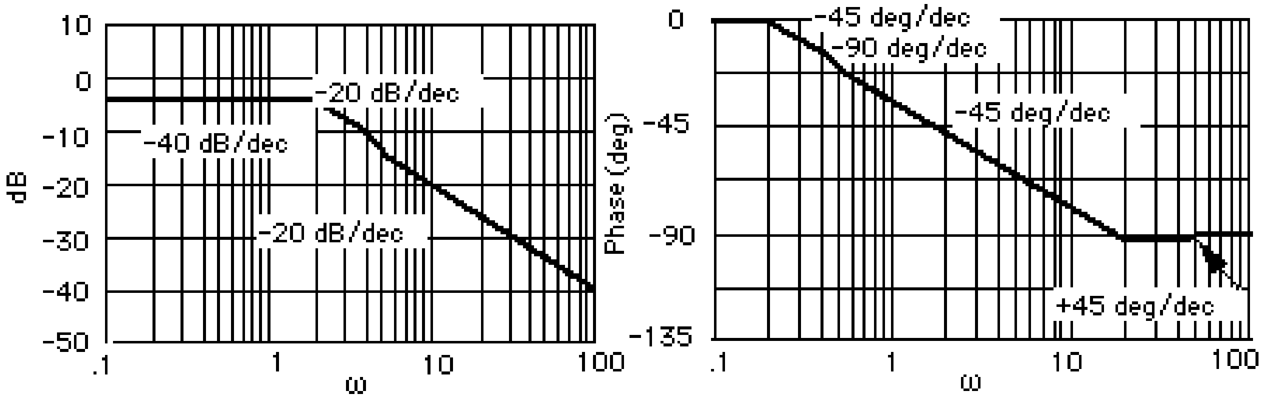

90°
120°
60°
150°
30°
180°
0°
5
10
15
20
210°
330°
240°
300°
270°

4.

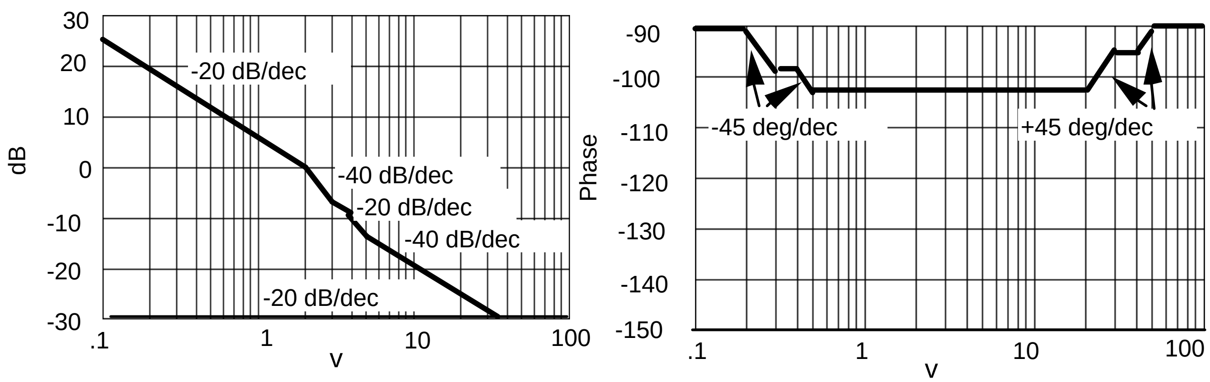
a.


10
0
-10
-20
-30
-40
-50
dB
.1
1
10
100
ω
-20 dB/dec
-40 dB/dec
-60 dB/dec
0
-45
-90
-135
-180
-225
-270
Phase (deg)
-45 deg/dec
-90 deg/dec
-45 deg/dec

b.

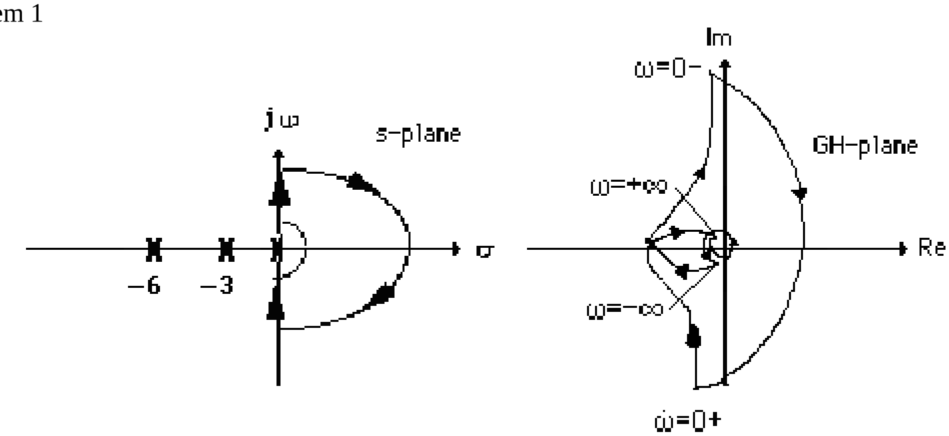

10
0
-10
-20
-30
-40
-50
dB
.1
1
10
100
ω
-20 dB/dec
-40 dB/dec
-20 dB/dec
0
-45
-90
-135
Phase (deg)
-45 deg/dec
-90 deg/dec
-45 deg/dec
+45 deg/dec

c.

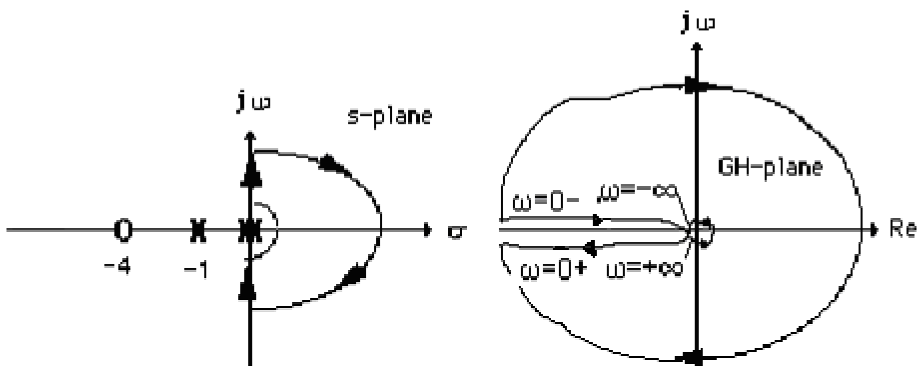

30
20
10
0
-10
-20
-30
dB
.1
1
10
100
v
-20 dB/dec
-40 dB/dec
-20 dB/dec
-40 dB/dec
-20 dB/dec
-90
-100
-110
-120
-130
-140
-150
Phase
-45 deg/dec
+45 deg/dec

5.

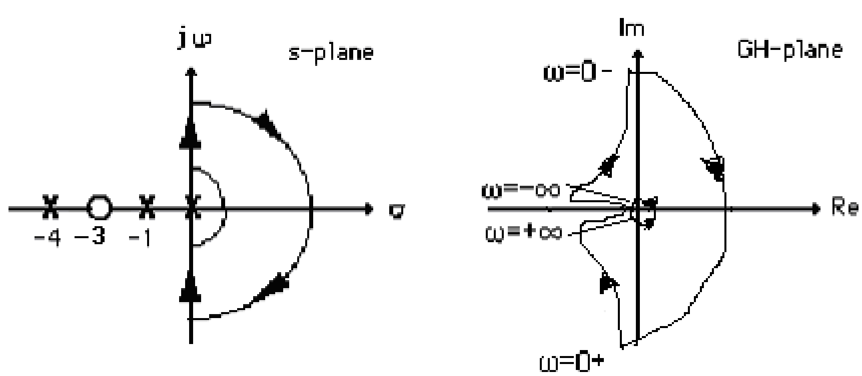
a. System 1



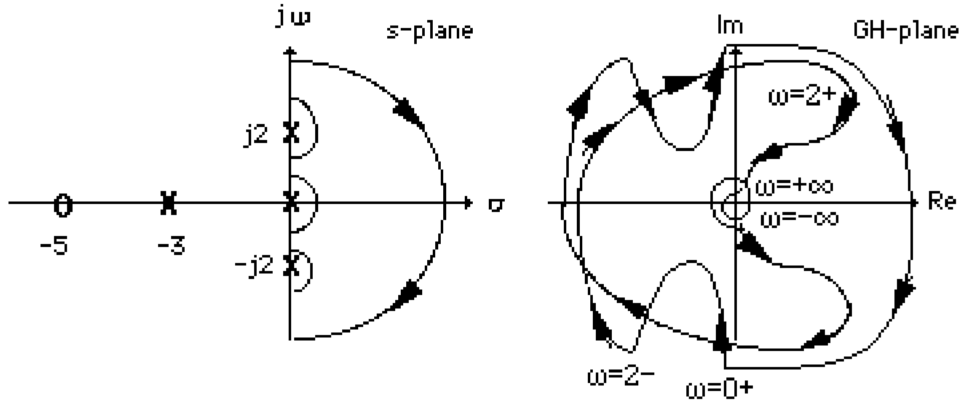
b. System 2



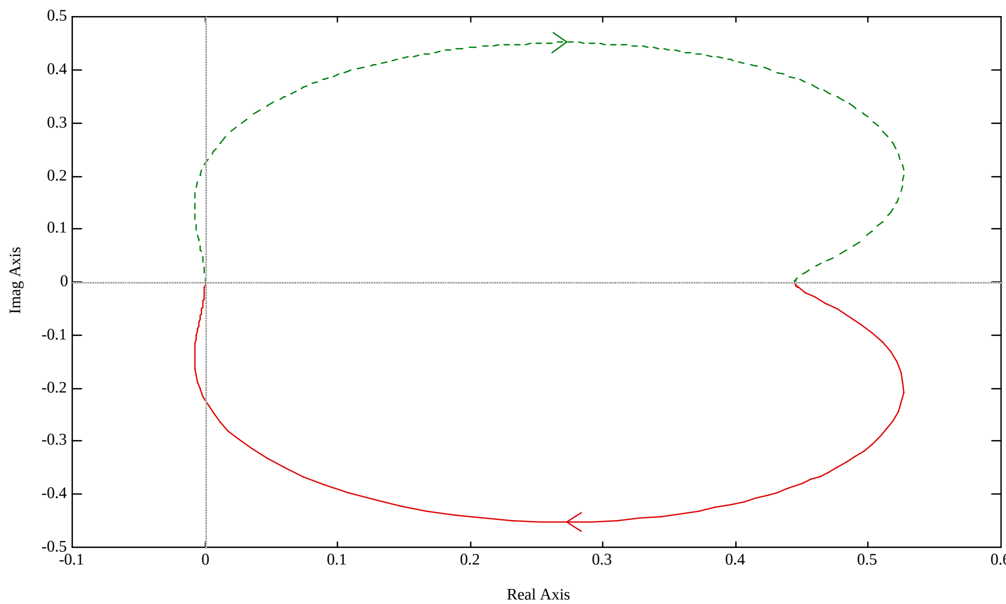
c. System 3



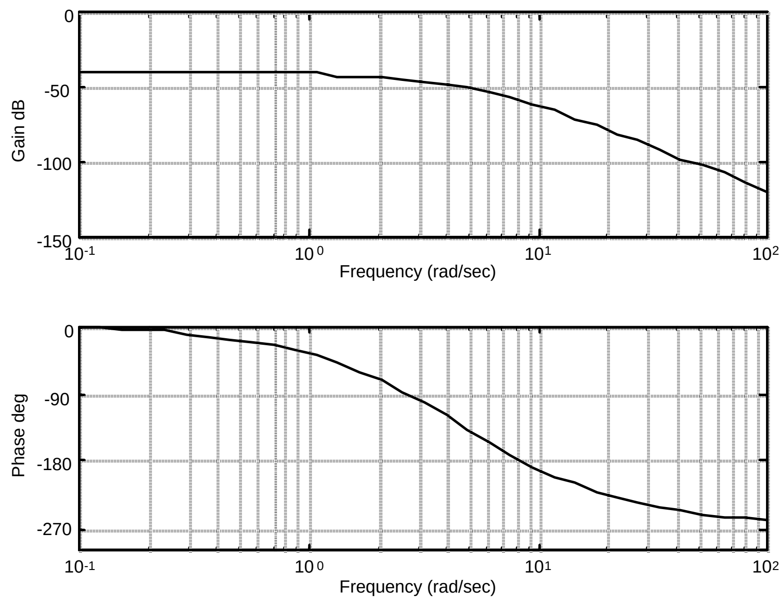
d.


jω
s-plane
j2
-5
-3
-j2
σ
Im
GH-plane
ω=2+
ω=+∞
ω=−∞
Re
ω=2−
ω=0+

6.


0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
Imag Axis
-0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
Real Axis

7.



8.

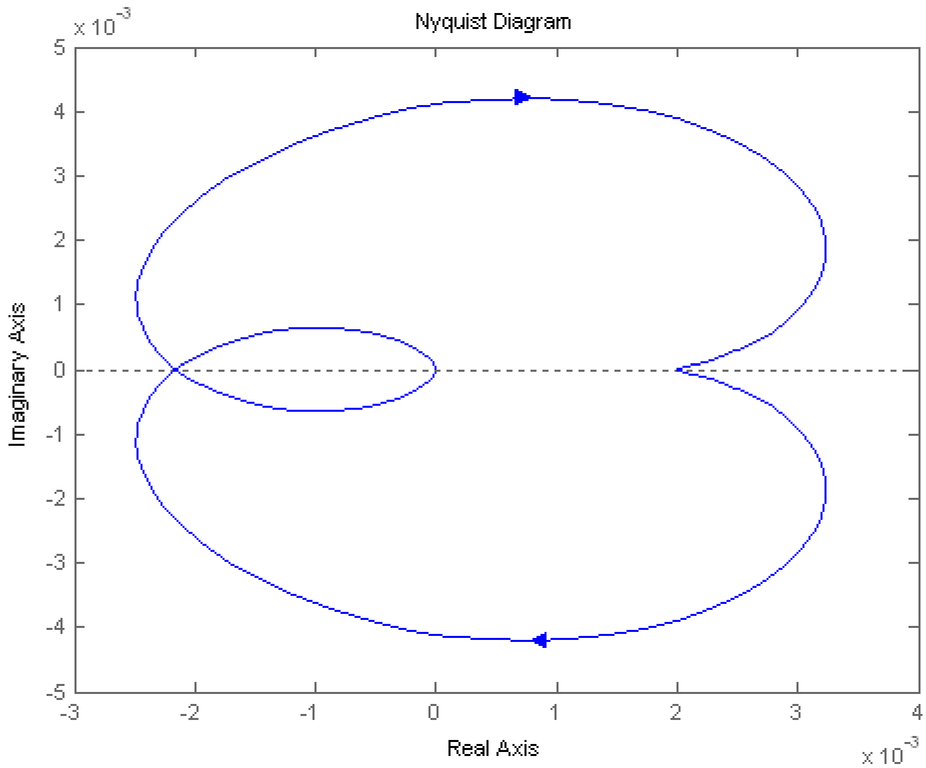
Program:

```
numg=[1 5];
deng=conv([1 6 100],[1 4 25]);
G=tf(numg,deng);
'G(s)'
Gzpk=zpk(G)
nyquist(G)
axis([-3e-3,4e-3,-5e-3,5e-3])
w=0:0.1:100;
[re,im]=nyquist(G,w);
for i=1:1:length(w)
M(i)=abs(re(i)+j*im(i));
A(i)=atan2(im(i),re(i))*(180/pi);
if 180-abs(A(i))<=1;
re(i);
im(i);
K=1/abs(re(i));
fprintf('\nw = %g',w(i))
fprintf(', Re = %g',re(i))
fprintf(', Im = %g',im(i))
fprintf(', M = %g',M(i))
fprintf(', Angle = %g',A(i))
fprintf(', K = %g',K)
Gm=20*log10(1/M(i));
fprintf(', Gm = %g',Gm)
break
end
end
```

Computer response:

```
ans =

G(s)


Zero/pole/gain:
              (s+5)
```

```
----------------------------------
(s^2  + 4s + 25) (s^2  + 6s + 100)


w = 10.1, Re = -0.00213722, Im = 2.07242e-005, M = 0.00213732, Angle =
179.444, K = 467.898, Gm = 53.4026

ans =

G(s)


Zero/pole/gain:
              (s+5)
----------------------------------
(s^2  + 4s + 25) (s^2  + 6s + 100)


w = 10.1, Re = -0.00213722, Im = 2.07242e-005, M = 0.00213732, Angle =
179.444, K = 467.898, Gm = 53.4026
```



9.

a. Since the real-axis crossing is at -0.3086, P = 0, N = 0. Therefore Z = P - N = 0. System is stable.

Derivation of real-axis crossing:

$$G(j\omega) = \left.\frac{50}{s(s+3)(s+6)}\right|_{s=j\omega} = \frac{50\left[-9\omega^2 - j\omega\left(18-\omega^2\right)\right]}{81\omega^4 + \left(18\omega - \omega^3\right)}.$$

Thus, the imaginary part = 0 at $\omega = \sqrt{18}$. Substituting this frequency into $G(j\omega)$, the real part is evaluated to be -0.3086.

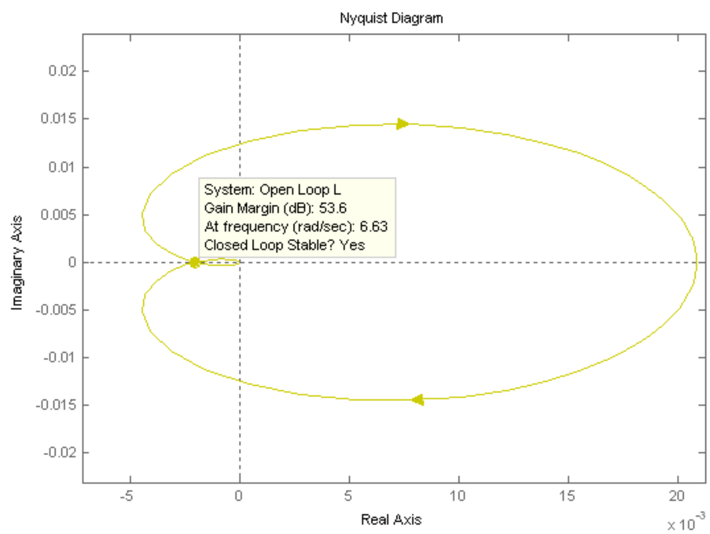
b. P = 0, N = -2. Therefore Z = P - N = 2. System is unstable.

c. P = 0, N = 0. Therefore Z = P - N = 0. System is stable

d. P = 0, N = -2. Therefore Z = P - N = 2. System is unstable.

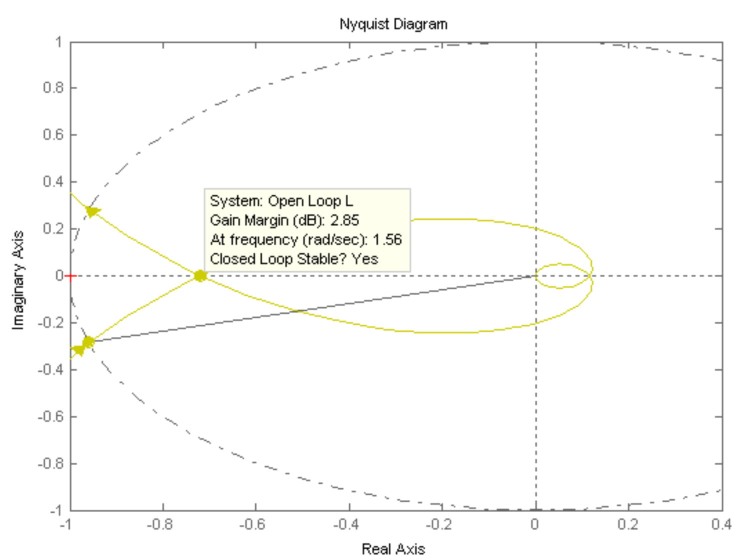
10.

System 1: For K = 1,



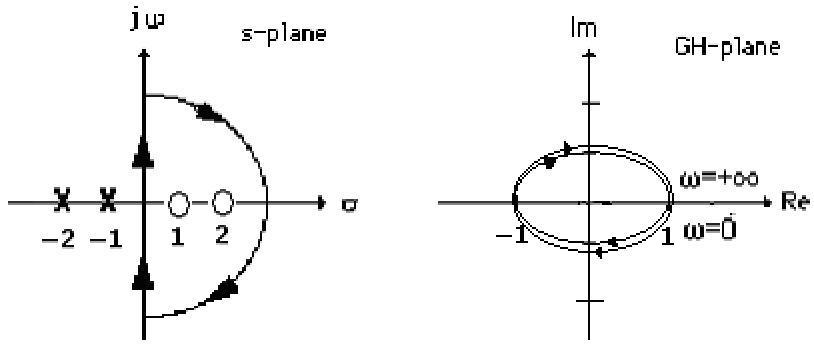
The Nyquist diagram intersects the real axis at -0.0021. Thus K can be increased to 478.63 before there are encirclements of -1. There are no poles encircles by the contour. Thus P = 0. Hence, Z = P - N, Z = 0 + 0 if K <478.63; Z = 0 –(-2) if K > 478.63. Therefore stability if 0 < K < 478.63.

System 2: For K = 1,



The Nyquist diagram intersects the real axis at -0.720. Thus K can be increased to 1.39 before there are encirclements of -1. There are no poles encircles by the contour. Thus P = 0. Hence, Z = P - N, Z = 0 + 0 if K <1.39; Z = 0 – (-2) if K > 1.39. Therefore stability if 0 < K < 1.39.

System 3: For K = 1,

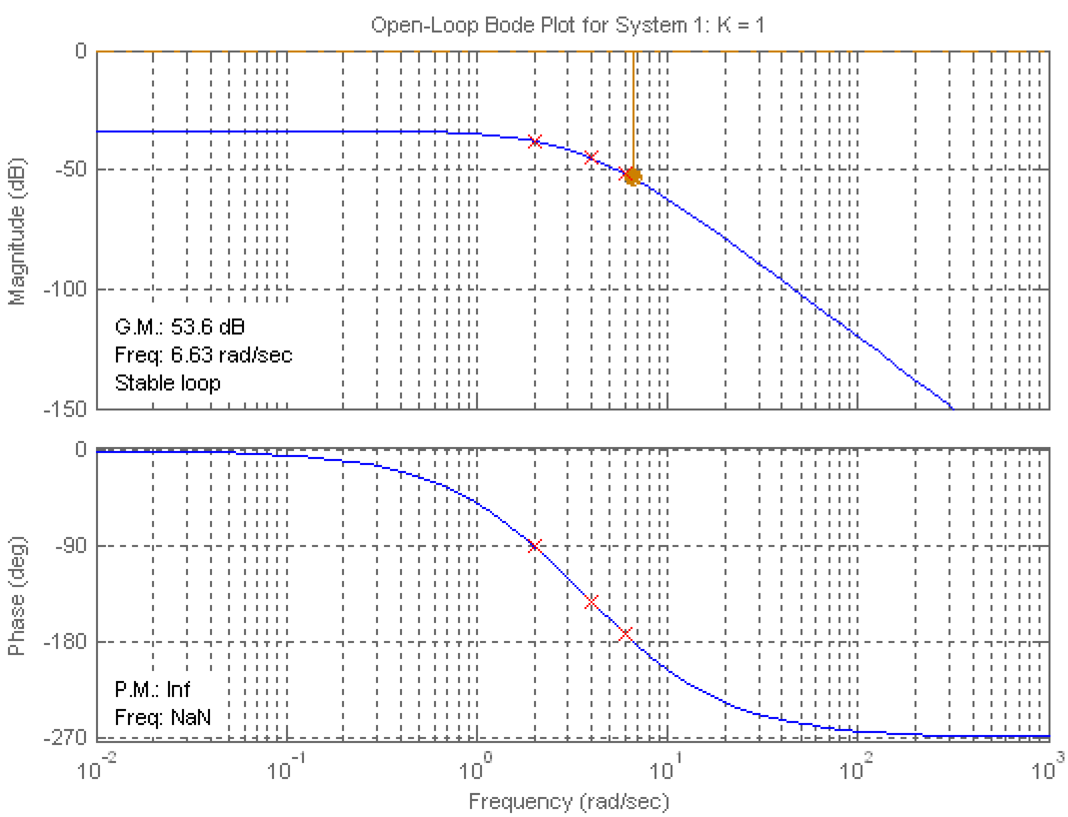


Stable if 0<K<1.

11.

Note: All results for this problem are based upon a non-asymptotic frequency response.

System 1: Plotting Bode plots for K = 1 yields the following Bode plot,



K = 1000:

For K = 1, phase response is 180^o at ω = 6.63 rad/s. Magnitude response is -53.6 dB at this frequency. For K = 1000, magnitude curve is raised by 60 dB yielding + 6.4 dB at 6.63 rad/s. Thus, the gain margin is

- 6.4 dB.

Phase margin: Raising the magnitude curve by 60 dB yields 0 dB at 9.07 rad/s, where the phase curve is 200.3^o. Hence, the phase margin is 180^o-200.3^o = - 20.3^o.

K = 100:

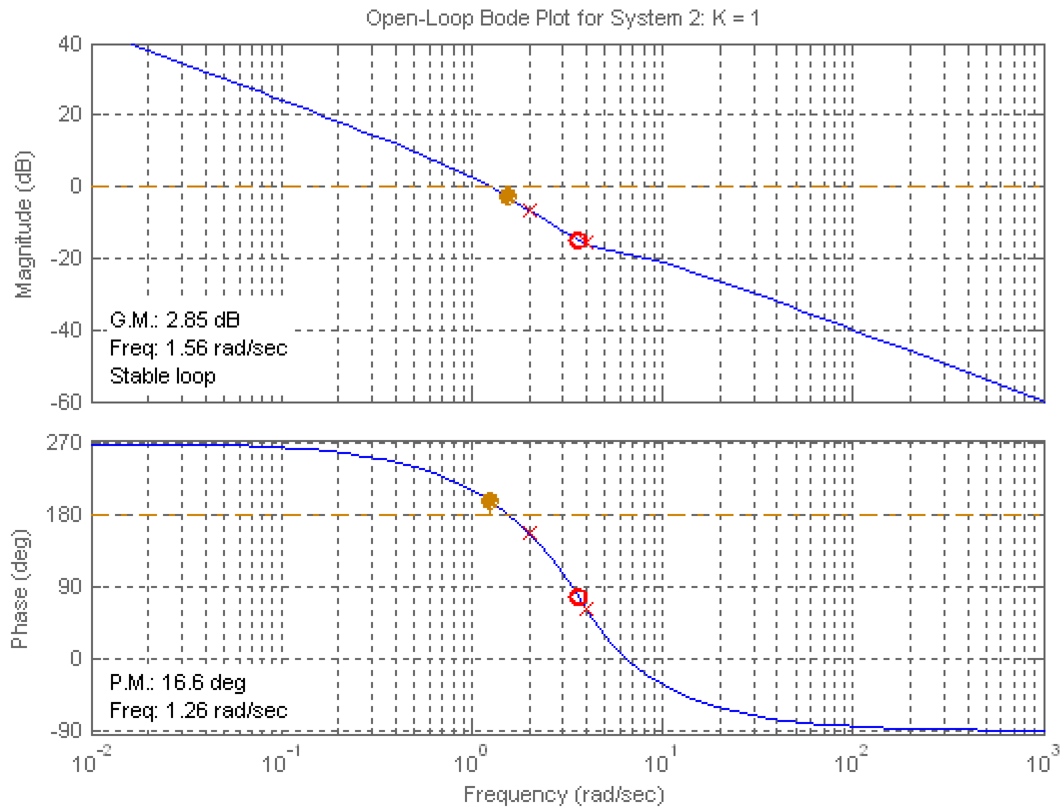
For K = 1, phase response is 180^o at ω = 6.63 rad/s. Magnitude response is -53.6 dB at this frequency. For K = 100, magnitude curve is raised by 40 dB yielding – 13.6 dB at 6.63 rad/s. Thus, the gain margin is 13.6 dB.

Phase margin: Raising the magnitude curve by 40 dB yields 0 dB at 2.54 rad/s, where the phase curve is 107.3^o. Hence, the phase margin is 180^o-107.3^o = 72.7^o.

K = 0.1:

For K = 1, phase response is 180^o at ω = 6.63 rad/s. Magnitude response is -53.6 dB at this frequency. For K = 0.1, magnitude curve is lowered by 20 dB yielding – 73.6 dB at 6.63 rad/s. Thus, the gain margin is 73.6 dB..

System 2: Plotting Bode plots for K = 1 yields



K = 1000:

For K = 1, phase response is 180° at ω = 1.56 rad/s. Magnitude response is -2.85 dB at this frequency. For K = 1000, magnitude curve is raised by 60 dB yielding + 57.15 dB at 1.56 rad/s. Thus, the gain margin is

– 57.15 dB.

Phase margin: Raising the magnitude curve by 54 dB yields 0 dB at 500 rad/s, where the phase curve is -91.03°. Hence, the phase margin is $180^{\circ}-91.03^{\circ} = 88.97^{\circ}$.

K = 100:

For K = 1, phase response is 180° at ω = 1.56 rad/s. Magnitude response is -2.85 dB at this frequency. For K = 100, magnitude curve is raised by 40 dB yielding + 37.15 dB at 1.56 rad/s. Thus, the gain margin is

– 37.15 dB.

Phase margin: Raising the magnitude curve by 40 dB yields 0 dB at 99.8 rad/s, where the phase curve is -84.3°. Hence, the phase margin is $180^{\circ}-84.3^{\circ} = 95.7^{\circ}$.

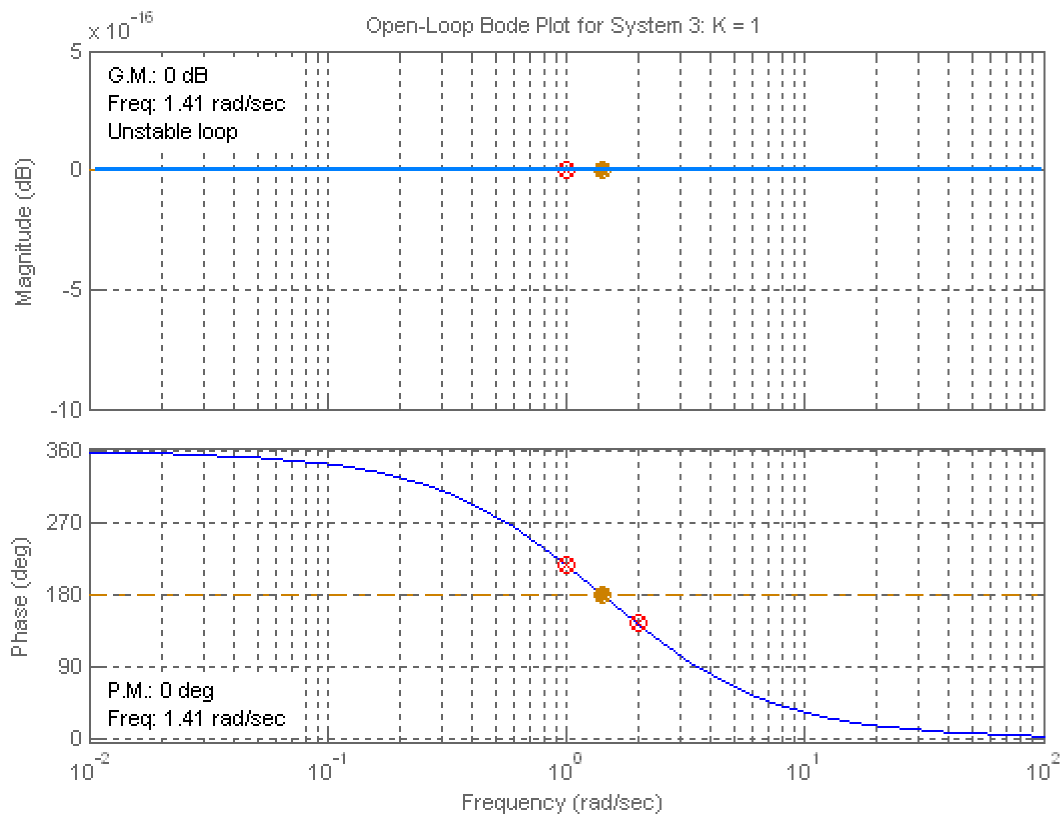
K = 0.1:

For K = 1, phase response is 180° at ω = 1.56 rad/s. Magnitude response is -2.85 dB at this frequency. For K = 0.1, magnitude curve is lowered by 20 dB yielding – 22.85 dB at 1.56 rad/s. Thus, the gain margin is

– 22.85 dB.

Phase margin: Lowering the magnitude curve by 20 dB yields 0 dB at 0.162 rad/s, where the phase curve is -99.8^{o}. Hence, the phase margin is 180^{o}-99.86^{o} = 80.2^{o}.

System 3: Plotting Bode plots for K = 1 yields



K = 1000:

For K = 1, phase response is 180^{o} at ω = 1.41 rad/s. Magnitude response is 0 dB at this frequency.

For K = 1000, magnitude curve is raised by 60 dB yielding 60 dB at 1.41 rad/s. Thus, the gain margin is - 60 dB.

Phase margin: Raising the magnitude curve by 60 dB yields no frequency where the magnitude curve is 0 dB. Hence, the phase margin is infinite.

K = 100:

For K = 1, phase response is 180^{o} at ω = 1.41 rad/s. Magnitude response is 0 dB at this frequency.

For K = 100, magnitude curve is raised by 40 dB yielding 40 dB at 1.41 rad/s. Thus, the gain margin is - 40 dB.

Phase margin: Raising the magnitude curve by 40 dB yields no frequency where the magnitude curve is 0 dB. Hence, the phase margin is infinite.

K = 0.1:

For K = 1, phase response is 180° at ω = 1.41 rad/s. Magnitude response is 0 dB at this frequency.

For K = 0.1, magnitude curve is lowered by 20 dB yielding -20 dB at 1.41 rad/s. Thus, the gain margin is 20 dB.

Phase margin: Lowering the magnitude curve by 20 dB yields no frequency where the magnitude curve is 0 dB. Hence, the phase margin is infinite.

12.

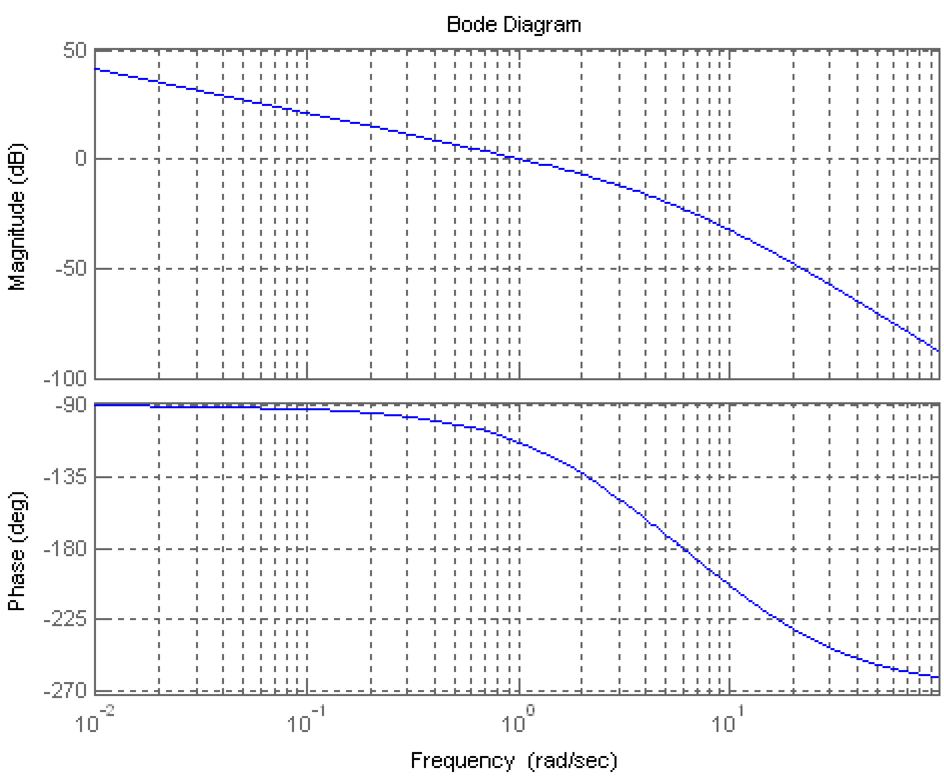
Program:

```
%Enter G(s)************************
numg=1;
deng=poly([0 -3 -12]);
'G(s)'
G=tf(numg,deng)
w=0.01:0.1:100;
%Enter K **************************
K=input('Type gain, K ');
bode(K*G,w)
pause
[M,P]=bode(K*G,w);
%Calculate Gain Margin**************
for i=1:1:length(P);
if P(i)<=-180;
fprintf('\nGain K = %g',K)
fprintf(', Frequency(180 deg) = %g',w(i))
fprintf(', Magnitude = %g',M(i))
fprintf(', Magnitude (dB) = %g',20*log10(M(i)))
fprintf(', Phase = %g',P(i))
Gm=20*log10(1/M(i));
fprintf(', Gain Margin (dB) = %g',Gm)
break
end
end
%Calculate Phase Margin**************
for i=1:1:length(M);
if M(i)<=1;
fprintf('\nGain K = %g',K)
fprintf(', Frequency (0 dB) = %g',w(i))
fprintf(', Magnitude = %g',M(i))
fprintf(', Magnitude (dB) = %g',20*log10(M(i)))
fprintf(', Phase = %g',P(i))
Pm=180+P(i);;
fprintf(', Phase Margin = %g',Pm)
break
end
end

'Alternate program using MATLAB margin function:'

clear
clf
%Bode Plot and Find Points
%Enter G(s)************************
numg=1;
deng=poly([0 -3 -12]);
'G(s)'
G=tf(numg,deng)
w=0.01:0.1:100;
```

```
%Enter K ***************************
K=input('Type gain, K ');
bode(K*G,w)
[Gm,Pm,Wcp,Wcg]=margin(K*G)
'Gm(dB)'
20*log10(Gm)
```

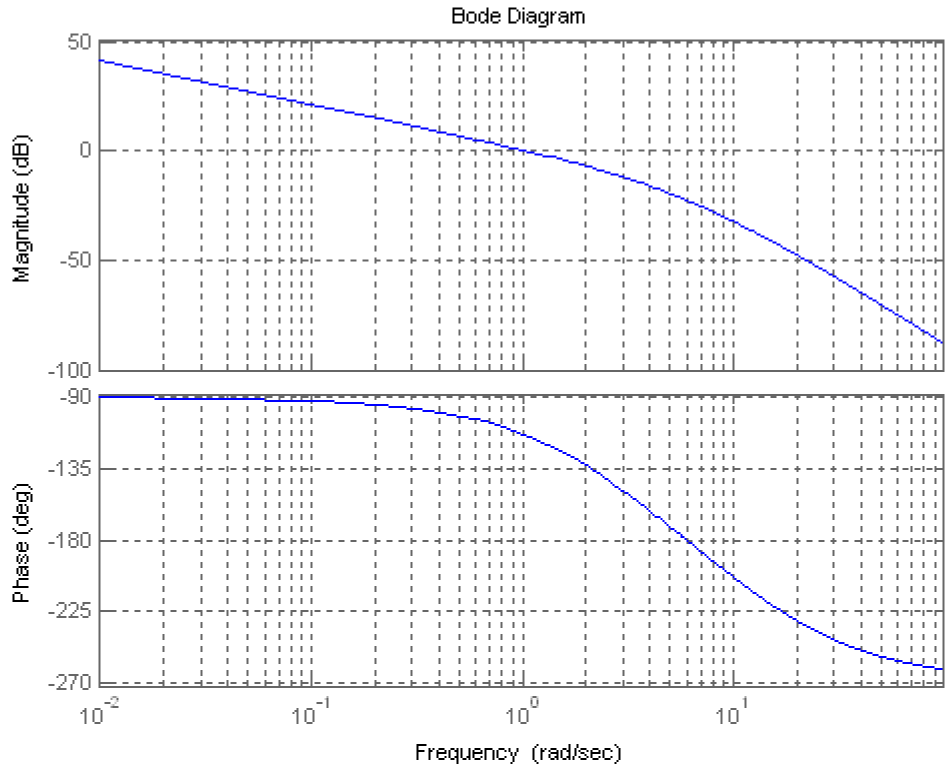
Computer response:

```
ans =

G(s)


Transfer function:
         1
-------------------
s^3 + 15 s^2 + 36 s

Type gain, K 40

Gain K = 40, Frequency(180 deg) = 6.01, Magnitude = 0.0738277, Magnitude
(dB) = -22.6356, Phase = -180.076, Gain Margin (dB) = 22.6356
Gain K = 40, Frequency (0 dB) = 1.11, Magnitude = 0.93481, Magnitude (dB) =
-0.585534, Phase = -115.589, Phase Margin = 64.4107
```



Alternate program using MATLAB margin function:

```
ans =

G(s)


Transfer function:
         1
-------------------
s^3 + 15 s^2 + 36 s

Type gain, K 40

Gm =

   13.5000


Pm =

   65.8119


Wcp =

     6


Wcg =

    1.0453


ans =

Gm(dB)


ans =

   22.6067
```



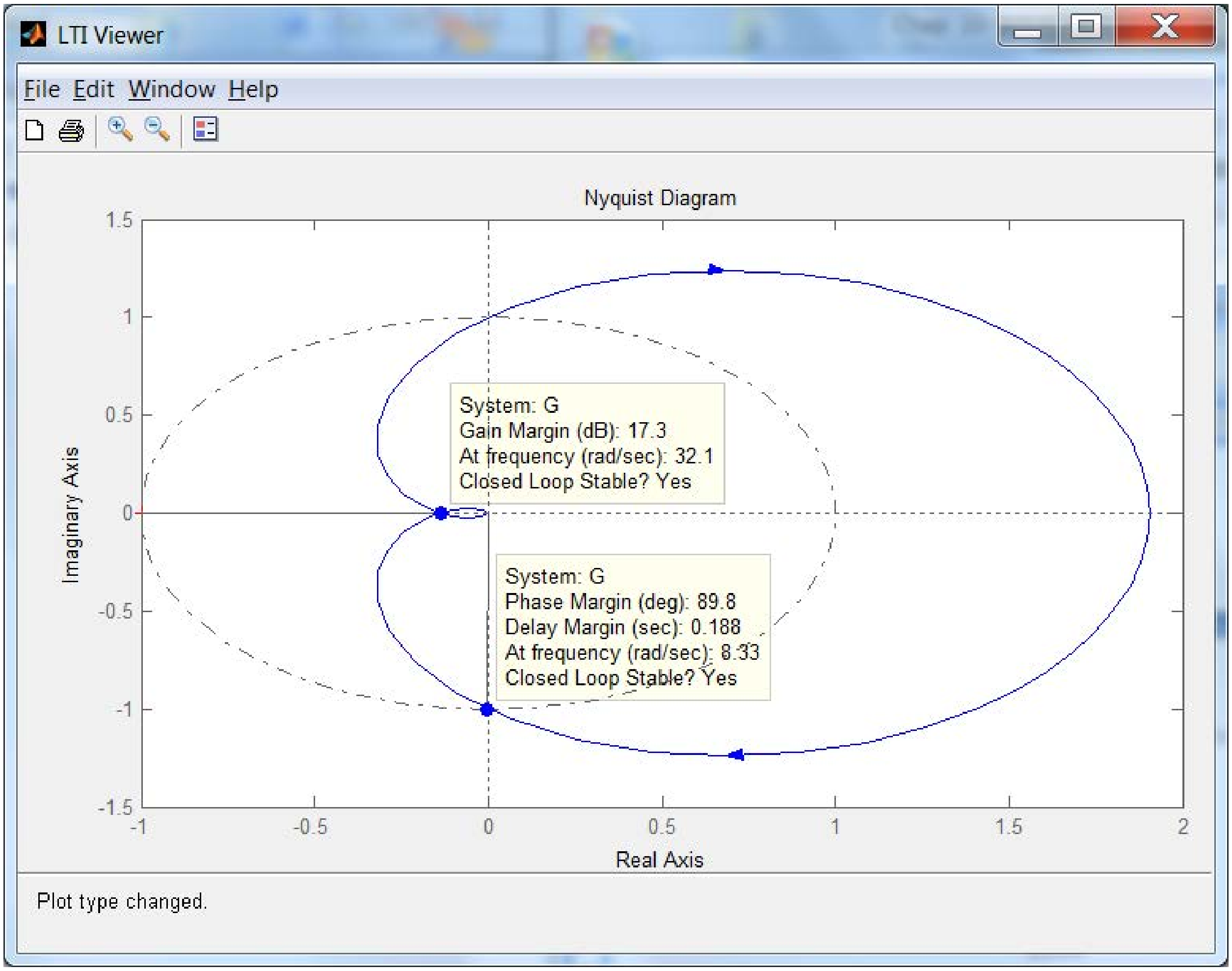
13.

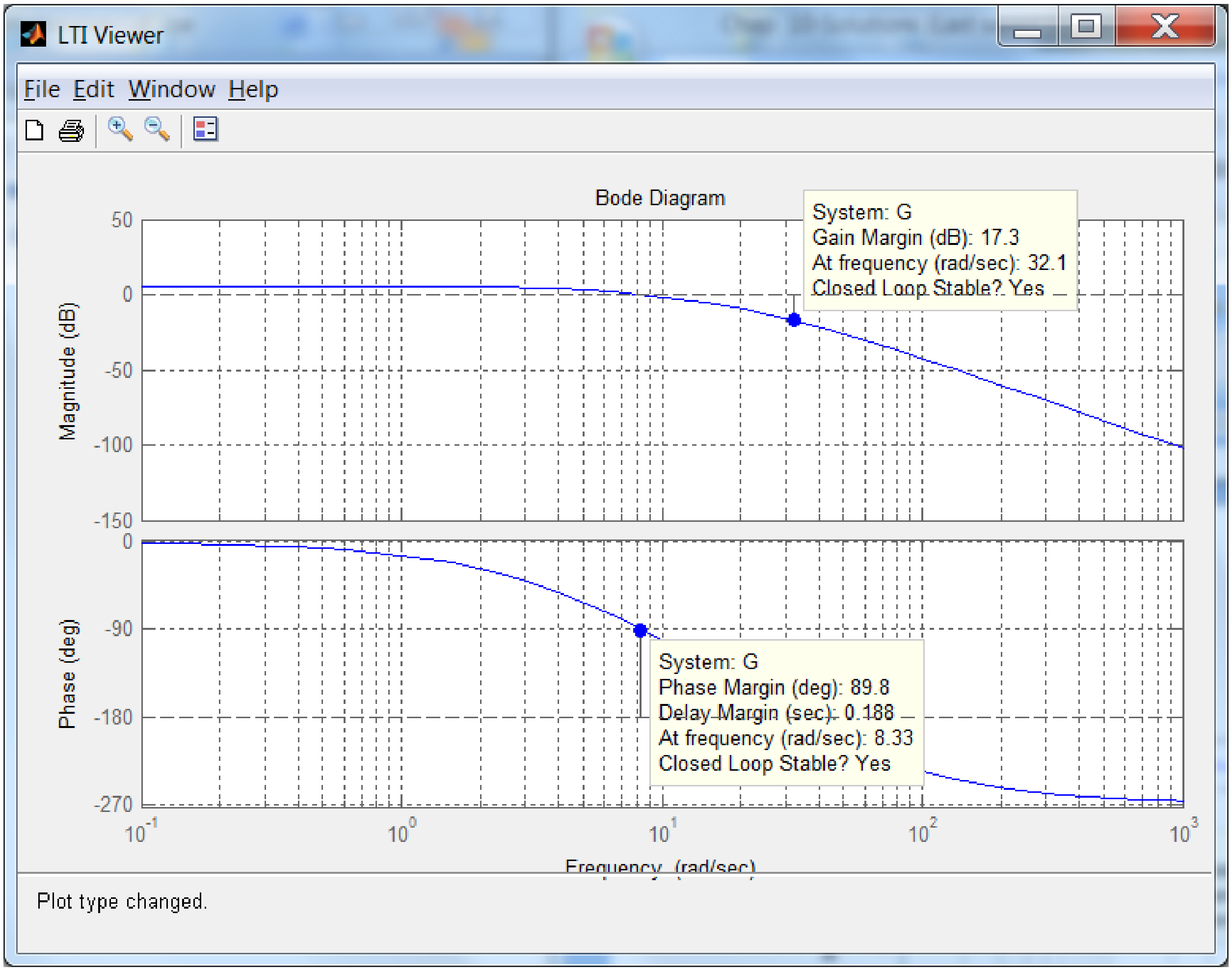
Program:

```
numg=8000;
deng=poly([-6 -20 -35]);
G=tf(numg,deng)
ltiview
```

Computer response:

```
Transfer function:
            8000
----------------------------
s^3 + 61 s^2 + 1030 s + 4200
```


LTI Viewer
File Edit Window Help
Nyquist Diagram
System: G
Gain Margin (dB): 17.3
At frequency (rad/sec): 32.1
Closed Loop Stable? Yes
System: G
Phase Margin (deg): 89.8
Delay Margin (sec): 0.188
At frequency (rad/sec): 8.33
Closed Loop Stable? Yes
Imaginary Axis
Real Axis
1.5
1
0.5
0
-0.5
-1
-1.5
-1
-0.5
0
0.5
1
1.5
2
Plot type changed.



14.

Squaring Eq. (10.51) and setting it equal to $\left(\frac{1}{\sqrt{2}}\right)^2$ yields

$$\frac{\omega_n^4}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2} = \frac{1}{2}$$

Simplifying,

$$\omega^4 + 2\omega_n^2(2\zeta^2 - 1)\omega^2 - \omega_n^4 = 0$$

Solving for ω^2 using the quadratic formula and simplifying yields,

$$\omega^2 = \omega_n^2 \left[-(2\zeta^2 - 1) \pm \sqrt{4\zeta^4 - 4\zeta^2 + 2} \right]$$

Taking the square root and selecting the positive term,

$$\omega = \omega_n\sqrt{(1 - 2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

15.

a. Using Eq. (10.55), ω_{BW} = 10.06 rad/s.

b. Using Eq. (10.56), ω_{BW} = 1.613 rad/s.

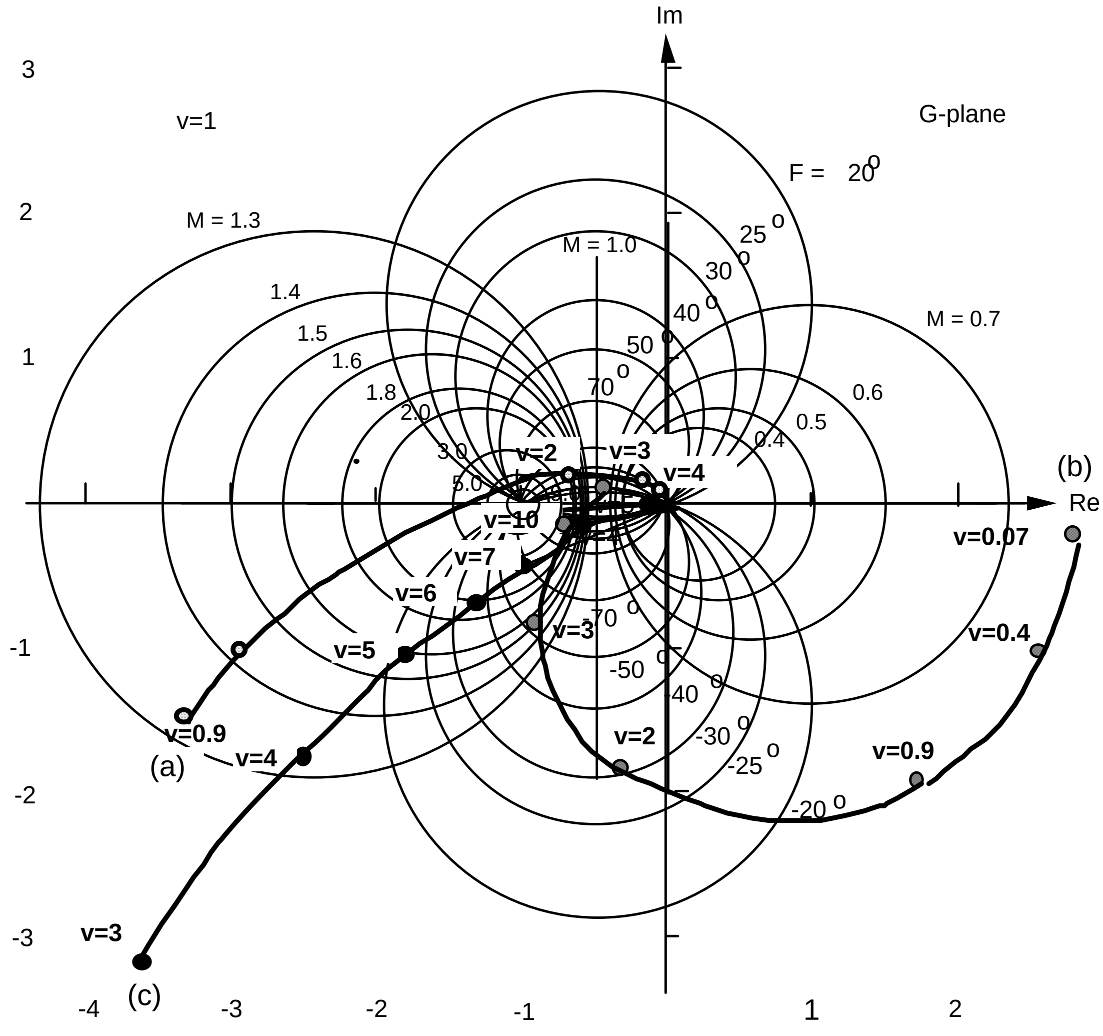
c. First find ζ. Since $T_s = \dfrac{4}{\zeta\omega_n}$ and $T_p = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$, $\dfrac{T_p}{T_s} = \dfrac{\zeta\pi}{4\sqrt{1-\zeta^2}}$. Solving for ζ with $\dfrac{T_p}{T_s} = 0.5$

yields ζ = 0.537. Using either Eq. (10.55) or (10.56) yields ω_{BW} = 2.29 rad/s.

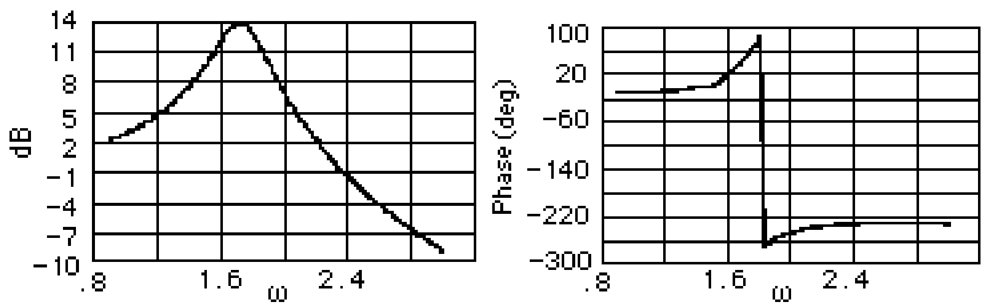
d. Using ζ = 0.3, $\omega_n T_r = 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1 = 1.3217$. Hence,

$\omega_n = \dfrac{1.3217}{T_r} = \dfrac{1.3217}{4} = 0.3304$ rad/s. Using Eq. (10.54) yields ω_{BW} = 0.4803 rad/s.

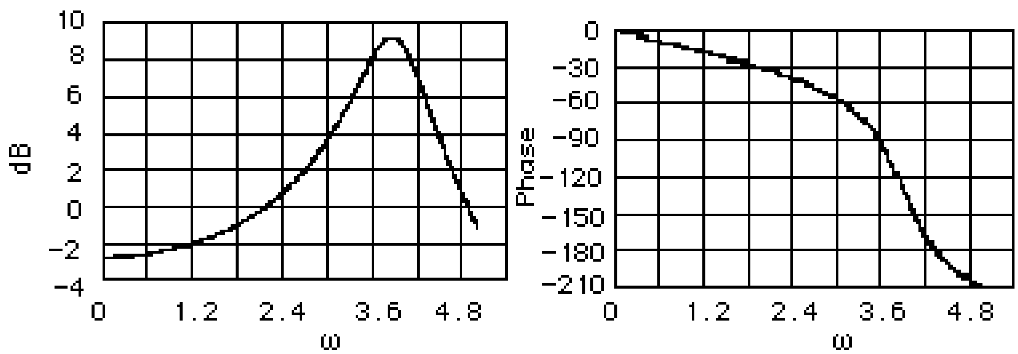
16.



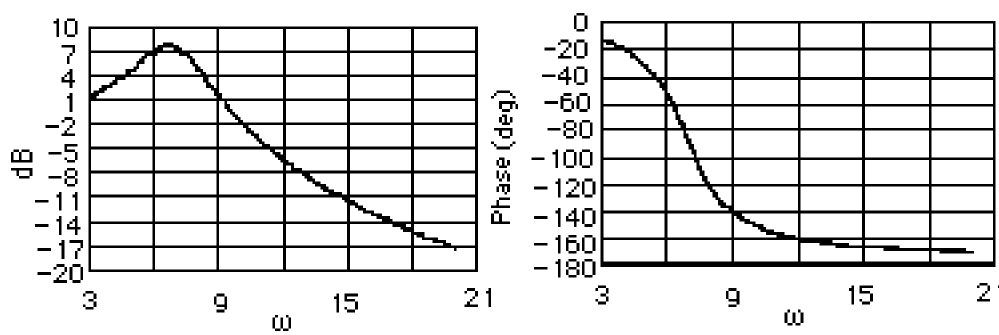
a.


14
11
8
5
2
−1
−4
−7
−10
dB
.8
1.6
2.4
ω
100
20
−60
−140
−220
−300
Phase (deg)
.8
1.6
2.4
ω

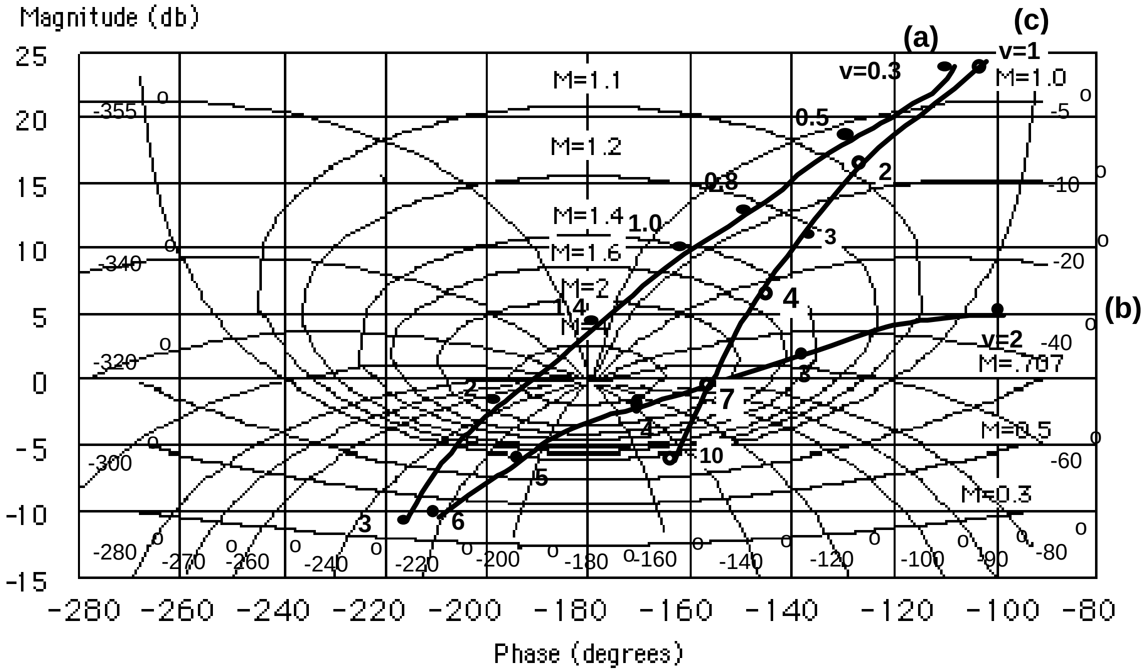
b.


10
8
6
4
2
0
−2
−4
dB
0
1.2
2.4
3.6
4.8
ω
0
−30
−60
−90
−120
−150
−180
−210
Phase
0
1.2
2.4
3.6
4.8
ω

c.


10
7
4
1
−2
−5
−8
−11
−14
−17
−20
dB
3
9
15
21
ω
0
−20
−40
−60
−80
−100
−120
−140
−160
−180
Phase (deg)
3
9
15
21
ω

17.



18.

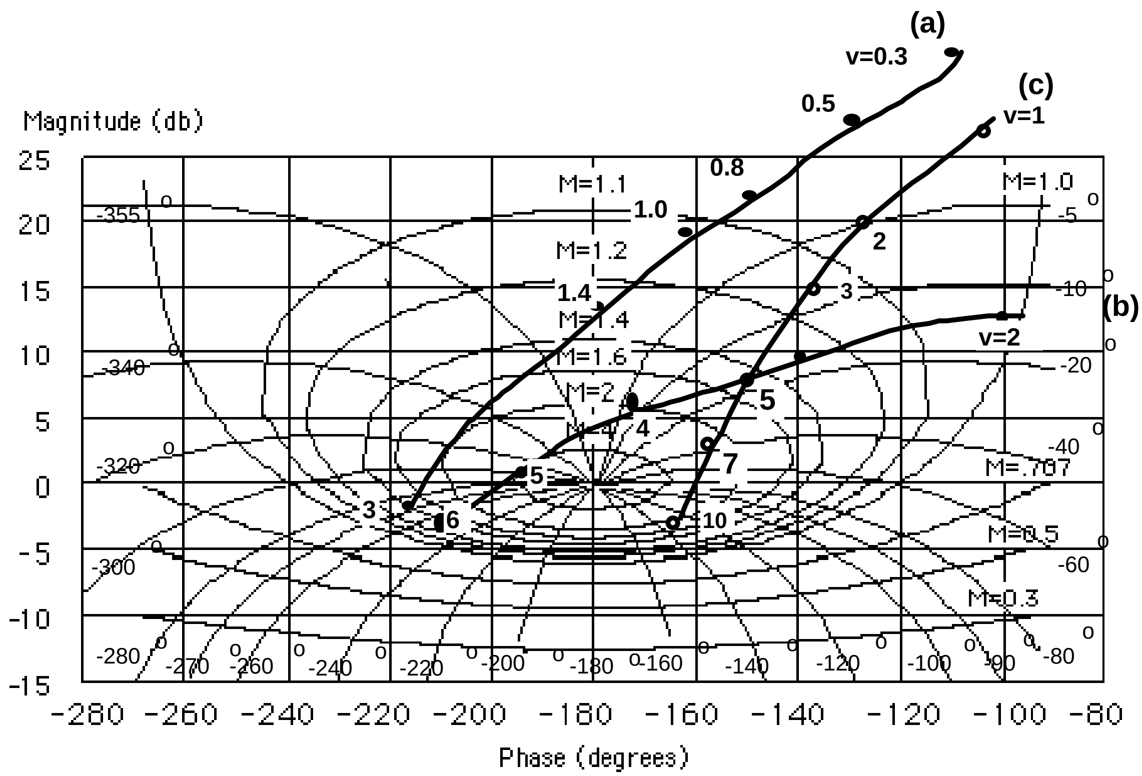
a. The polar plot is approximately tangent to M = 5. Using Figure 10.40, the student would estimate 72% overshoot. However, notice that the polar plot intersects the negative real axis at a magnitude greater than unity. Hence, the system is actually unstable and the estimated percent overshoot is not correct.

b. The polar plot is approximately tangent to M = 3. Using Figure 10.40, we estimate 58% overshoot.

c. The polar plot is approximately tangent to M = 2.5. Using Figure 10.40, we estimate 52% overshoot.

19.

Raise each curve in Problem 17 by (a) 9.54 dB, (b) 7.96 dB, and (c) 3.52 dB, respectively.



Systems (a) and (b) are both unstable since the open-loop magnitude is greater than unity when the open-loop phase is 180^{o}. System (c) is tangent to approximately M = 3. Using Figure 10.40, we estimate 58% overshoot.

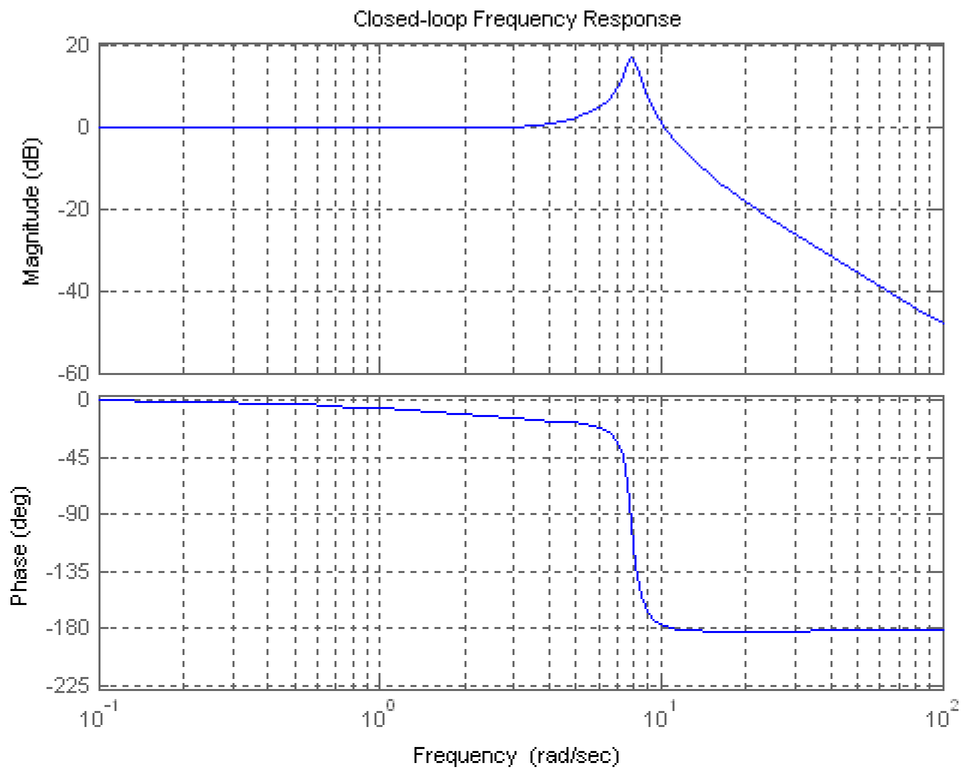
20.

Program:

```
%Enter G(s)************************
numg=[1 5];
deng=[1 4 25 0];
'G(s)'
G=tf(numg,deng)
%Enter K *************************
K=input('Type gain, K ');
'T(s)'
T=feedback(K*G,1)
bode(T)
title('Closed-loop Frequency Response')
[M,P,w]=bode(T);
[Mp i]=max(M);
Mp
MpdB=20*log10(Mp)
wp=w(i)
for i=1:1:length(M);
if M(i)<=0.707;
fprintf('Bandwidth = %g',w(i))
break
end
end
```

Computer response:

```
ans =

G(s)

Transfer function:

      s + 5

------------------

s^3 + 4 s^2 + 25 s


Type gain, K 40

ans =

T(s)

Transfer function:

       40 s + 200

------------------------

s^3 + 4 s^2 + 65 s + 200


Mp =

    6.9745

MpdB =

   16.8702

wp =

    7.8822

Bandwidth = 11.4655
```



21.

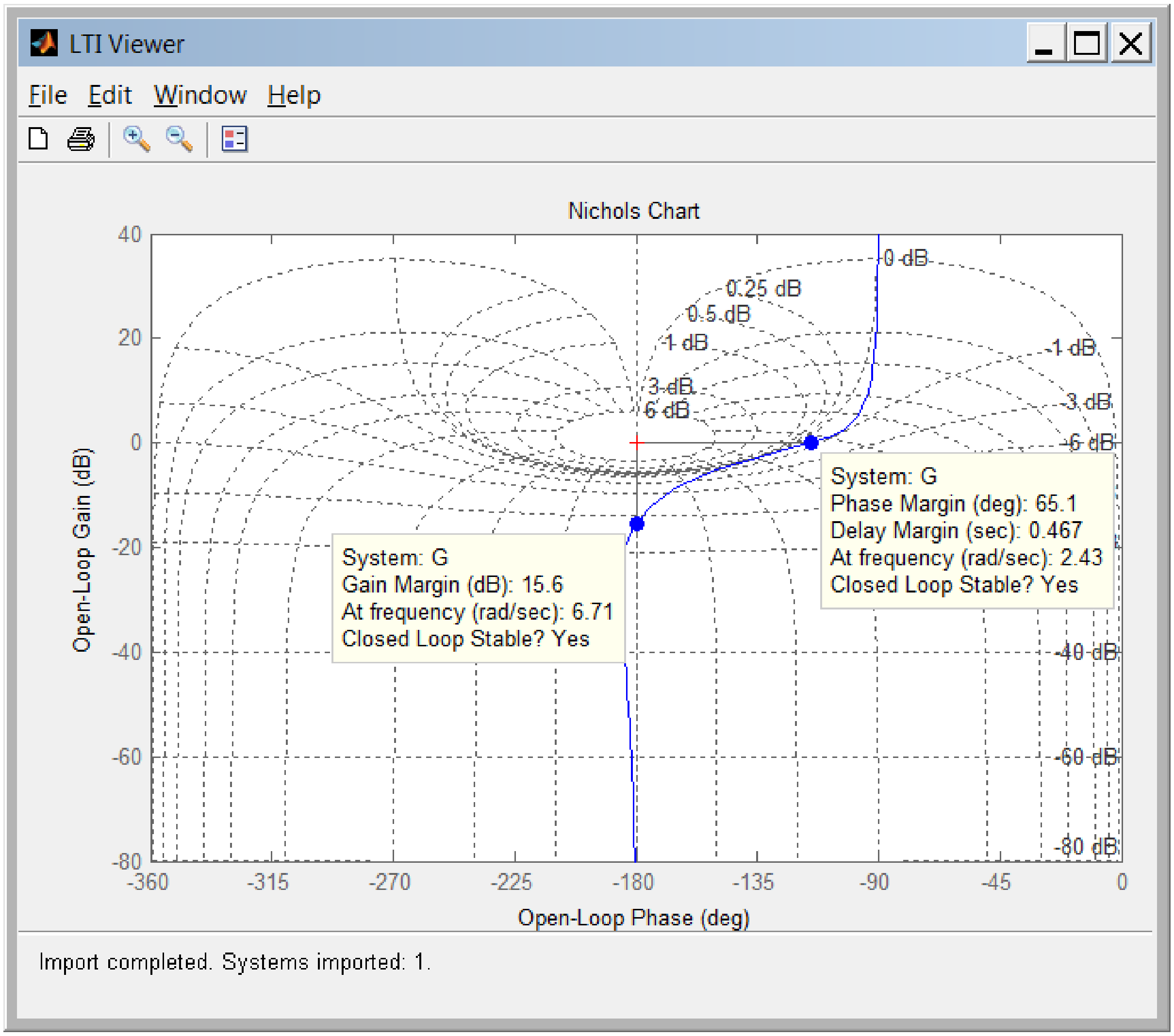
Program:

```
numg=[5 30];
deng=[1 4 15 0];
G=tf(numg,deng)
bode(G)                                %Make a Bode plot.
title('Open-Loop Frequency Response')
                                       %Add a title to the Bode plot.
[Gm,Pm,Wcp,Wcg]=margin(G);             %Find margins and margin
                                       %frequencies.
'Gain margin(dB); Phase margin(deg.); 0 dB freq. (r/s);'
'180 deg. freq. (r/s)'                 %Display label.
margins=[20*log10(Gm),Pm,Wcg,Wcp]      %Display margin data.
ltiview
```

Computer response:

```
Transfer function:
     5 s + 30
------------------
s^3 + 4 s^2 + 15 s


ans =

Gain margin(dB); Phase margin(deg.); 0 dB freq. (r/s);
```

```
ans =

180 deg. freq. (r/s)


margins =

   15.5649    65.1103    2.4319    6.7088
```



22.

Program:

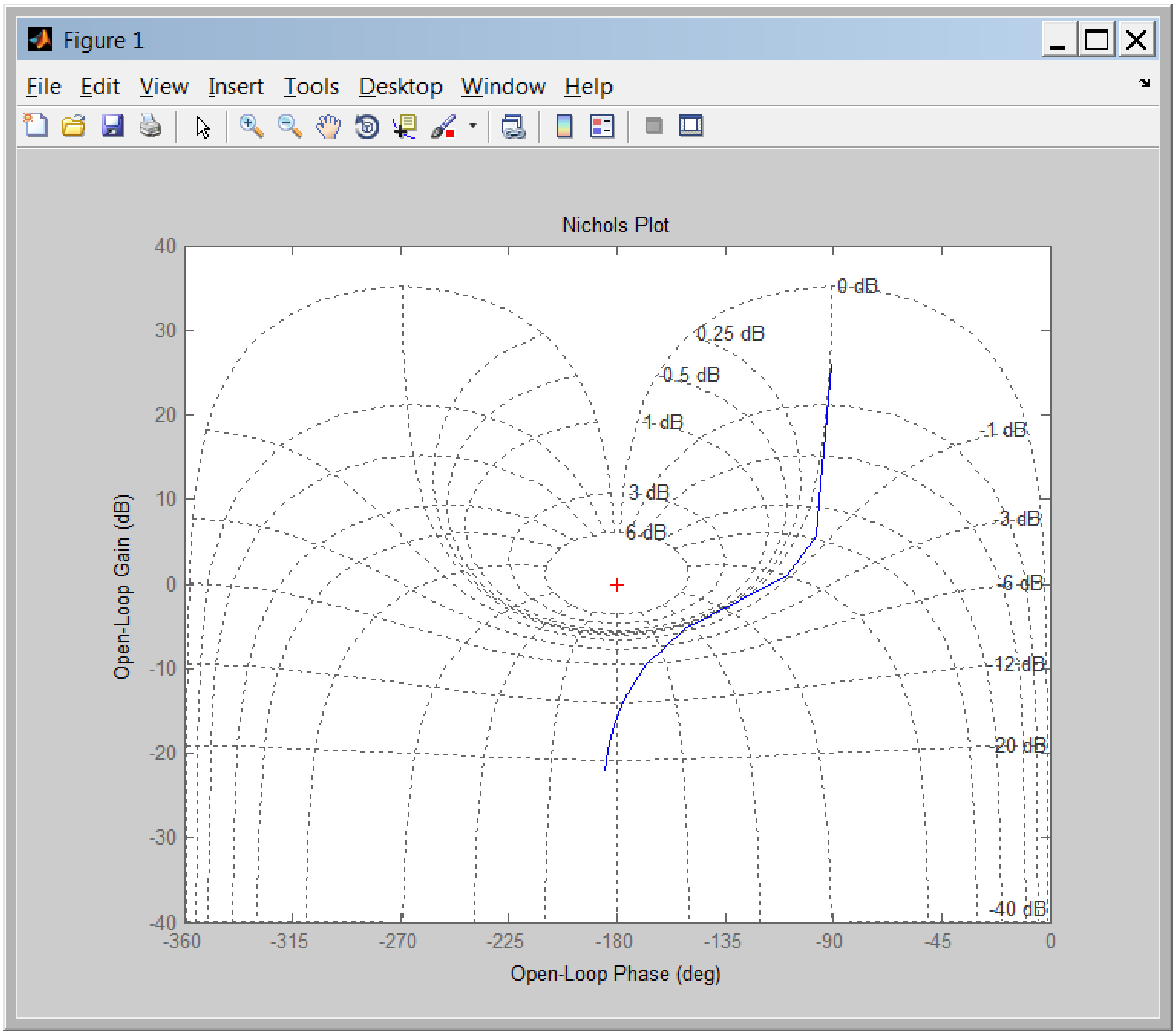
```
%Enter G(s)************************************
numg=5*[1 6];
deng=[1 4 15 0];
'Open-Loop System'
'G(s)'
G=tf(numg,deng)
clf
w=.10:1:10;
nichols(G,w)
ngrid
title('Nichols Plot')
```

```
[M,P]=nichols(G,w);
for i=1:1:length(M);
if M(i)<=0.45;
BW=w(i);
break
end
end
pause
MpdB=input('Enter Mp in dB from Nichols Plot  ');
Mp=10^(MpdB/20);
z2=roots([4,-4,(1/Mp^2)]);%Since Mp=1/sqrt(4z^2(1-z^2))
z1=sqrt(z2);
z=min(z1);
Pos=exp(-z*pi/(sqrt(1-z^2)));
Ts=(4/(BW*z))*sqrt((1-z^2)+sqrt(4*z^4-4*z^2+2));
Tp=(pi/(BW*sqrt(1-z^2)))*sqrt((1-z^2)+sqrt(4*z^4-4*z^2+2));
'Closed-Loop System'
'T(s)'
T=feedback(G,1)
bode(T)
title('Closed-Loop Frequency Response Plots')
fprintf('\nDamping Ratio = %g',z)
fprintf(', Percent Overshoot = %g',Pos*100)
fprintf(', Bandwidth = %g',BW)
fprintf(', Mp (dB) = %g',MpdB)
fprintf(', Mp = %g',Mp)
fprintf(', Settling Time = %g',Ts)
fprintf(', Peak Time = %g',Tp)
pause
step(T)
title('Closed-Loop Step Response')
```

Computer response:

```
ans =

Open-Loop System


ans =

G(s)


Transfer function:
     5 s + 30
------------------
s^3 + 4 s^2 + 15 s

Enter Mp in dB from Nichols Plot  0

ans =

Closed-Loop System


ans =

T(s)


Transfer function:
```

```
       5 s + 30
-----------------------
s^3 + 4 s^2 + 20 s + 30


Damping Ratio = 0.707107, Percent Overshoot = 4.32139, Bandwidth = 5.1, Mp
(dB) = 0, Mp = 1, Settling Time = 1.35847, Peak Time = 1.06694>>
```



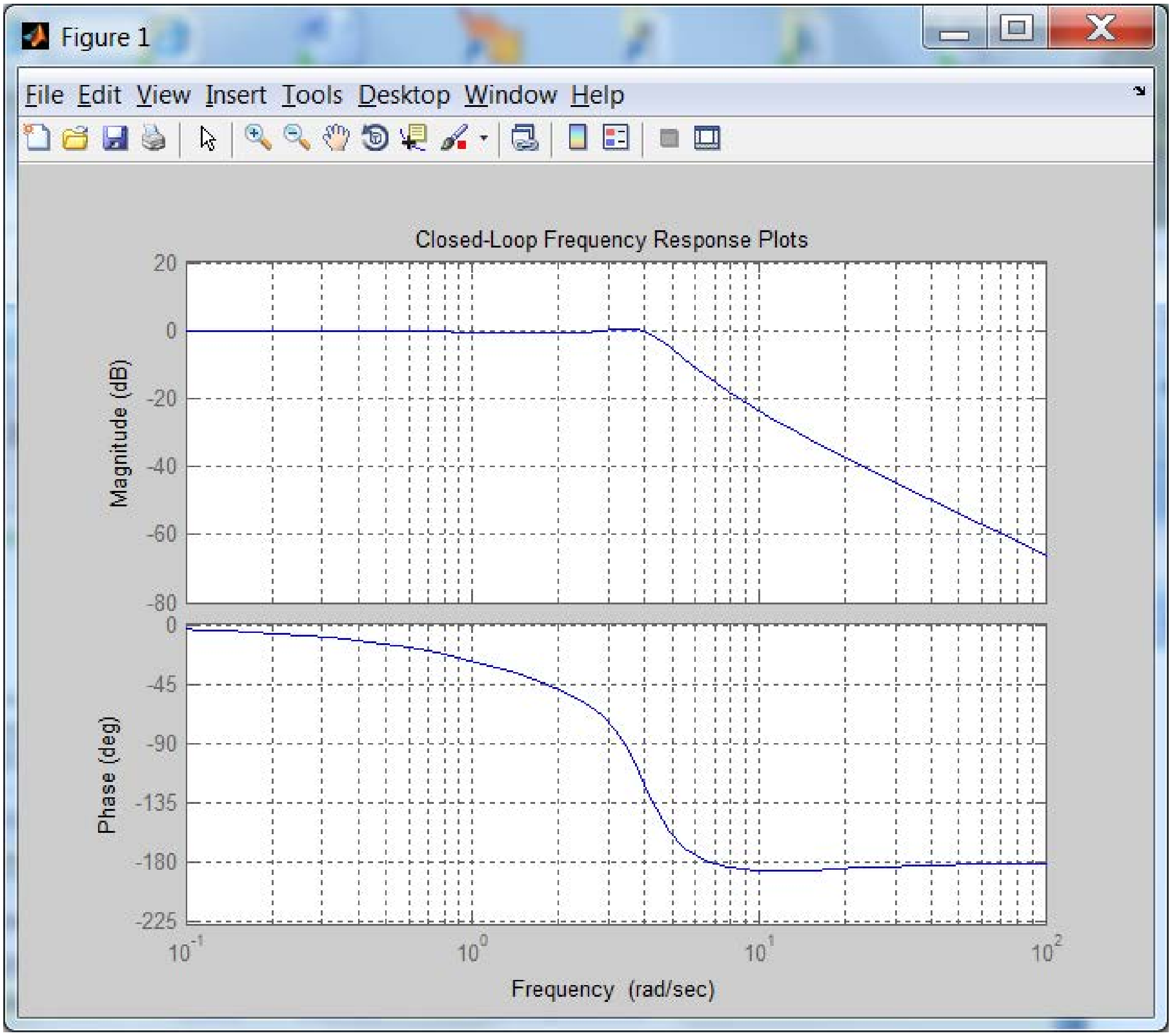
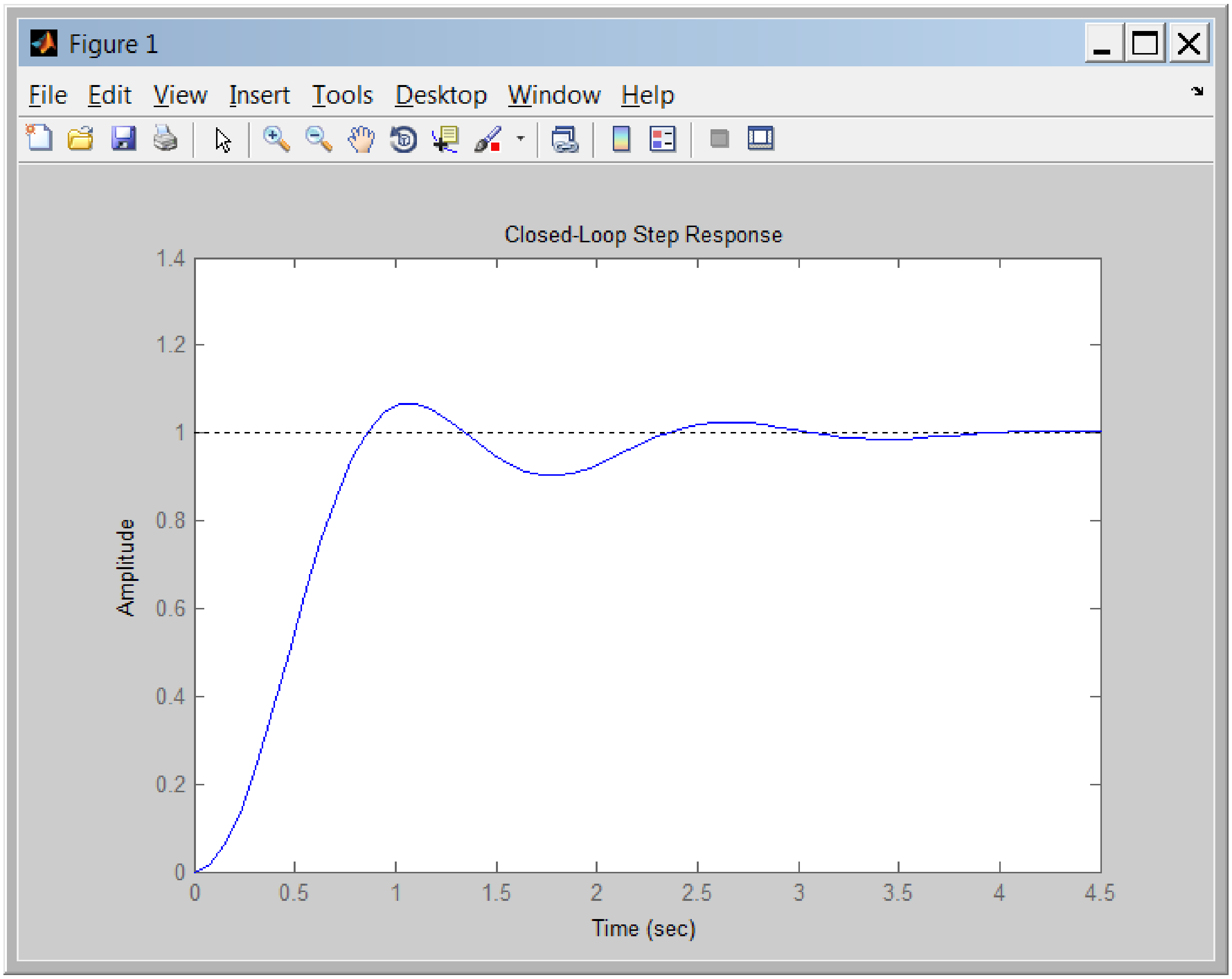


Figure 1
File Edit View Insert Tools Desktop Window Help
Closed-Loop Frequency Response Plots
Magnitude (dB)
20
0
-20
-40
-60
-80
Phase (deg)
0
-45
-90
-135
-180
-225
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/sec)



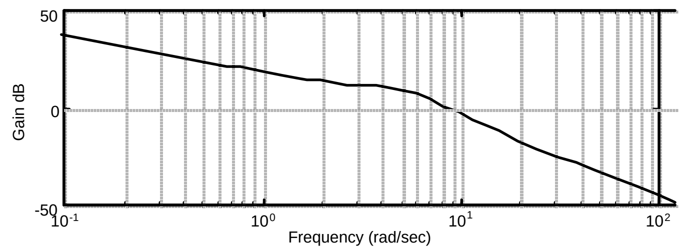
23.

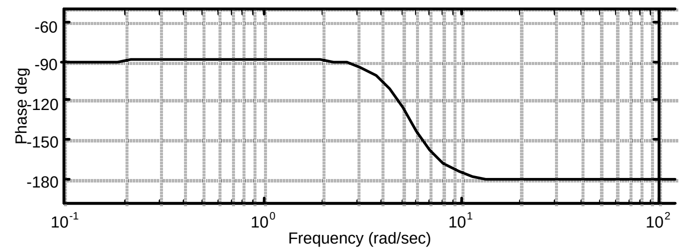
System 1: Using non-asymptotic frequency response plots, the zero dB crossing is at 9.7 rad/s at a phase of -163.2^{o}. Therefore the phase margin is $180^{o} - 163.2^{o} = 16.8^{o}$. $|G(j\omega)|$ is down 7 dB at 14.75 rad/s. Therefore the bandwidth is 14.75 rad/s. Using Eq. (10.73), $\zeta = 0.15$. Using Eq. (4.38), %OS = 62.09%. Eq. (10.55) yields $T_s = 2.76$ s, and Eq. (10.56) yields $T_p = 0.329$ s.

System 2: Using non-asymptotic frequency response plots, the zero dB crossing is at 6.44 rad/s at a phase of -150.73^{o}. Therefore the phase margin is $180^{o} - 150.73^{o} = 29.27^{o}$. $|G(j\omega)|$ is down 7 dB at 10.1 rad/s. Therefore the bandwidth is 10.1 rad/s. Using Eq. (10.73), $\zeta = 0.262$. Using Eq. (4.38), %OS = 42.62%. Eq. (10.55) yields $T_s = 2.23$ s, and Eq. (10.56) yields $T_p = 0.476$ s.

24.

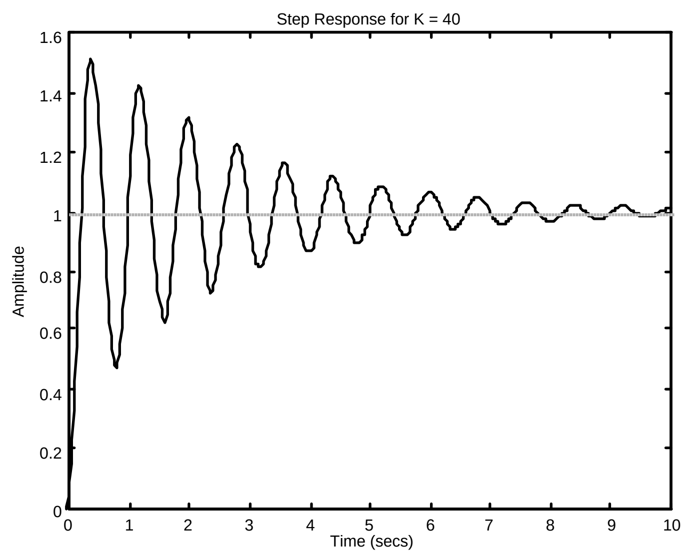
a.





b. Zero dB frequency = 7.8023; Looking at the phase diagram at this frequency, the phase margin is 8.777 degrees. Using Eq. (10.73) or Figure 10.48, ζ = 0.08. Thus, %OS = 77.7.

c.



25.

From the Bode plots: Gain margin = 14.96 dB; phase margin = 49.57^0; 0 dB frequency = 2.152 rad/s; 180^0 frequency = 6.325 rad/s; bandwidth(@-7 dB point) = 3.8 rad/s. From Eq. (10.73) ζ = 0.48; from Eq. (4.38) %OS = 17.93; from Eq. (10.55) T_s = 2.84 s; from Eq. (10.56) T_p = 1.22 s.

26.

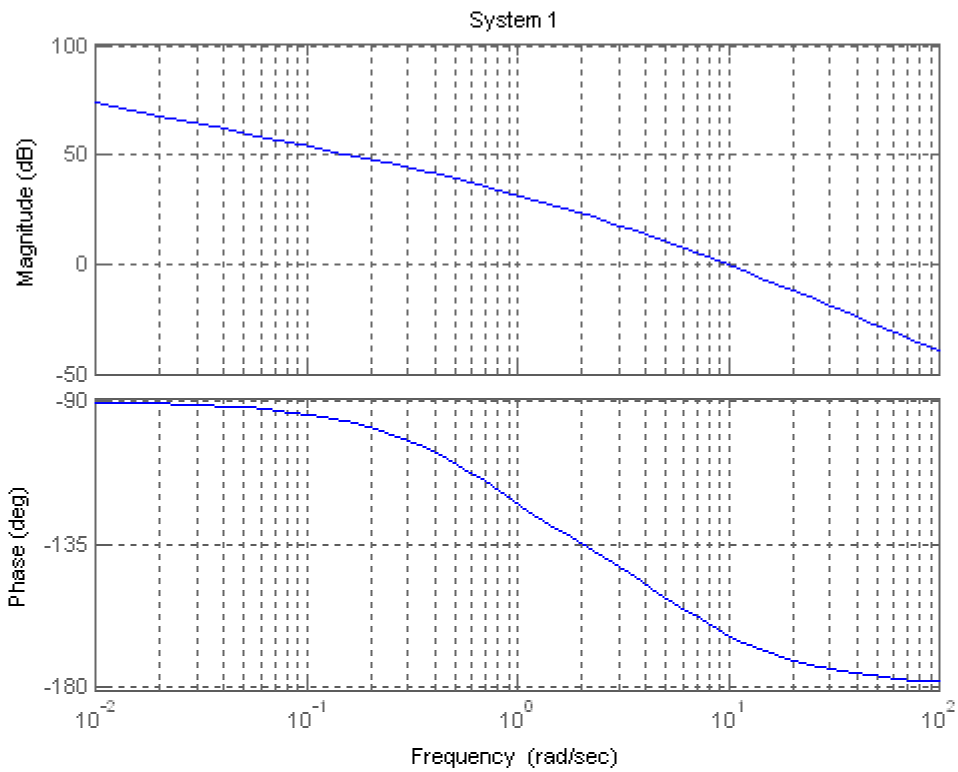
Program:
```
G=zpk([-2],[0 -1 -4],100)
%G=zpk([-3 -5],[0 -2 -4 -6],50)
G=tf(G)
```

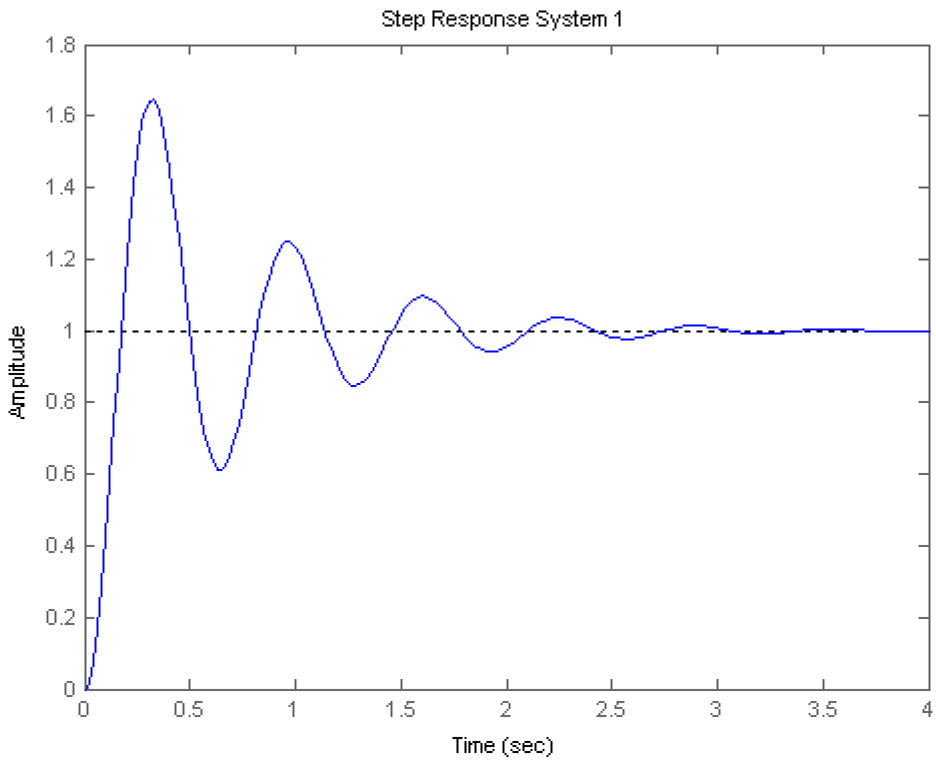
```
bode(G)
title('System 1')
%title('System 2')
pause
%Find Phase Margin
[Gm,Pm,Wcg,Wcp]=margin(G);
w=1:.01:20;
[M,P,w]=bode(G,w);
%Find Bandwidth
for k=1:1:length(M);
   if 20*log10(M(k))+7<=0;
   'Mag'
   20*log10(M(k))
   'BW'
   wBW=w(k)
   break
end
end
%Find Damping Ratio,Percent Overshoot, Settling Time, and Peak Time
for z= 0:.01:10
Pt=atan(2*z/(sqrt(-2*z^2+sqrt(1+4*z^4))))*(180/pi);
if (Pm-Pt)<=0
   z;
   Po=exp(-z*pi/sqrt(1-z^2));
   Ts=(4/(wBW*z))*sqrt((1-2*z^2)+sqrt(4*z^4-4*z^2+2));
   Tp=(pi/(wBW*sqrt(1-z^2)))*sqrt((1-2*z^2)+sqrt(4*z^4-4*z^2+2));
fprintf('Bandwidth = %g ',wBW)
fprintf('Phase Margin = %g',Pm)
fprintf(', Damping Ratio = %g',z)
fprintf(', Percent Overshoot = %g',Po*100)
fprintf(',Settling Time = %g',Ts)
fprintf(', Peak Time = %g',Tp)
   break
end
end
T=feedback(G,1);
step(T)
title('Step Response System 1')
%title('Step Response System 2')
```

Computer response:

```
Zero/pole/gain:
  100 (s+2)
-------------
s (s+1) (s+4)


Transfer function:
   100 s + 200
-----------------
s^3 + 5 s^2 + 4 s


ans =

Mag


ans =

   -7.0007
```

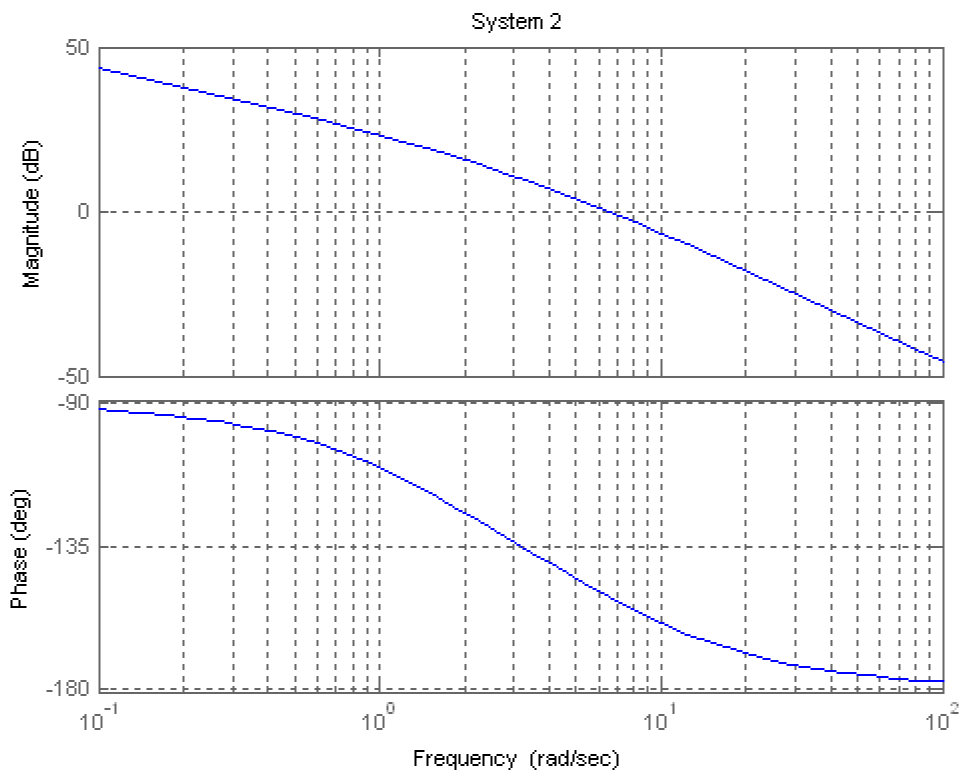
```
ans =

BW


wBW =

   14.7500

Bandwidth = 14.75 Phase Margin = 16.6617, Damping Ratio = 0.15, Percent
Overshoot = 62.0871,Settling Time = 2.76425, Peak Time = 0.329382
```

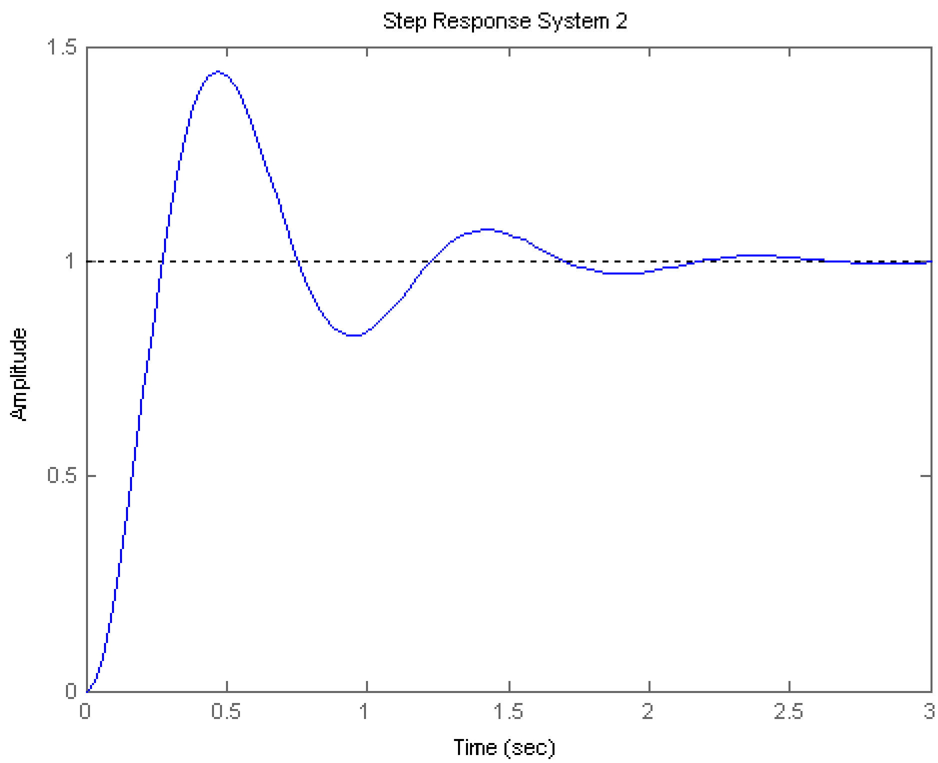





```
Zero/pole/gain:

  50 (s+3) (s+5)

-------------------

s (s+2) (s+4) (s+6)

Transfer function:

    50 s^2 + 400 s + 750

----------------------------

s^4 + 12 s^3 + 44 s^2 + 48 s

ans =

Mag

ans =

   -7.0026

ans =
```

```
BW

wBW =

   10.1100

Bandwidth = 10.11 Phase Margin = 29.2756, Damping Ratio = 0.27, Percent Overshoot

= 41.439,Settling Time = 2.1583, Peak Time =

0.475337
```





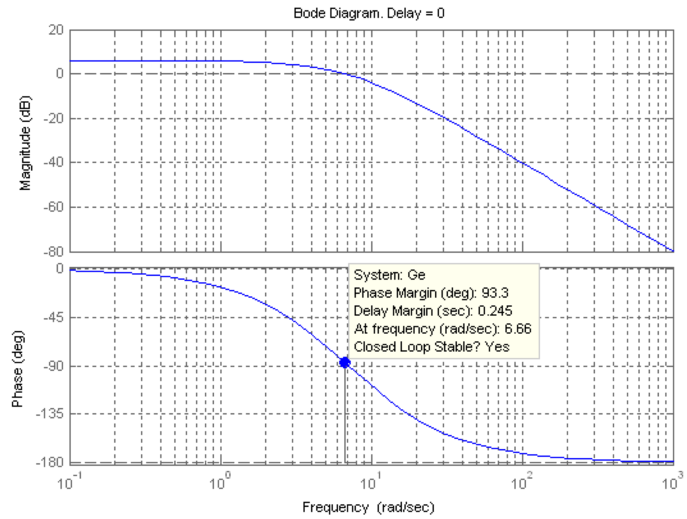
27.

The phase margin of the given system is 20^{o}. Using Eq. (10.73), $\zeta = 0.176$. Eq. (4.38) yields 57% overshoot. The system is Type 1 since the initial slope is - 20 dB/dec. Continuing the initial slope down to the 0 dB line yields $K_V = 4$. Thus, steady-state error for a unit step input is zero; steady state error for a unit ramp input is $\frac{1}{K_v} = 0.25$; steady-state error for a parabolic input is infinite.

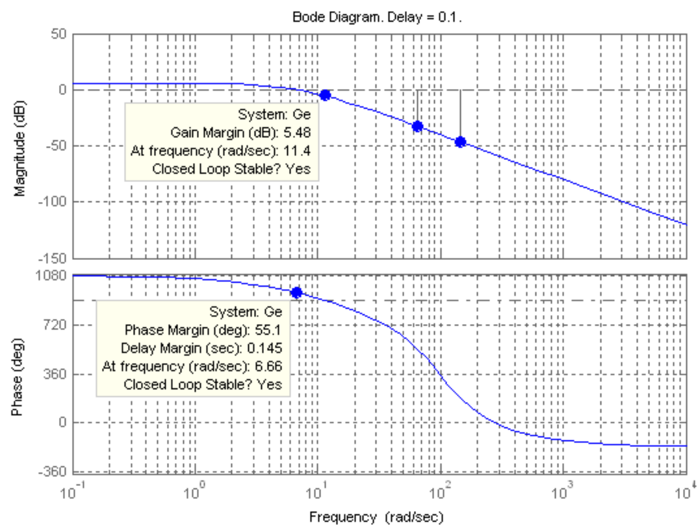
28.

The magnitude response is the same for all time delays and crosses zero dB at 0.5 rad/s. The following is a plot of the magnitude and phase responses for the given time delays:

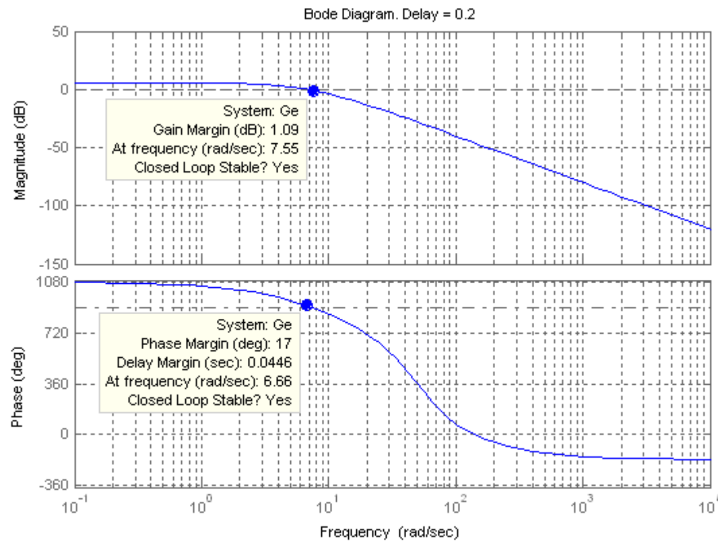
a.



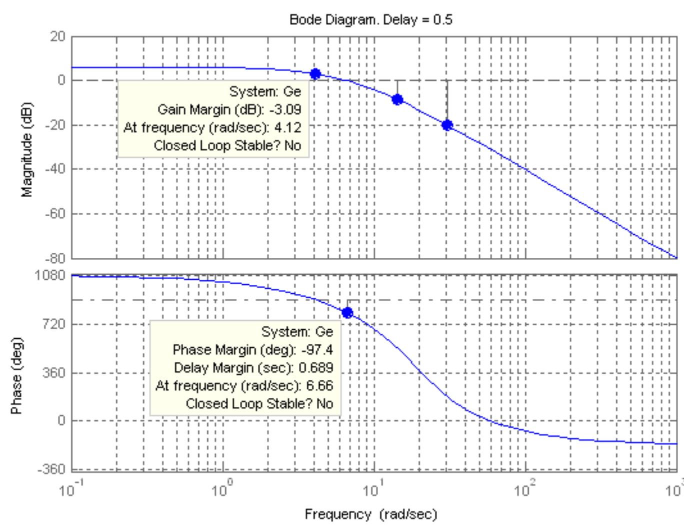
For T = 0, $\Phi_M = 93.3^o$; System is stable.



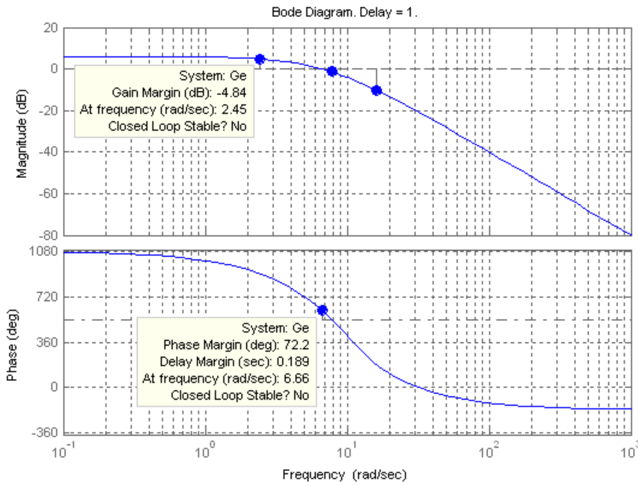
For T = 0.1, $\Phi_M = 55.1^o$; System is stable.



For T = 0.2, $\Phi_M = 17^o$; System is stable.



For T = 0.5, $\Phi_M = -97^o$; System is unstable.



For T = 1, $\Phi_M = 72.2^o$; System is unstable because the gain margin is -4.84 dB.

b.

For T = 0, the phase response reaches 180^o at infinite frequency. Therefore the gain margin is infinite. The system is stable.

For T = 0.1, the phase response is -180^o at 11.4 rad/s. The magnitude response is -5.48 dB at 11.4 rad/s. Therefore, the gain margin is 5.48 dB. The system is stable.

For T = 0.2, the phase response is -180^o at 7.55 rad/s. The magnitude response is -1.09 dB at 7.55 rad/s. Therefore, the gain margin is 1.09 dB and the system is stable.

For T = .5, the phase response is -180^o at 4.12 rad/s. The magnitude response is +3.09 dB at 4.12 rad/s. Therefore, the gain margin is – 3.09 dB and the system is unstable.

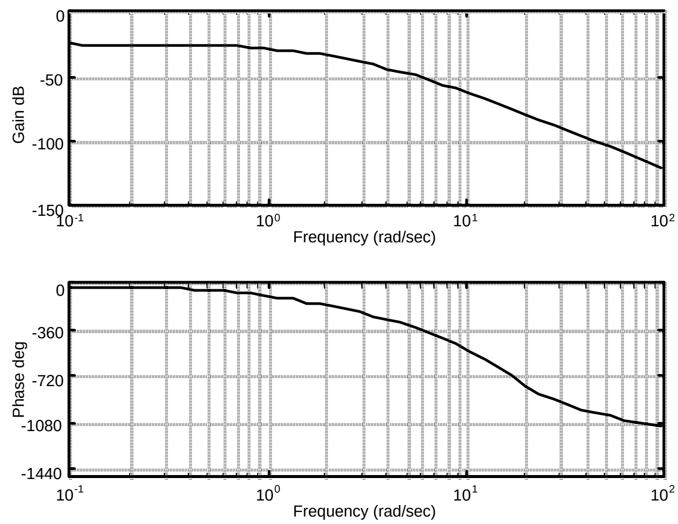
For T = 1, the phase response is -180^o at 2.45 rad/s. The magnitude response is +4.84 dB at 2.45 rad/s. Therefore, the gain margin is -4.84 dB and the system is unstable.

c.
T = 0; T = 0.1; T = 0.2

d.
T = 0.5, -3.09 dB; T = 1, - 4.84 dB;

29.

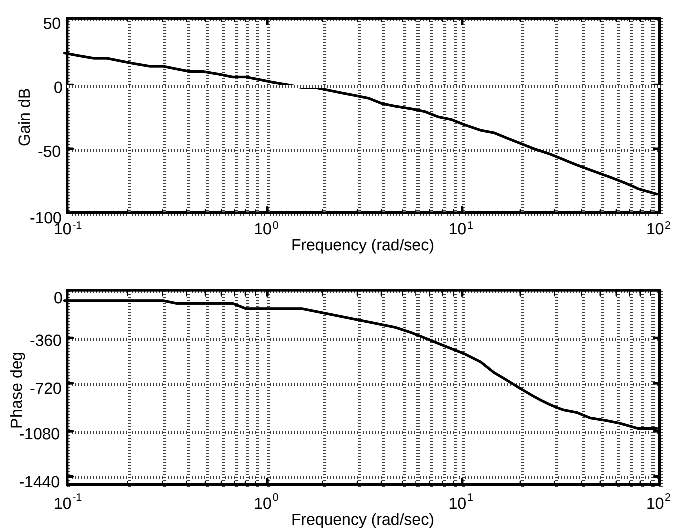
The Bode plots for K = 1 and 0.5 second delay is:



The phase is -180^{o} at 2.12 rad/s. At this frequency, the gain is -34.76 dB. Thus the gain can be raised by 34.76 dB = 54.71. Hence for stability, 0<K<54.71.

30.

The Bode plots for K = 40 and a delay of 0.5 second is shown below.



The magnitude curve crosses zero dB at a frequency of 1.0447 rad/s. At this frequency, the phase plot shows a phase margin of 35.74 degrees. Using Eq. (10.73) or Figure 10.48, ζ = 0.33. Thus, %OS = 33.3.

31.

Program:

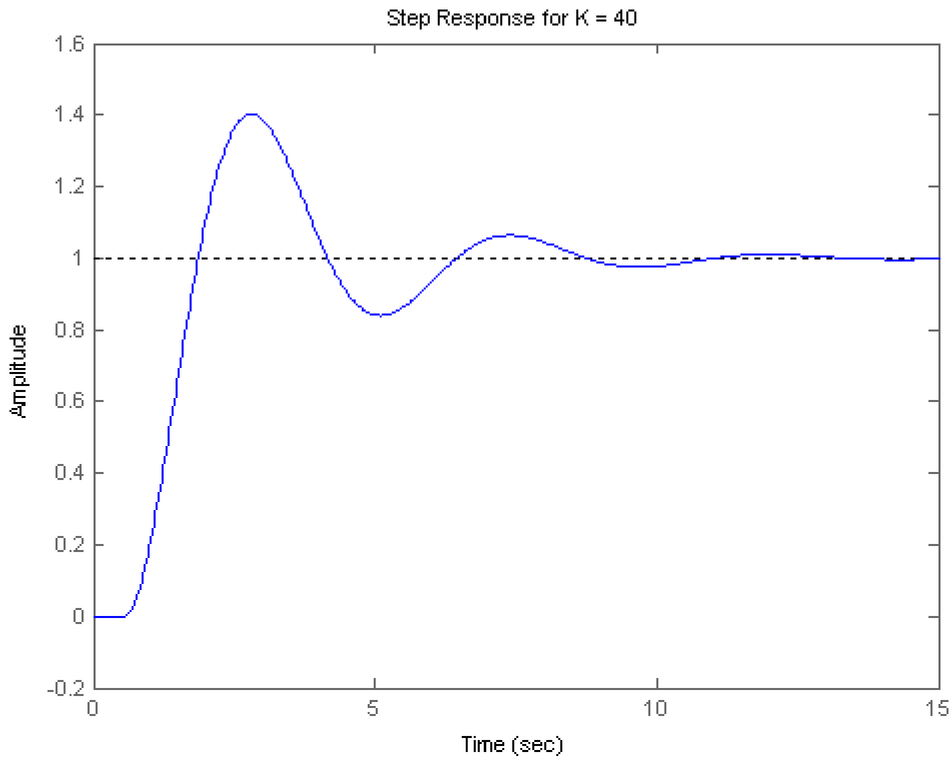
```
%Enter G(s)************************
numg1=1;
deng1=poly([0 -3 -12]);
'G1(s)'
G1=tf(numg1,deng1)
[numg2,deng2]=pade(0.5,5);
```

```
'G2(s) (delay)'
G2=tf(numg2,deng2)
'G(s)=G1(s)G2(s)'
G=G1*G2
%Enter K **************************
K=input('Type gain, K ');
T=feedback(K*G,1);
step(T)
title(['Step Response for K = ',num2str(K)])
```

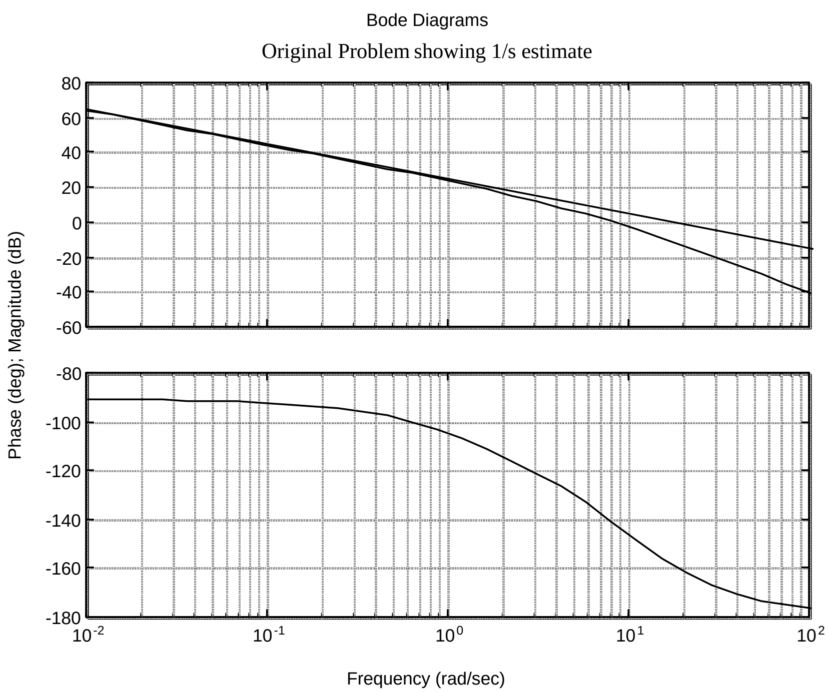
Computer response:

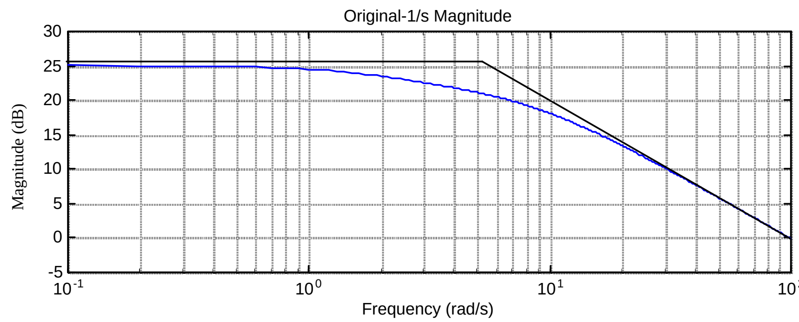
```
ans =

G1(s)

Transfer function:

         1

------------------

s^3 + 15 s^2 + 36 s

ans =

G2(s) (delay)

Transfer function:

-s^5 + 60 s^4 - 1680 s^3 + 2.688e004 s^2 - 2.419e005 s + 9.677e005

-----------------------------------------------------------------

s^5 + 60 s^4 + 1680 s^3 + 2.688e004 s^2 + 2.419e005 s + 9.677e005

ans =

G(s)=G1(s)G2(s)

Transfer function:

    -s^5 + 60 s^4 - 1680 s^3 + 2.688e004 s^2 - 2.419e005 s + 9.677e005

-------------------------------------------------------------------------

s^8 + 75 s^7 + 2616 s^6 + 5.424e004 s^5 + 7.056e005 s^4 + 5.564e006 s^3

                                             + 2.322e007 s^2 + 3.484e007 s
```

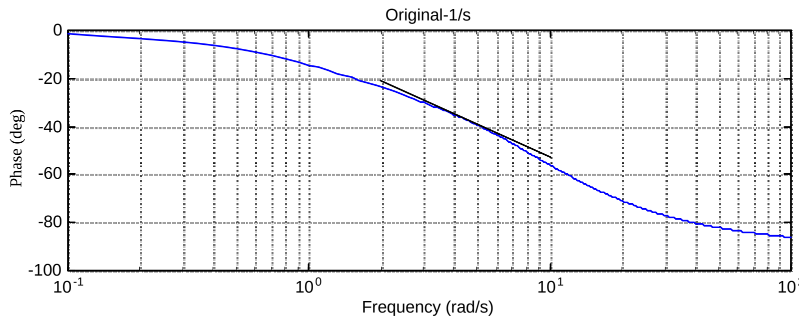
```
Type gain, K 40
```

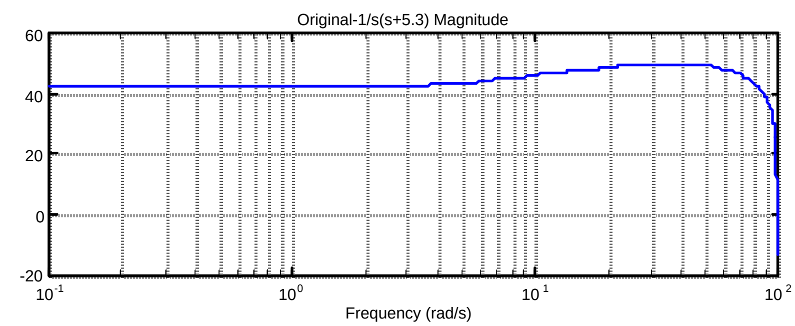


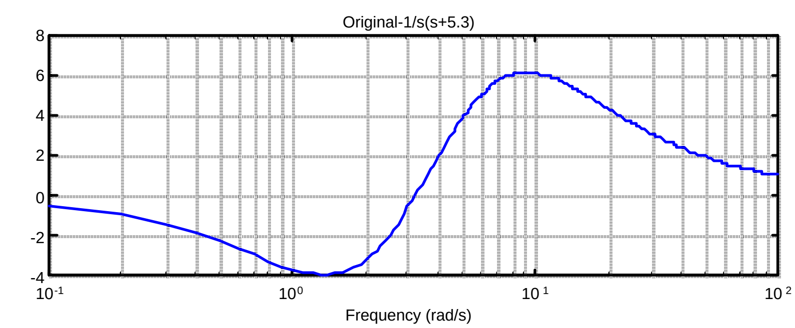
32.











Estimated K = 41 dB = 112. Therefore, final estimate is $G(s) = \dfrac{112}{s(s+5.3)}$.

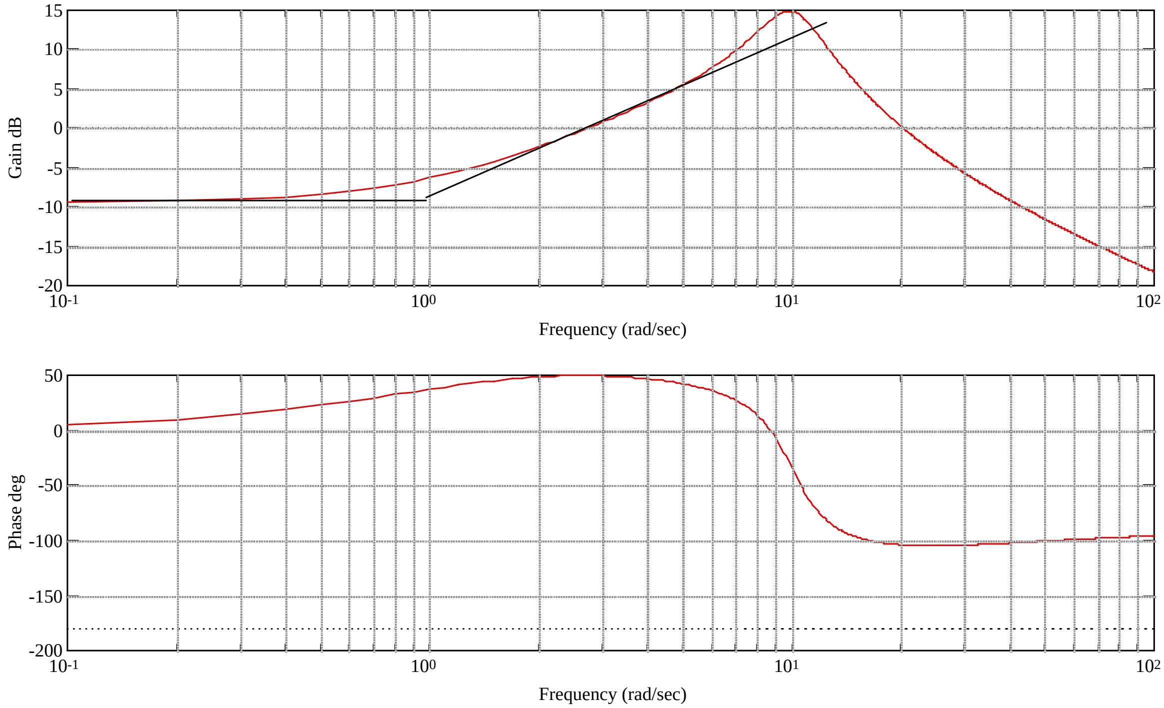
33.

Program:

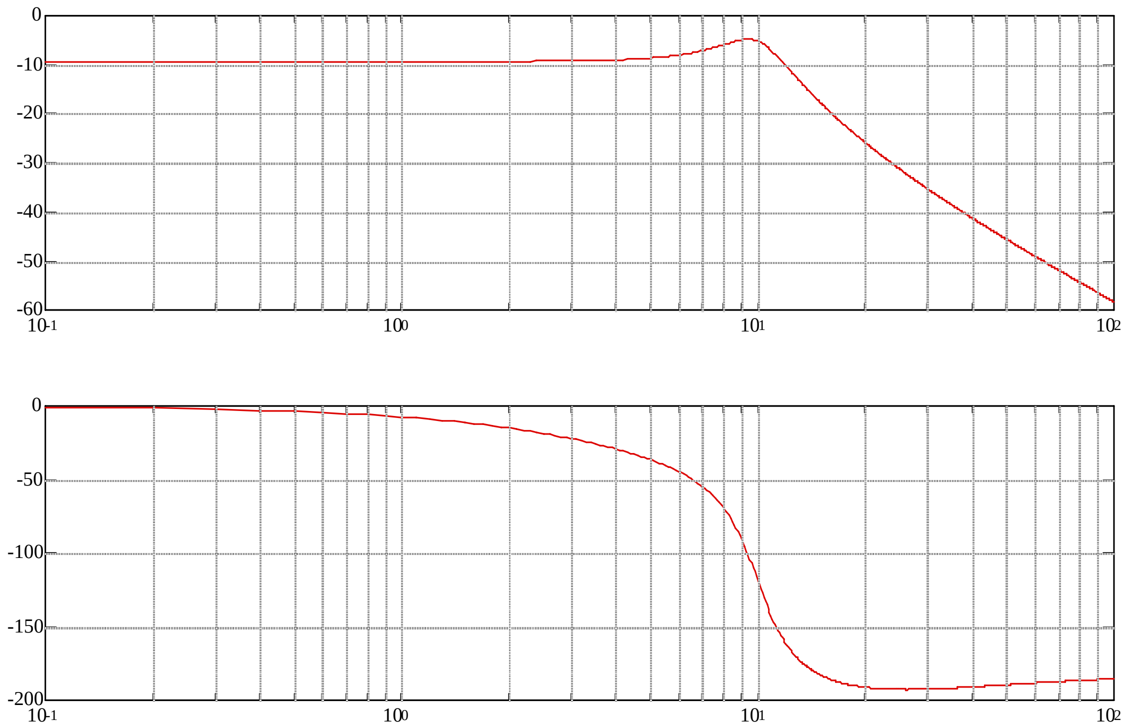
```
%Generate total system Bode plots - numg0,deng0 - M0,P0
clf
numg0=12*poly([-1 -20]);
deng0=conv([1 7],[1 4 100]);
G0=tf(numg0,deng0);
w=0.1:0.1:100;
```

```
[M0,P0]=bode(G0,w);
M0=M0(:,:);
P0=P0(:,:);
[20*log10(M0),P0,w];
bode(G0,w)
pause
%Subtract (s+1) [numg1,deng1] and generate Bode plot-M2,P2
numg1=[1 1];
deng1=1;
G1=tf(numg1,deng1);
[M1,P1]=bode(G1,w);
M1=M1(:,:);
P1=P1(:,:);
M2=20*log10(M0)-20*log10(M1);
P2=P0-P1;
clf
subplot(2,1,1)
semilogx(w,M2)
grid
subplot(2,1,2)
semilogx(w,P2)
grid
pause
%Subtract10^2/(s^2+2*0.3*10s+10^2) [numg2,deng2] and generate Bode plot-
M4,P4
numg2=100;
deng2=[1 2*0.3*10 10^2];
G2=tf(numg2,deng2);
[M3,P3]=bode(G2,w);
M3=M3(:,:);
P3=P3(:,:);
M4=M2-20*log10(M3);
P4=P2-P3;
clf
subplot(2,1,1)
semilogx(w,M4)
grid
subplot(2,1,2)
semilogx(w,P4)
grid
pause
%Subtract(8.5/23)(s+23)/(s+8.5) [numg3,deng3] and generate Bode plot-M6,P6
numg3=(8.5/23)*[1 23];
deng3=[1 8.5];
G3=tf(numg3,deng3);
[M5,P5]=bode(G3,w);
M5=M5(:,:);
P5=P5(:,:);
M6=M4-20*log10(M5);
P6=P4-P5;
clf
subplot(211)
semilogx(w,M6)
grid
subplot(212)
semilogx(w,P6)
grid
```

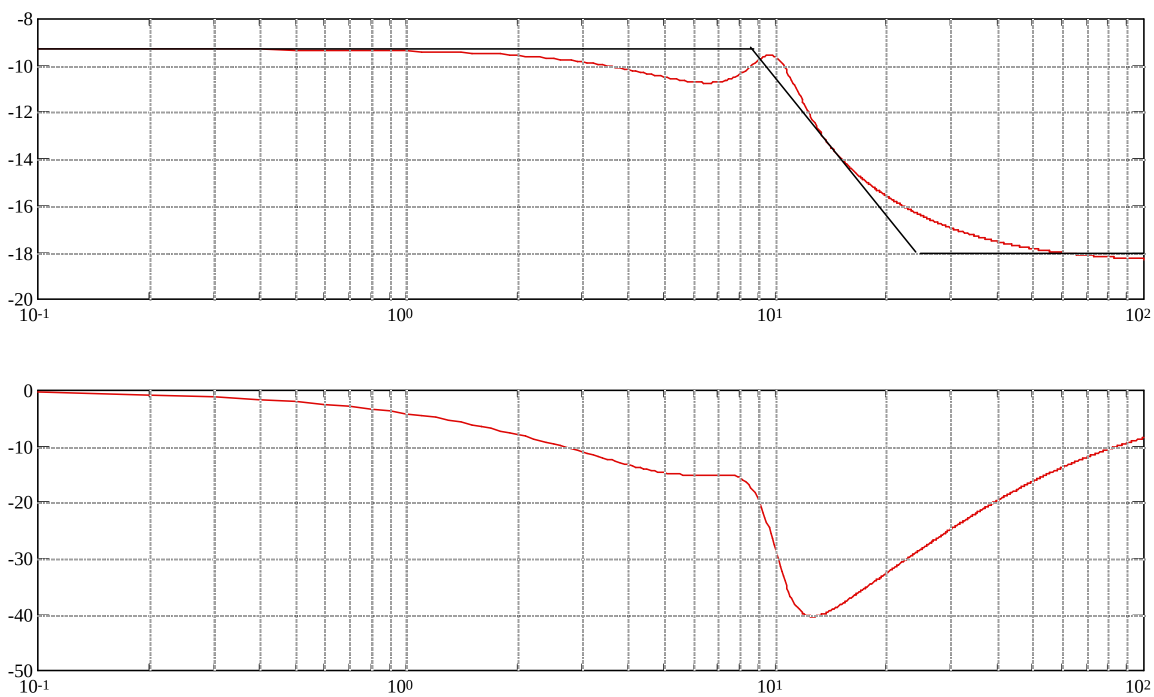
Computer responses and analysis:



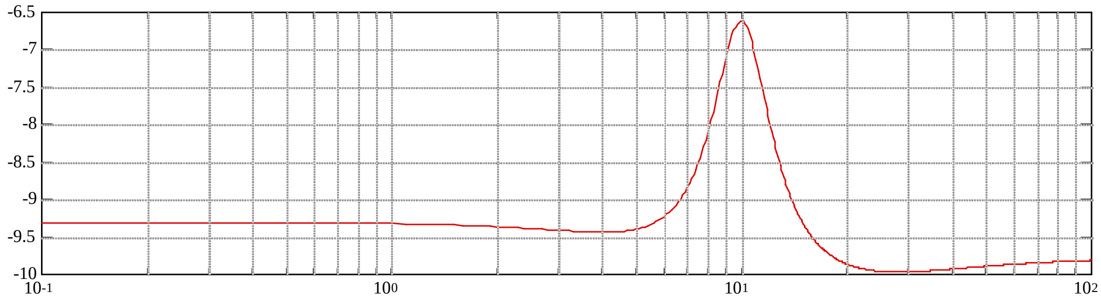
Original data showing estimate of a component, (s+1)

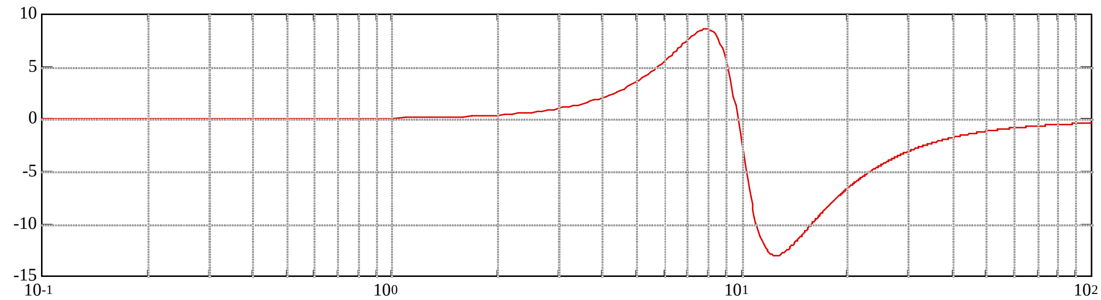


Original data minus (s+1) showing estimate of (10^2/(s^2+2*0.3*10s+10^2))



Original data minus (s+1)(10^2/(s^2+2*0.3*10s+10^2)) showing estimate of (8.5/23)(s+23)/(s+8.5)



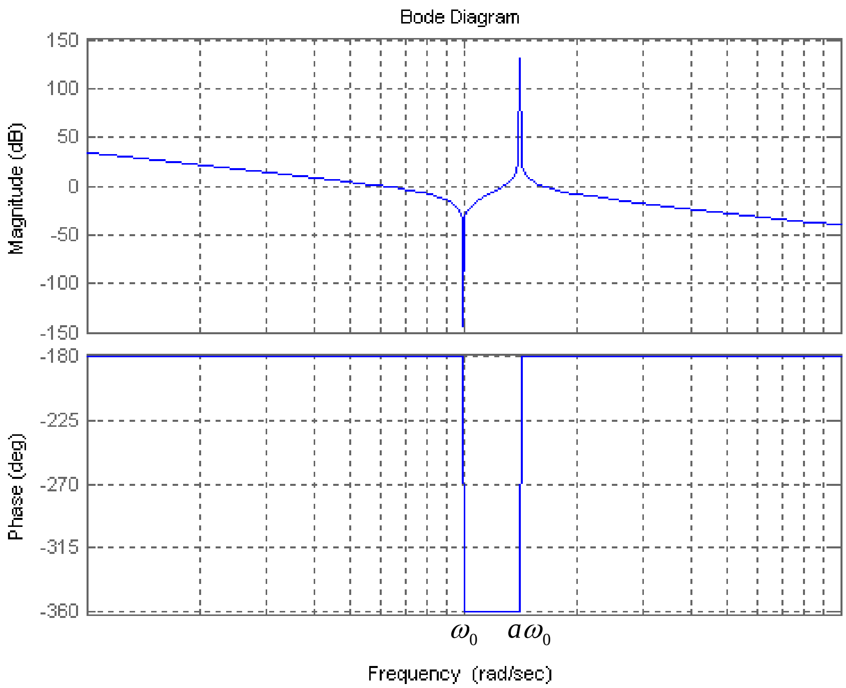


Original data minus final estimate of $G(s) = (s+1)*\frac{100}{s^2+6s+100}*\frac{8.5}{23}\frac{s+23}{s+8.5}$

Thus the final estimate is $G(s) = (s+1)*\frac{100}{s^2+6s+100}*\frac{8.5}{23}\frac{s+23}{s+8.5}*K$. Since the original plot starts from -10 dB, 20 log K = -10, or K = 0.32.

34.

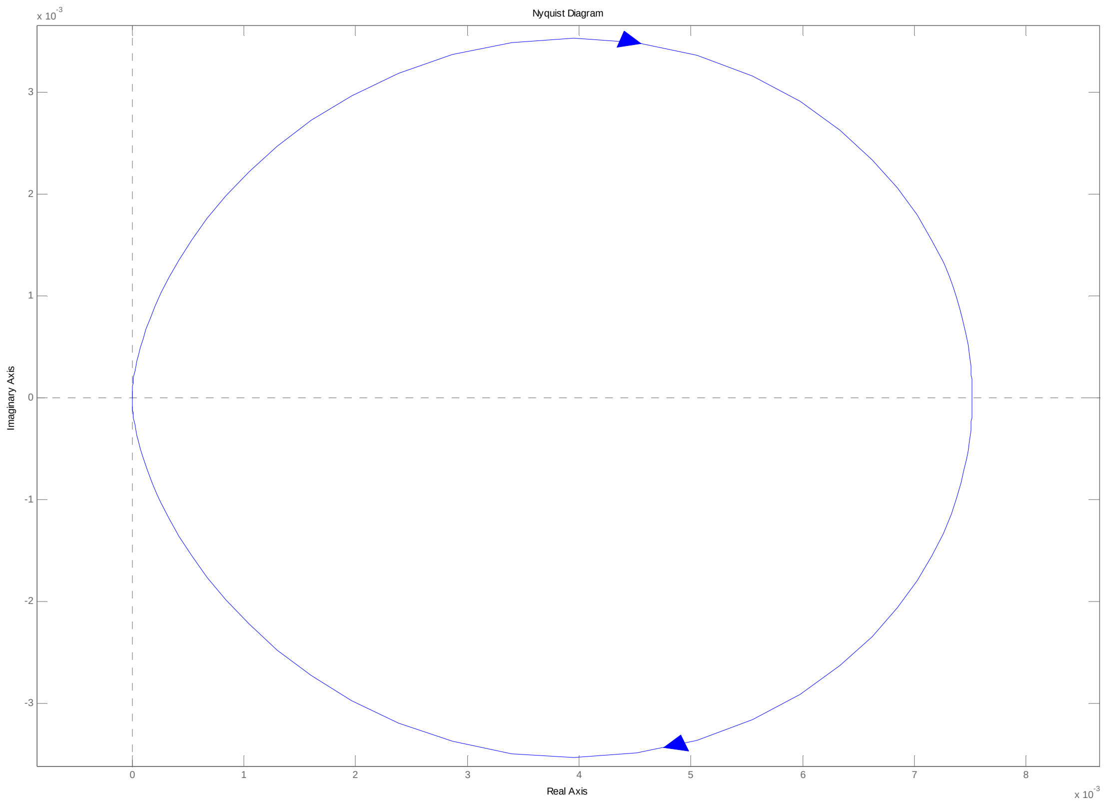
The Bode plot is:



The frequency response is $P(j\omega) = \dfrac{\omega_0^2 - \omega^2}{-\omega^2(a\omega_0^2 - \omega^2)}$, there is no imaginary part so the phase will be either -180° or 0°. As $\omega \to 0$, $P(j\omega) < 0$ so its phase is -180°. When $\omega = \omega_0$, $P(j\omega) = 0$. Then for $\omega_0 < \omega < a\omega_0$ the phase is 0°. When $\omega = a\omega_0$, $P(j\omega) = \infty$. And for $\omega > a\omega_0$ $P(j\omega) < 0$. $P(j\omega)$ decreases in the intervals in which $\omega < \omega_0$ and $\omega > a\omega_0$, and will increase when $\omega_0 < \omega < a\omega_0$.

35.

a. The Nyquist plot is:



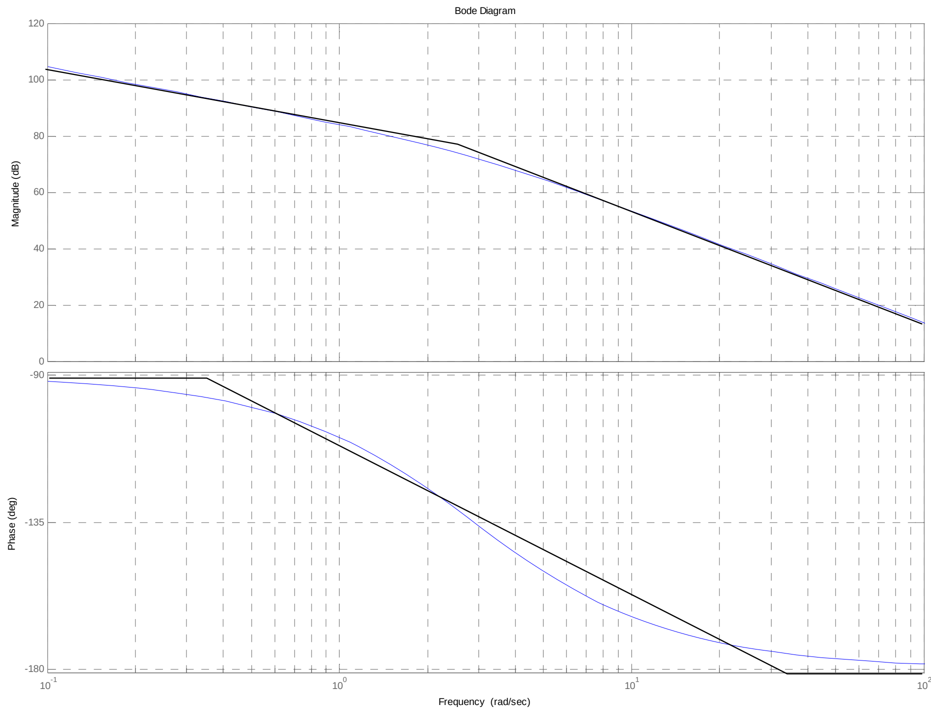
b. The gain margin is infinite as the plot never crosses the 180° line. The phase margin is undefined since $|G(j\omega)| < 1$ at all frequencies.

c. The system is closed loop stable for all $0 < K < \infty$.

36.

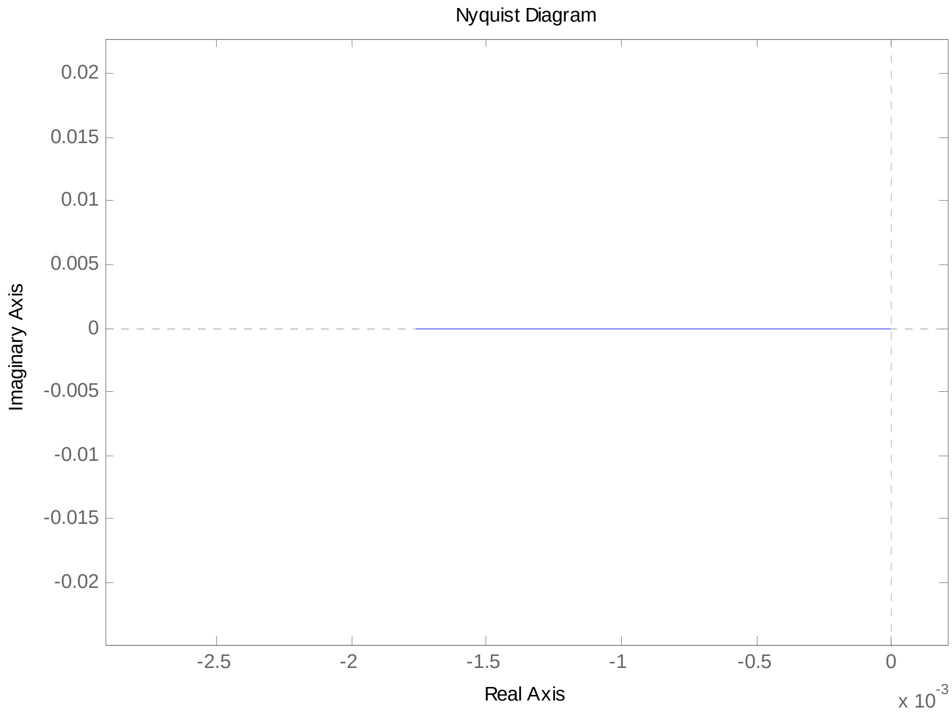
The exact Bode plot and the asymptotic approximations are shown on the following figure. The magnitude asymptotes are obtained by noting that when $\omega \to 0$, $P(s) \approx \dfrac{16782}{s}$. So when $\omega = 0.1$, $|P(j0.1)| = 167820 = 104.5db$ and the slope of the line is -6db/oct. This means that $|P(j0.2)| = 104.5db - 6db = 98.5db$. So a line is drawn between these two points until $\omega = 2.89$ is reached. For higher frequencies the slope is -12db/oct, so the line is continued.

To plot the phase asymptote at very low frequencies the phase is -180° due to the integrator until 2.89/10=0.289rad/sec. At very high frequencies from 2.89*10=28.9 rad/sec and up the phase will be -270° due to the plant's pole contribution. A line is drawn between 0.289 and 28.9 rad/sec with -135° at 2.89 rad/sec.

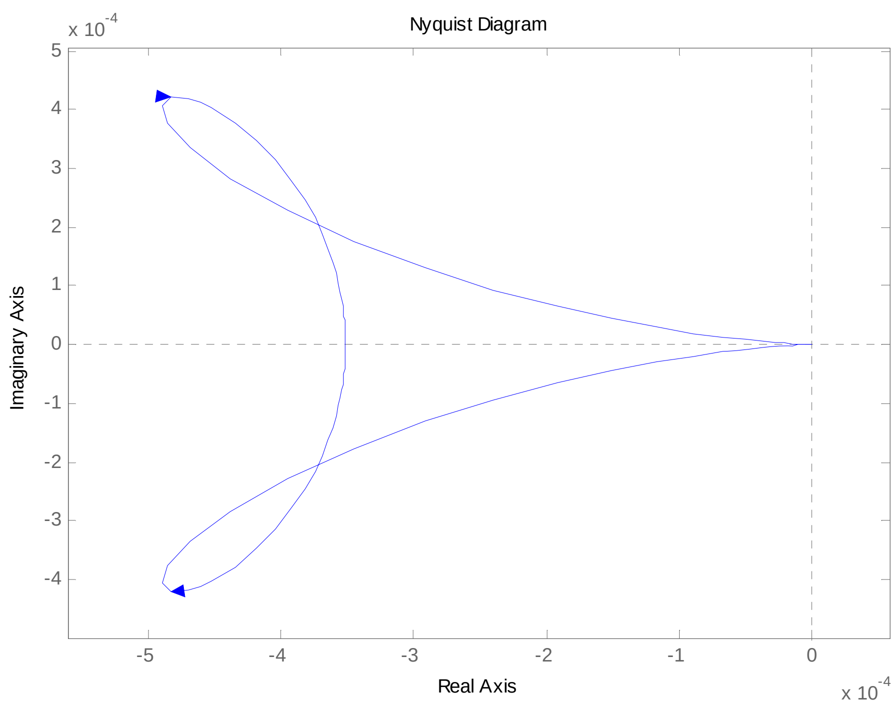


37.

a. The frequency response $P(j\omega)=\dfrac{1300}{\omega^2+860^2}$ is a real quantity. The Nyquist diagram is shown next. Since the open loop transfer function has one RHP pole, P=1 and Z=P-N=1 so the system will be closed loop unstable.

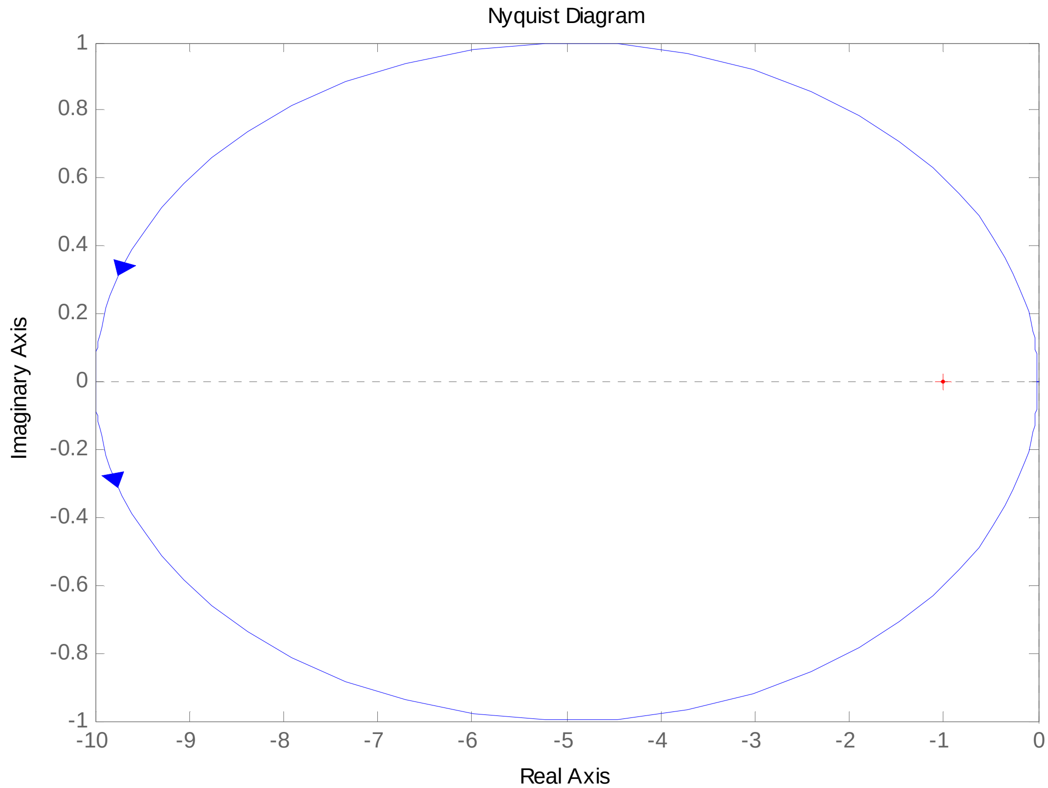


b. In this case for $K=1$ there are no encirclements of the -1 point so the system is closed loop unstable. However for large values of K there will be a negative encirclement of the -1 point so Z=P-N=0 and the system is closed loop stable. The real axis crossover occurs at -3.5X10-4 so the range of stability is $K > \dfrac{1}{3.5\times 10^{-4}} = 2851$



38.

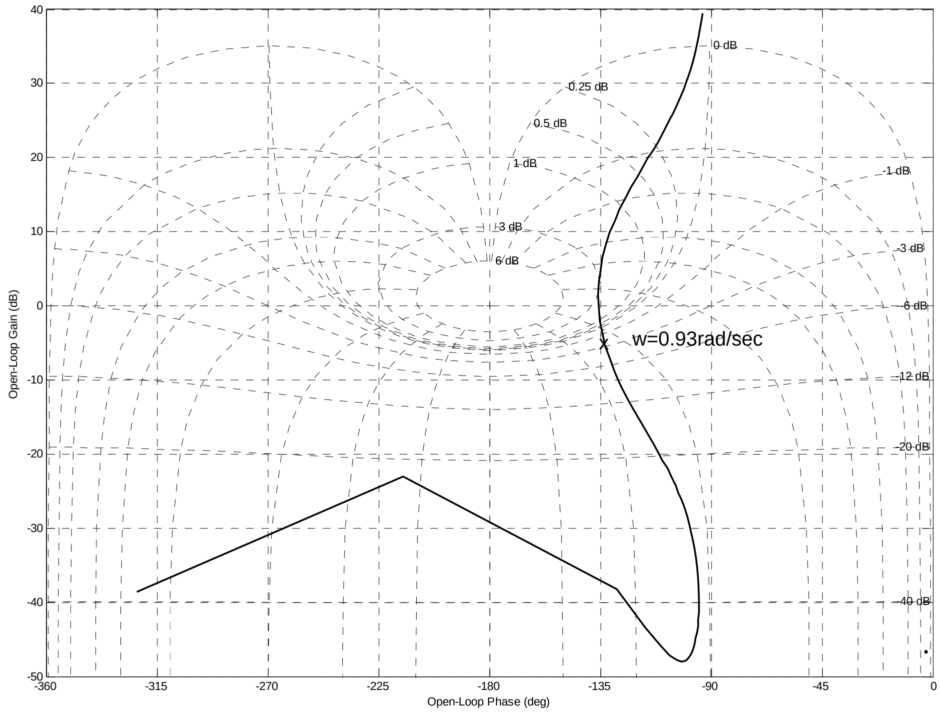
The Nyquist diagram with $K = 1$ is shown next.



As shown the system is closed loop stable, since the system has one open loop RHP pole so P=1, N=1 and Z=P-N=1-1=0. Increasing the gain will not affect stability. However smaller K will eventually make the system unstable. As seen from the figure the system becomes marginally stable when $K = 0.1$. Ths syetem then is closed loop stable for $0.1 < K < \infty$.

39.

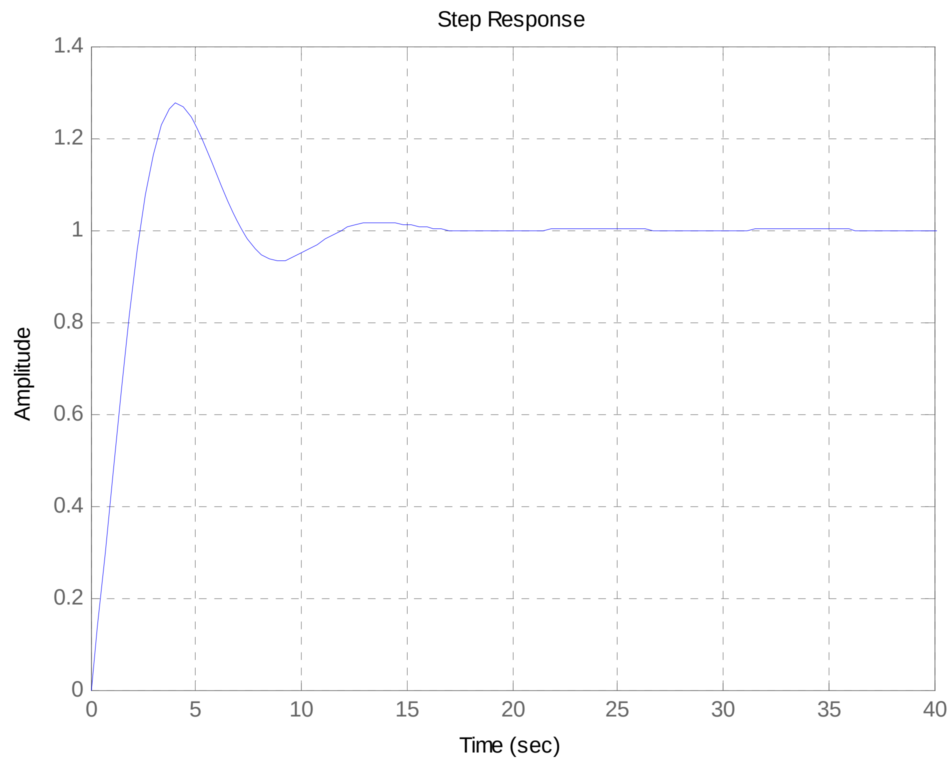
a. The Nichols Chart is shown below.
b. It can be seen there that $M_p = 3db = 1.41$. From figure 10.40 the %OS≈35%. From the Nichols chart, the phase margin is $\Phi_M = 180^\circ - 136^\circ = 44^\circ$, and from figure 10.48 in the text this corresponds to a $\xi \approx 0.4$ damping ratio. It follows from figure 10.41a that since the open loop bandwidth is $\omega_{BW} = 0.93\dfrac{rad}{\sec}$, the settling time is $T_s \approx \dfrac{15}{\omega_{BW}} = 16.1\sec$. The system is type 1 so $c_{final} = 1$.



c.

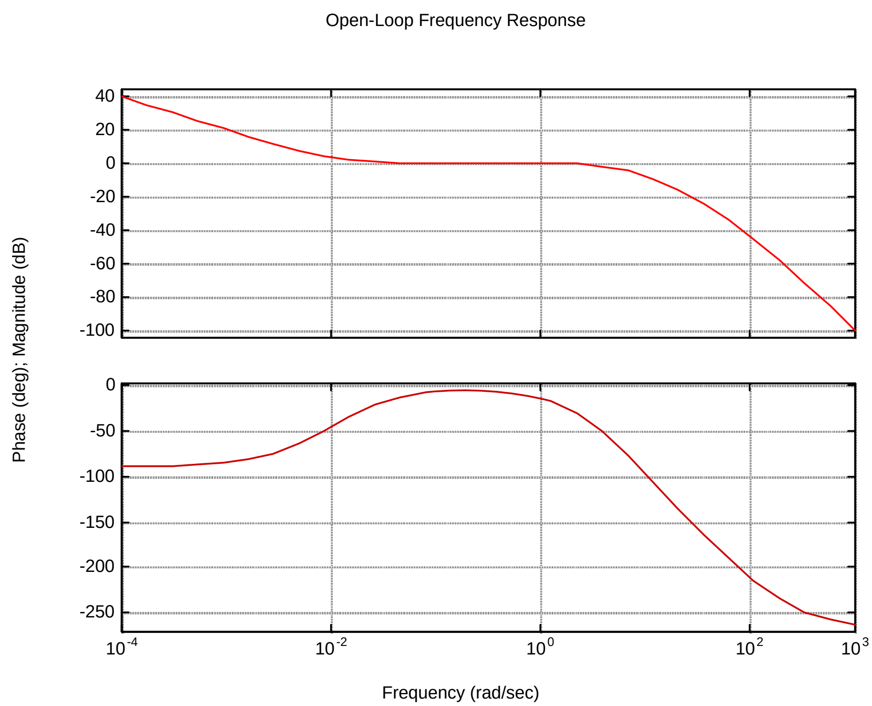
```
>> syms s

>> s=tf('s');

>> P=0.63/(1+0.36*s/305.4+s^2/305.4^2)/(1+0.04*s/248.2+s^2/248.2^2);

 >> G=0.5*(s+1.63)/s/(s+0.27);

 >> L=G*P;

>> T=L/(1+L)

 >> step(T)
```

It can be seen in the following figure that we slightly overestimated the %OS, but the settling time is close to the predicted.


Step Response
Amplitude
Time (sec)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
0
5
10
15
20
25
30
35
40

40.

a.

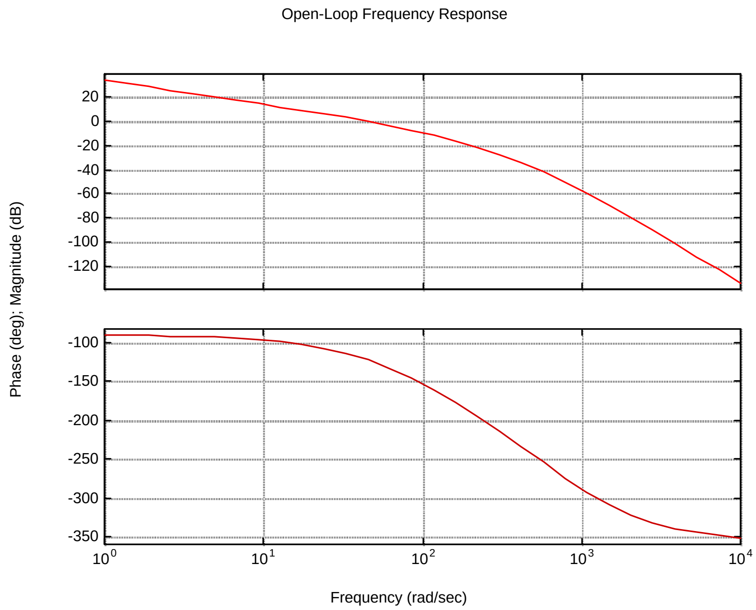


From the Bode plot: Gain margin = 29 52 dB; phase margin = 157.5^0; 0 dB frequency = 1.63 rad/s; 180^0 frequency = 49.8 rad/s.

b. System is stable since it has 180^0 of phase with a magnitude less than 0 dB.

41.

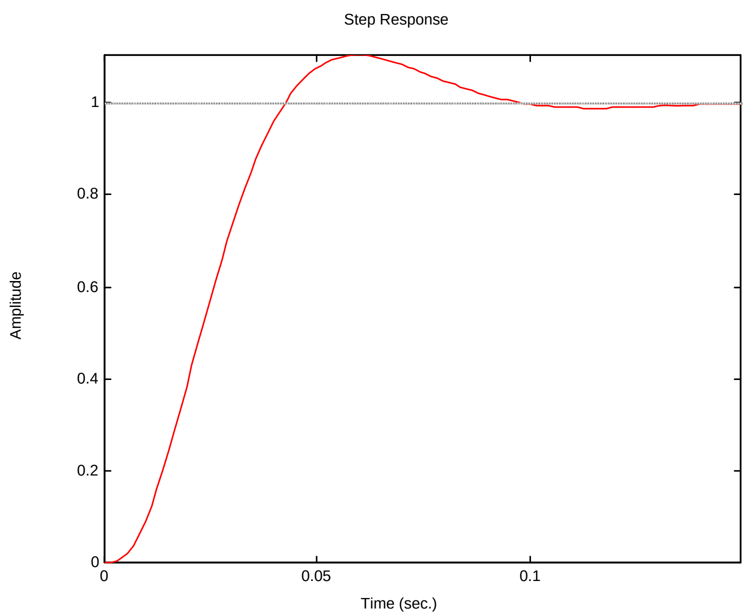
a.



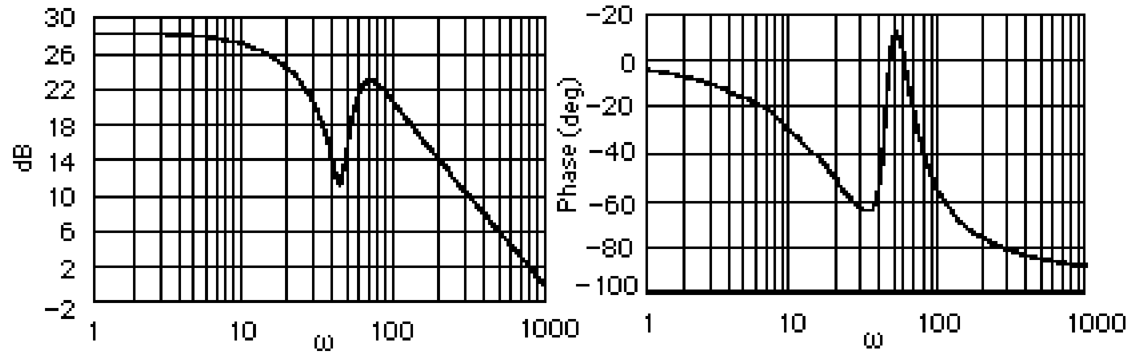
From the Bode plot: Gain margin = 17.1 dB; phase margin = 57.22^0; 0 dB frequency = 45.29 rad/s; 180^0 frequency = 169.03 rad/s; bandwidth(@-7 dB open-loop) = 85.32 rad/s.

b. From Eq. (10.73) ζ = 0.58; from Eq. (4.38) %OS = 10.68; from Eq. (10.55) T_s = 0.0949 s; from Eq. (10.56) T_p = 0.0531 s.

c.



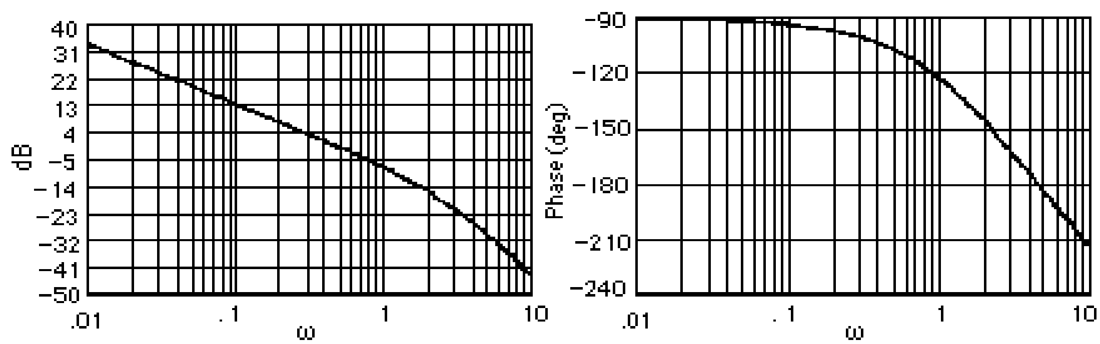
42.



Resonance at 70 rad/s.

43.

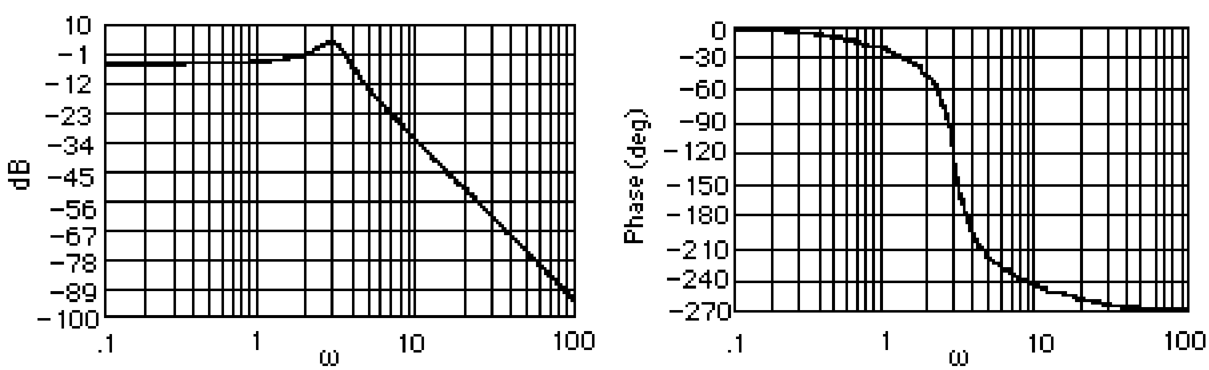
$G(s) = \dfrac{10}{s(s+2)(s+10)}$. Plotting the Bode plots,



The gain is zero dB at 0.486 rad/s and the phase angle is -106.44. Thus, the phase margin is 180^{o} - 106.44^{o} = 73.56^{o} . Using Eq. (10.73), $\zeta = 0.9$. Using Eq. (4.38), %OS = 0.15%.

44.

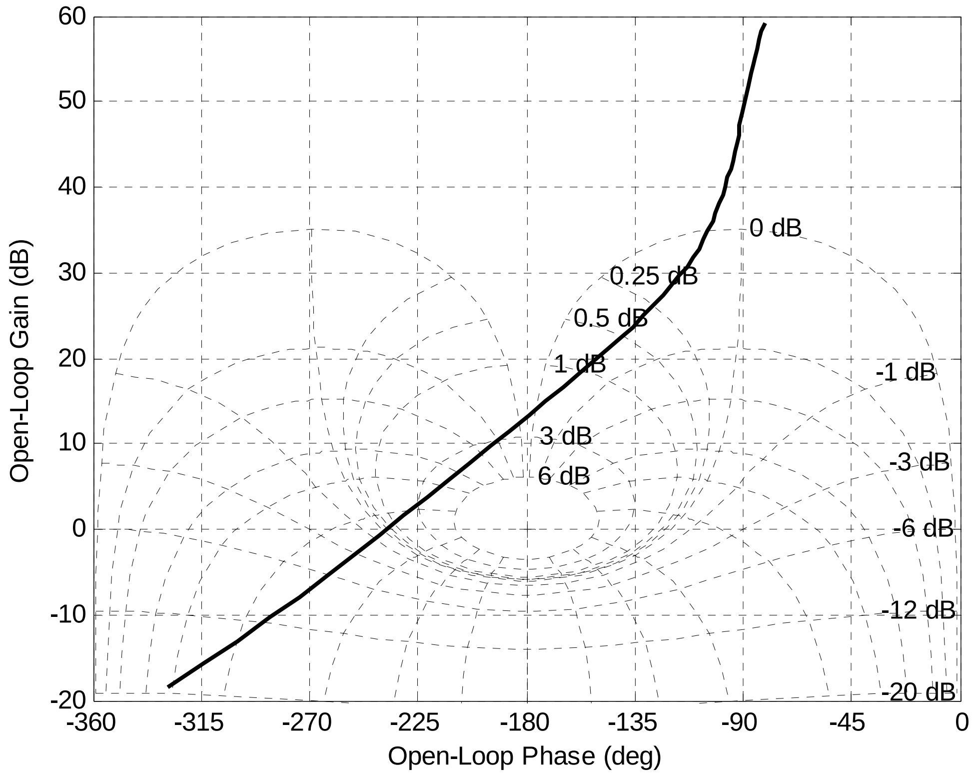
$G(s) = \dfrac{22.5}{(s+4)(s^2+0.9s+9)}$. Plotting the Bode plots,



The phase response is 180° at ω = 3.55 rad/s, where the gain is -1.17 dB. Thus, the gain margin is 1.17 dB. Unity gain is at ω = 2.094 rad/s, where the phase is - 49.85° and at ω = 3.452 rad/s, where the phase is -173.99°. Hence the phase margin is measured at ω = 3.452 rad/s and is 180°- 173.99° = 6.01°. Using Eq. (10.73), ζ = 0.0525. Eq. (4.38) yields %OS = 84.78%.

45.

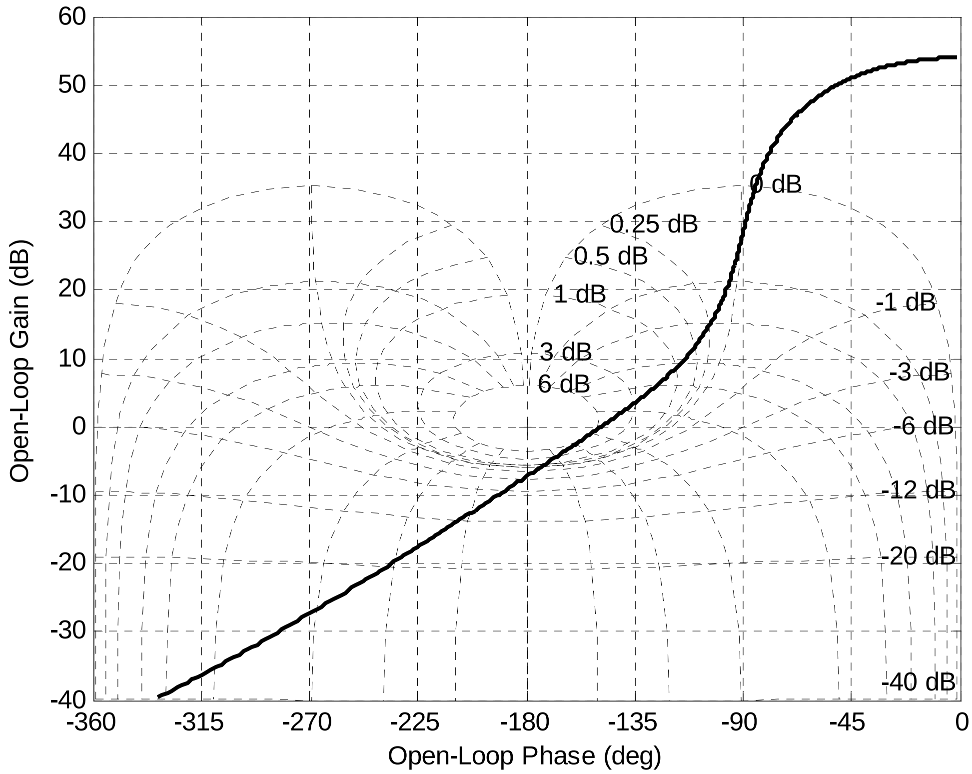
a. The Nichols Chart, shown below, shows that for $L=1$ the system is closed loop unstable since the number of open loop RHP poles P=0 and the chart crosses the -180° line above the 0db line.



b. It can be observed from the diagram that the -180° crossover occurs when $|L(j\omega)| = 13.25db = 4.5$ so the closed loop stability range is $0 < L < \dfrac{1}{4.5} = 0.221$.

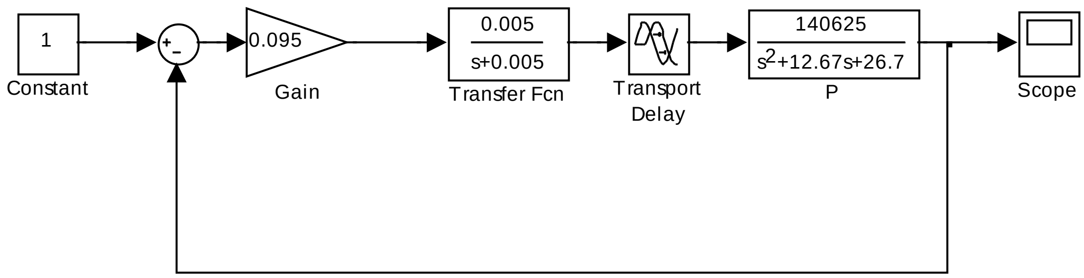
c. The Nichols Chart below shows that $L(0) = G(0)P(0) \approx 58db$, so for a unit step input $c_{final} \approx \dfrac{794.33}{1+794.33} = 0.998$. Also $M_p \approx 6db = 2$ which from figure 10.40 in the text corresponds to a %OS=47%. The phase margin is $\Phi_M = 180^\circ - 148.5^\circ = 31.5^\circ$ so from figure

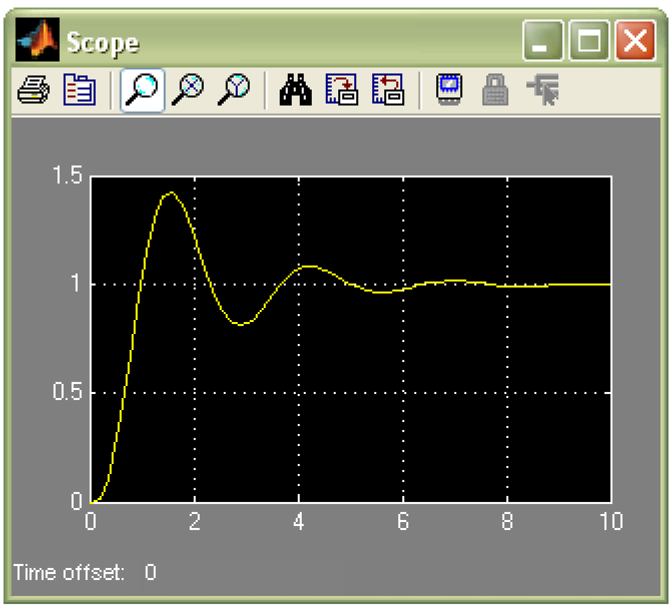
10.48 the corresponding damping factor is $\xi = 0.3$. The open loop bandwidth is $\omega_{BW} = 2.5\dfrac{rad}{\sec}$ so from figure 10.41a $\omega_{BW} T_s \approx 20$ and $T_s \approx 8\sec$



d.

The block diagram and simulation are:

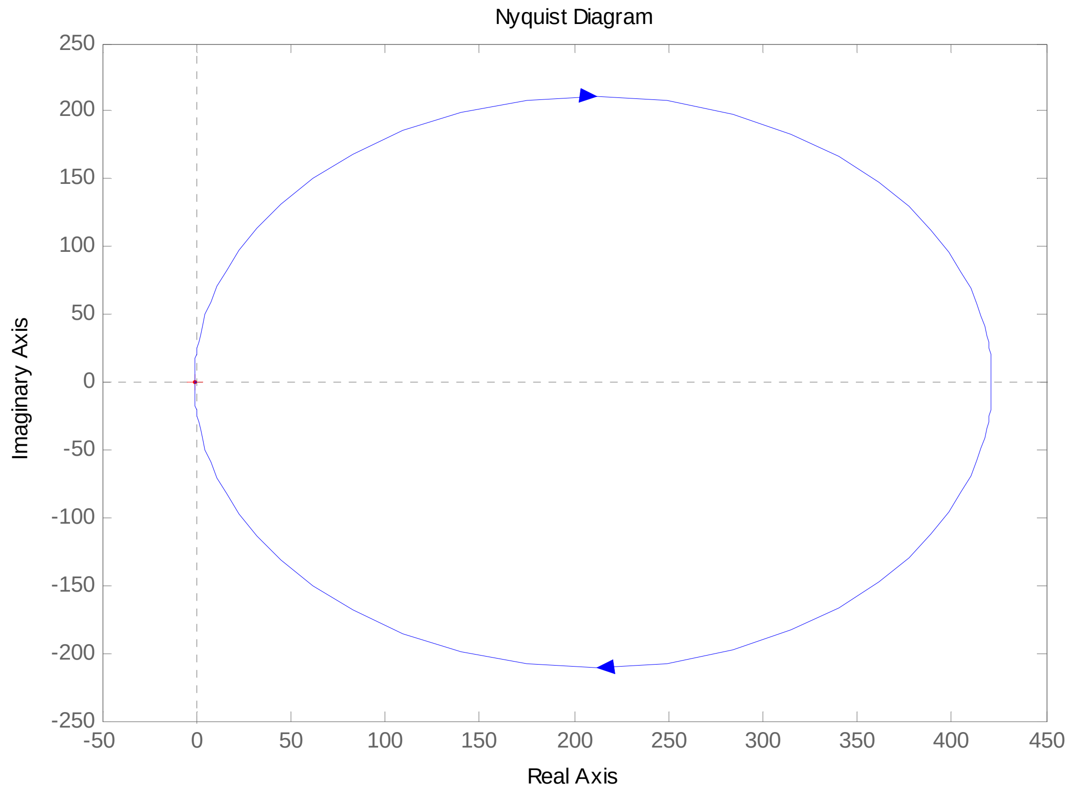




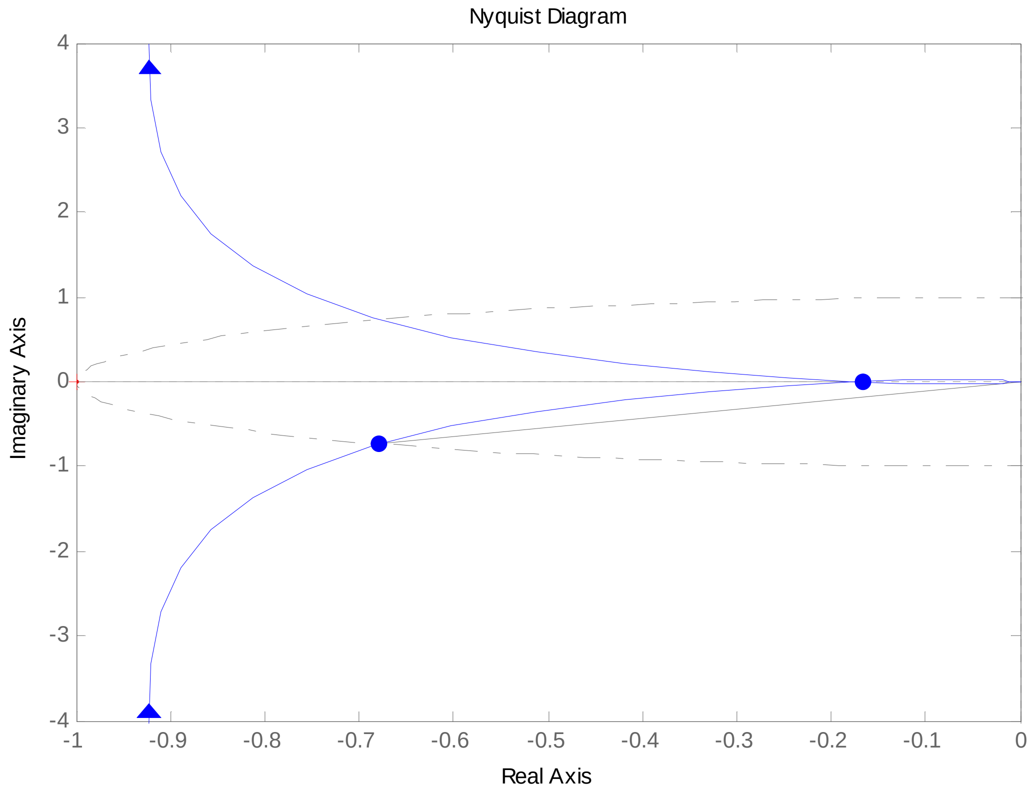
The simulation verifies the predictions.

46.

b. The Nyquist diagram is:



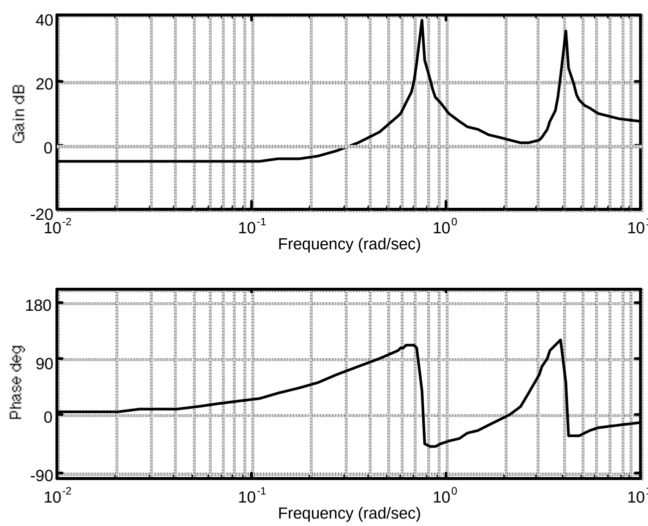
A closed view of the -1 point shows below that $|L(j1.74)| = 1$ and $\arg(L(j1.74)) = -132.8^\circ$ so the phase margin without delay is $\Phi_M = 180^\circ - 132.8^\circ = 47.2^\circ = 0.824rad$



c. The maximum allowable time delay is obtained by noting solving $1.74T < 0.824$ or $T < 0.47\,\text{sec}$

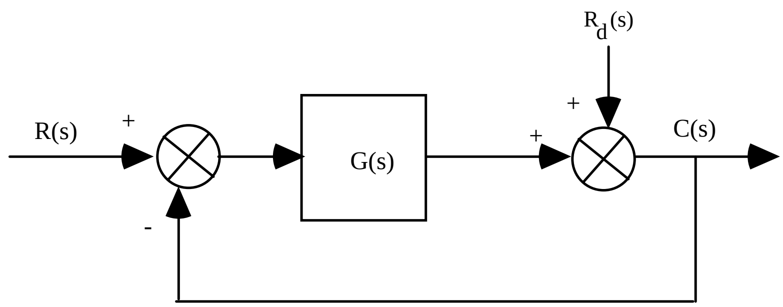
47.

a.



The frequencies that will be reduced occur at the peaks of the magnitude plot. The frequencies at the peaks are 4.14 rad/s and 0.754 rad/s.

b. Consider a system with a disturbance, R_d at the output of a system:



The transfer function relating C(s) to R_d(s) is $\frac{C(s)}{R_d(s} = \frac{1}{1+G(s)}$. Therefore,

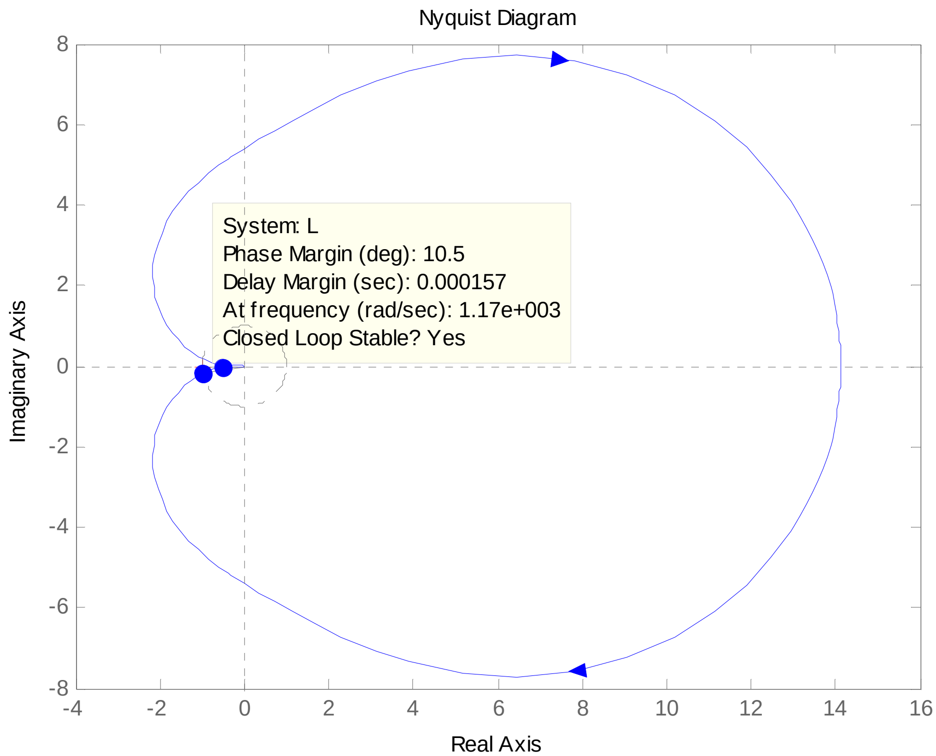
$$C(s) = \frac{1}{1+\frac{N_G}{D_G}} * \frac{N_{Rd}}{D_{Rd}} = \frac{D_G}{D_G + N_G} * \frac{N_{Rd}}{D_{Rd}}$$

Thus, if the poles of G(s) match the poles of R_d ($D_G = D_{Rd}$) there will be cancellation and the dynamics of the disturbance will be reduced. Thus, if the dynamics of R_d is oscillation, add poles in cascade with G(s) that have the same dynamics. Since the poles yield large gain at these bending frequencies a zero is placed near the poles so that the filter will have minimal effect on the transient response (similar to placing a zero near a pole for a lag compensator). This arrangement of poles and zeros is called a dipole. Also note that a high gain at the bending frequency yields negative feedback for the output to subtract from R_d. Care should be exercised through analysis and simulation to be sure that the system's response to an input, other than the disturbance, is not adversely affected by the additional poles.

48.

a. The following code will generate the Nyquist diagram:

```
>> s=tf('s');

>> P=1.163e8/(s^3+962.5*s^2+5.958e5*s+1.16e8);

>> G=78.575*(s+436)^2/(s+132)/(s+8030);

>> L=G*P;

>> nyquist(L)
```



As seen in the figure the system is closed loop stable.

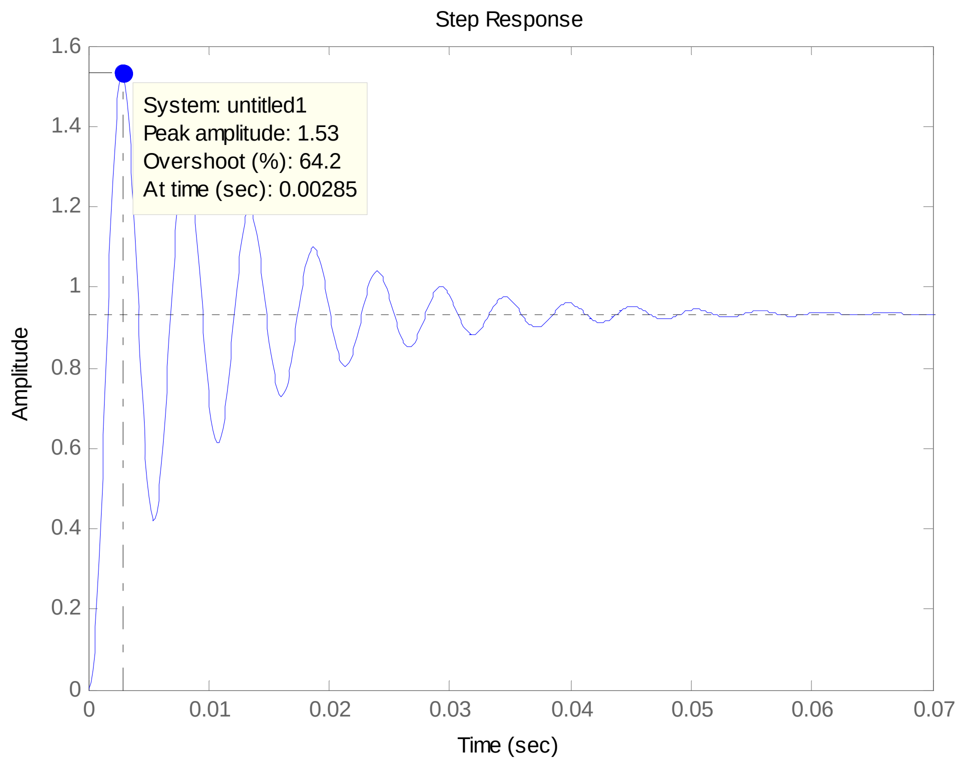
b. It can be seen in the figure that the phase margin is 10.5 degrees.

c. From Figure 10.48 in the text, a 10.5 degree phase margin corresponds approximately to a damping factor of 0.16. This in turn means that

$$\%OS = 100e\frac{\xi\pi}{1-\xi^2} = 60\%$$

d. To simulate the response of the system to a unit step, we can use:

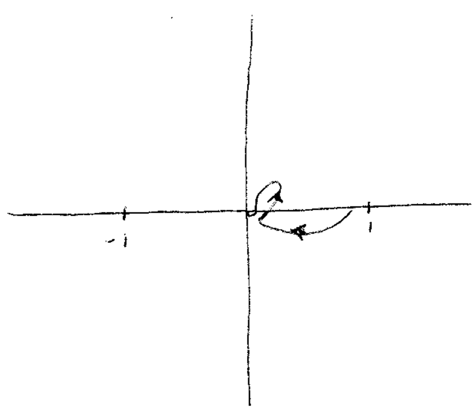
```
>> step(L/(1+L))
```



It can be seen that the actual overshoot is 64.2%.

49.

a. The Nyquist plot will approximately loop as follows (for positive ω(:

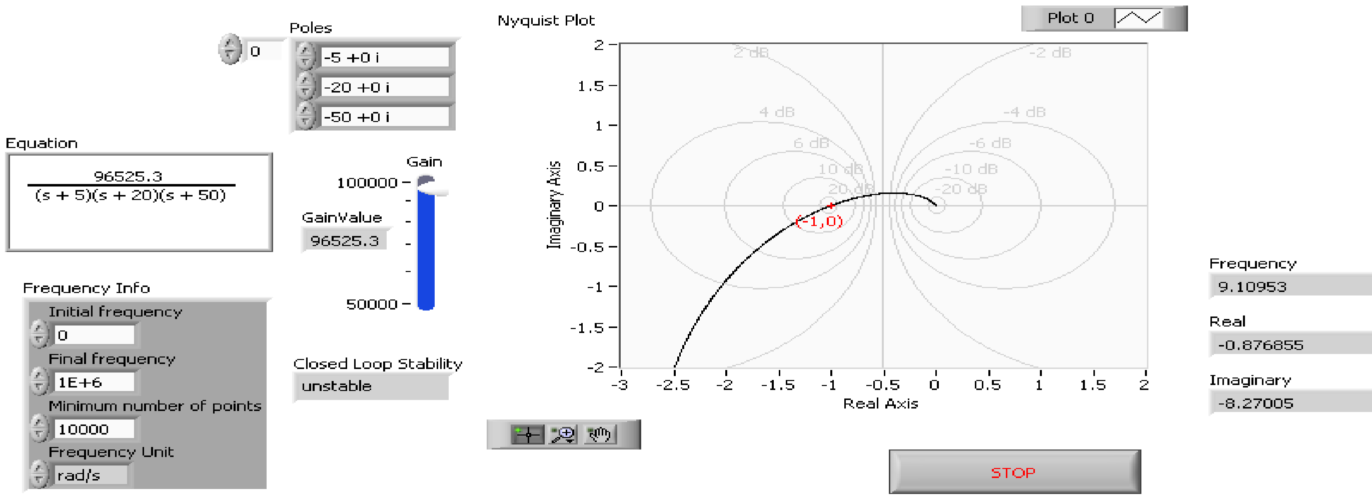


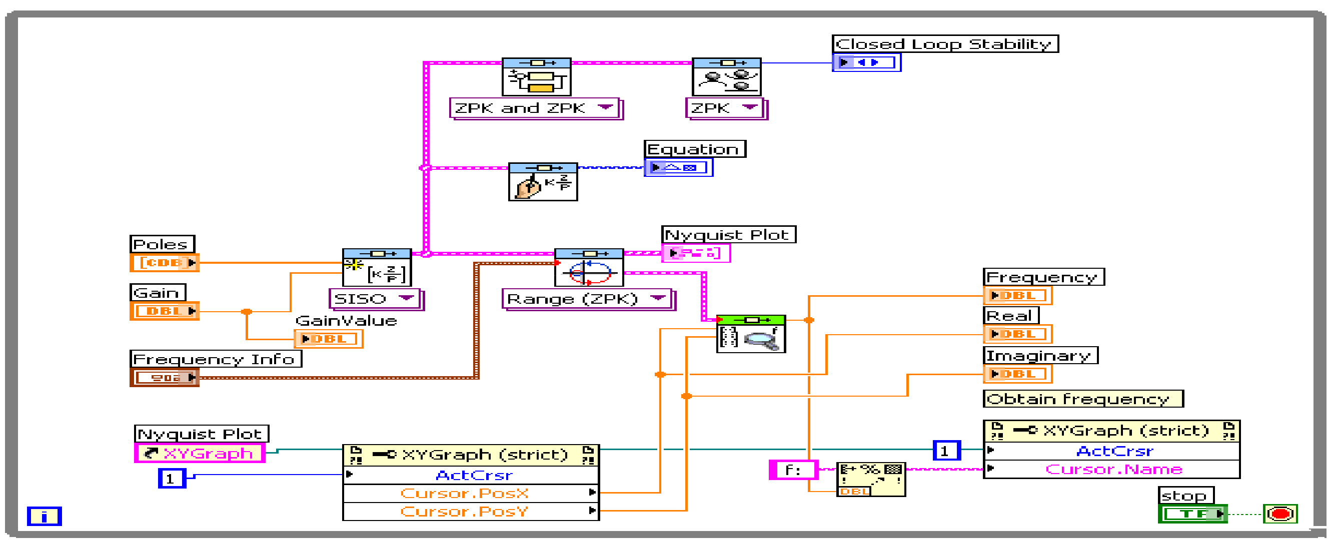
b. For K=1, it can be seen that there are no encirclements of the -1 point.

c. The system will be stable for 0<K<∞, because the phase never exceeds -90 degrees, so there can be no encirclements of the -1 point.

50.

Front Panel and Block Diagram for "CDEx Nyquist Analysis.vi" (modified)





Front Panel and Block Diagram for Nyquist plot and frequency response data
Front Panel

ENTER OPEN LOOP NUMERATOR/DENOMINATOR POLYNOMIALS

Numerator

0	10000	0	0

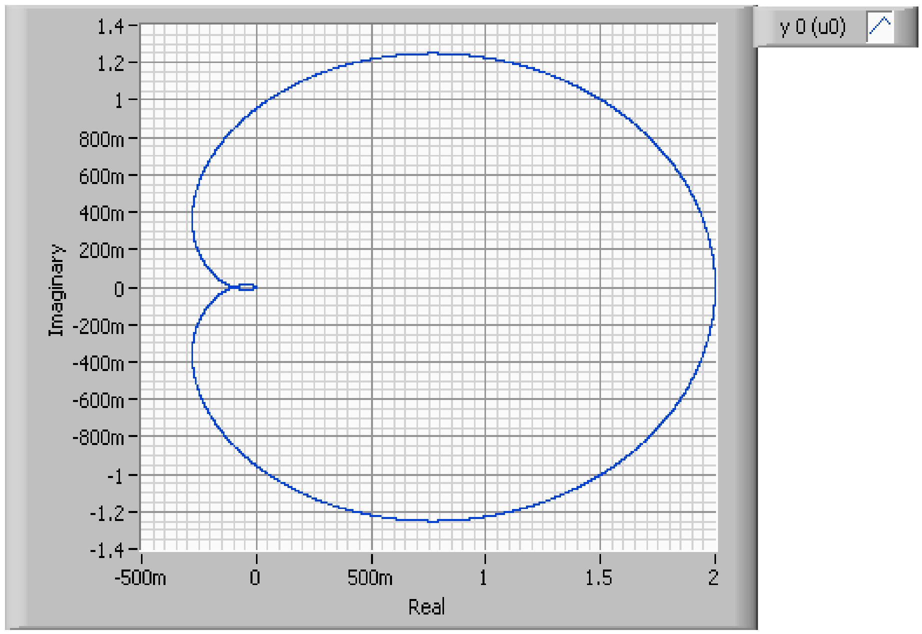
Denominator

0	5000	1350	75	1

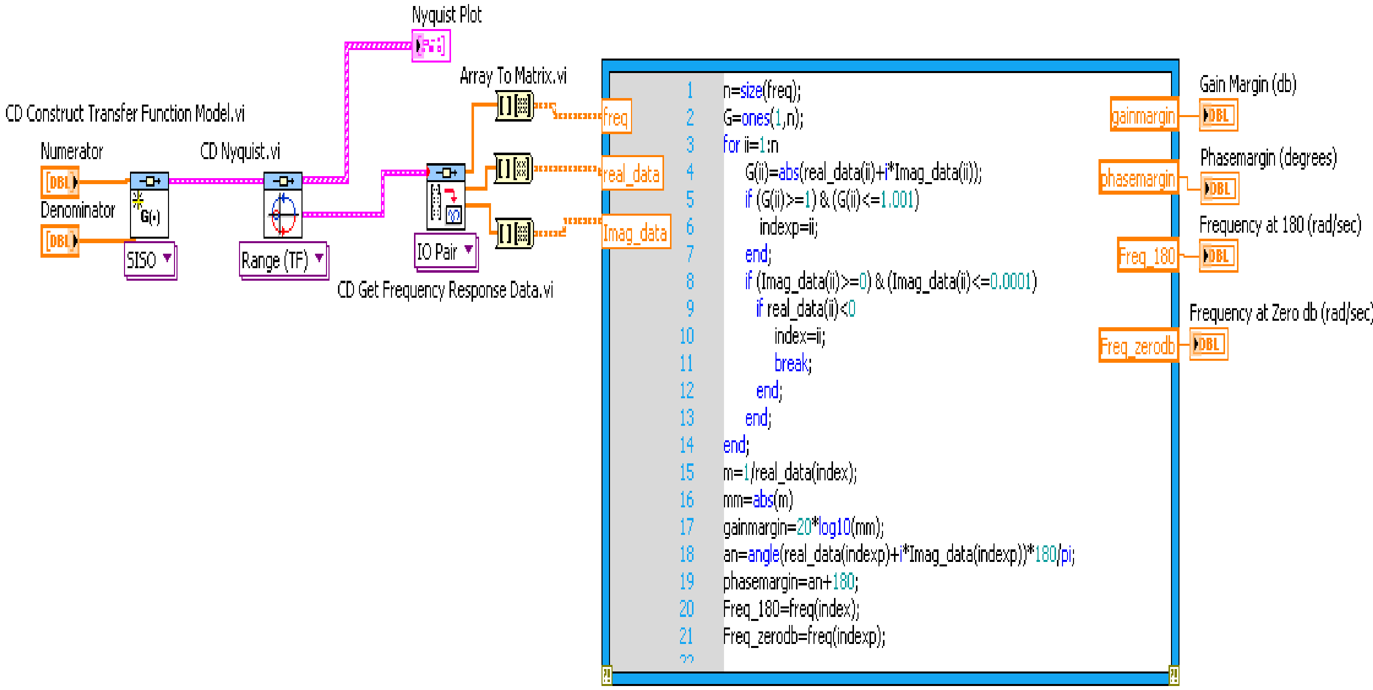
Gain Margin (db)	Phasemargin (degrees)
19.6802	92.9377

Frequency at 180 (rad/sec)	Frequency at Zero db (rad/sec)
36.7669	7.73551

Nyquist Plot

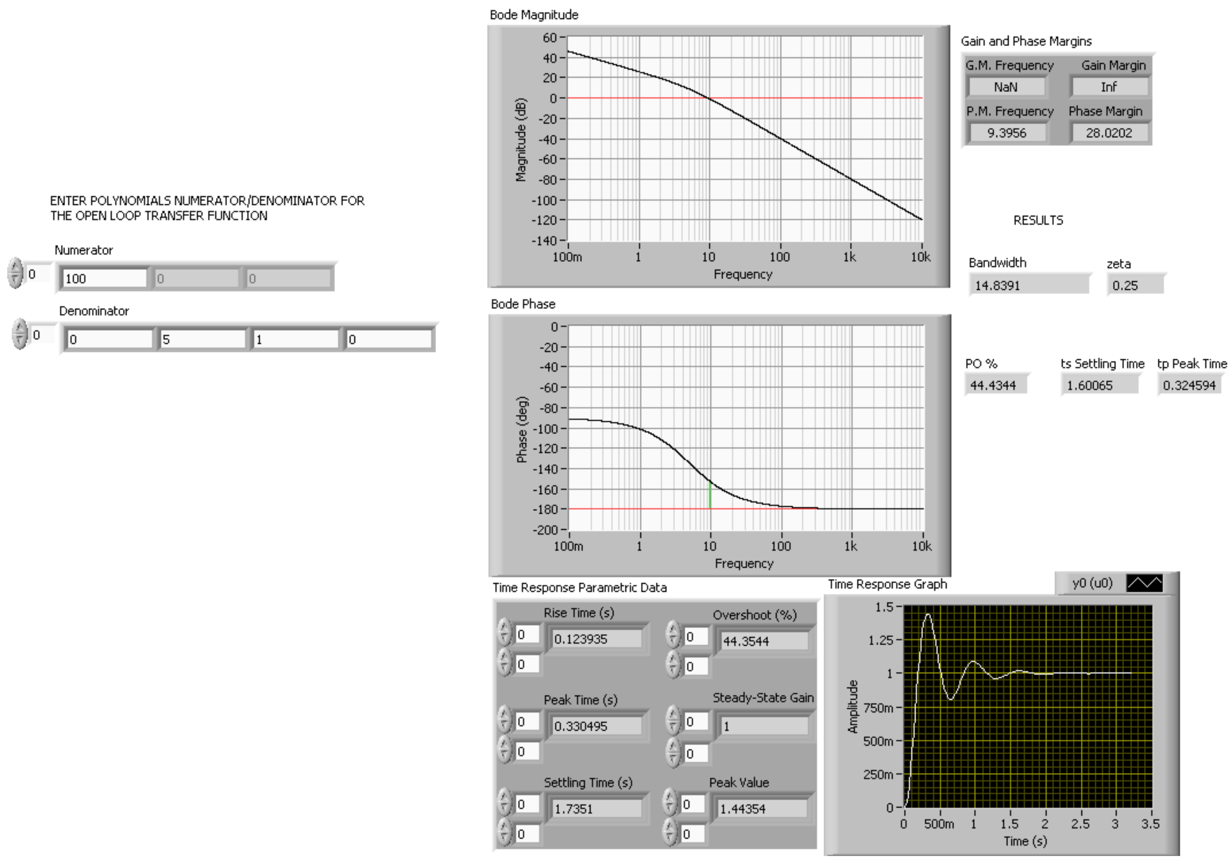


Block Diagram



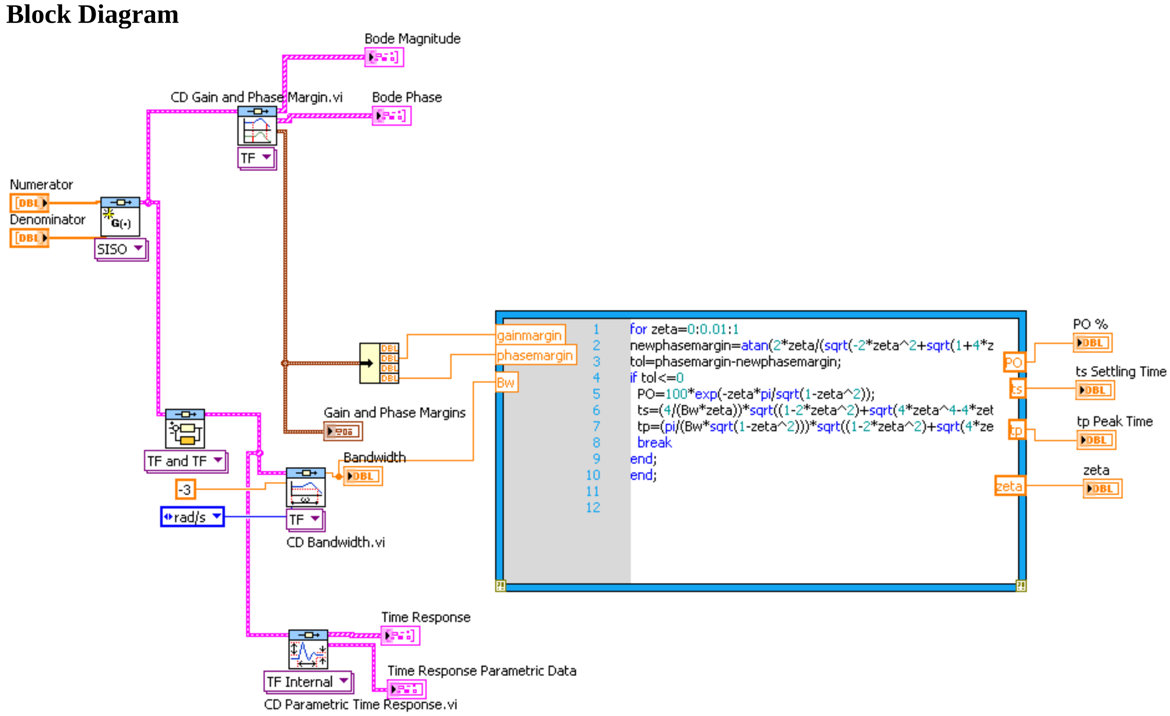
51.

Front Panel



Note: Notice minor differences in estimating the transient domain specifications from the open loop frequency response with respect to the close loop parametric data. That is: PO% is 44.43% (44.35% from close loop data), settling time is 1.6 sec (1.7 secs from close loop data), and finally, peak time is 0.32 secs (0.33 secs from close loop data).

Block Diagram



The details of the MathScript node code follow:

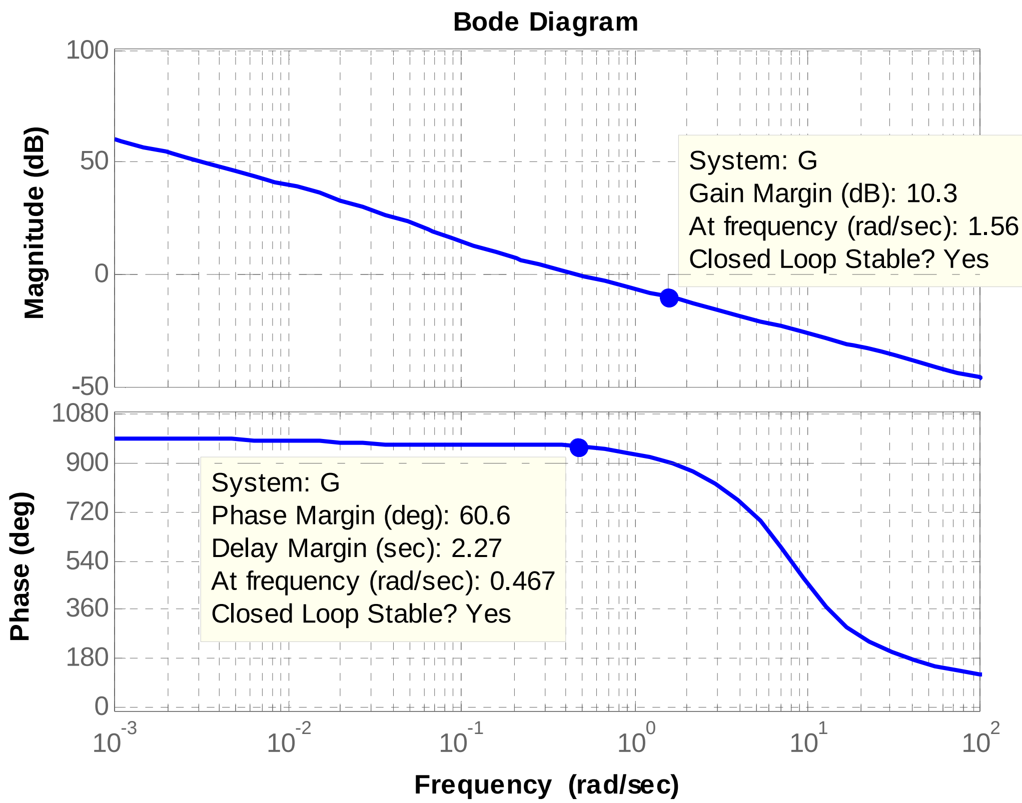
```
for zeta=0:0.01:1
newphasemargin=atan(2*zeta/(sqrt(-2*zeta^2+sqrt(1+4*zeta^4))))*(180/pi);
tol=phasemargin-newphasemargin;
if tol<=0
  PO=100*exp(-zeta*pi/sqrt(1-zeta^2));
  ts=(4/(Bw*zeta))*sqrt((1-2*zeta^2)+sqrt(4*zeta^4-4*zeta^2+2));
  tp=(pi/(Bw*sqrt(1-zeta^2)))*sqrt((1-2*zeta^2)+sqrt(4*zeta^4-4*zeta^2+2));
  break
end;
end;
```

52.

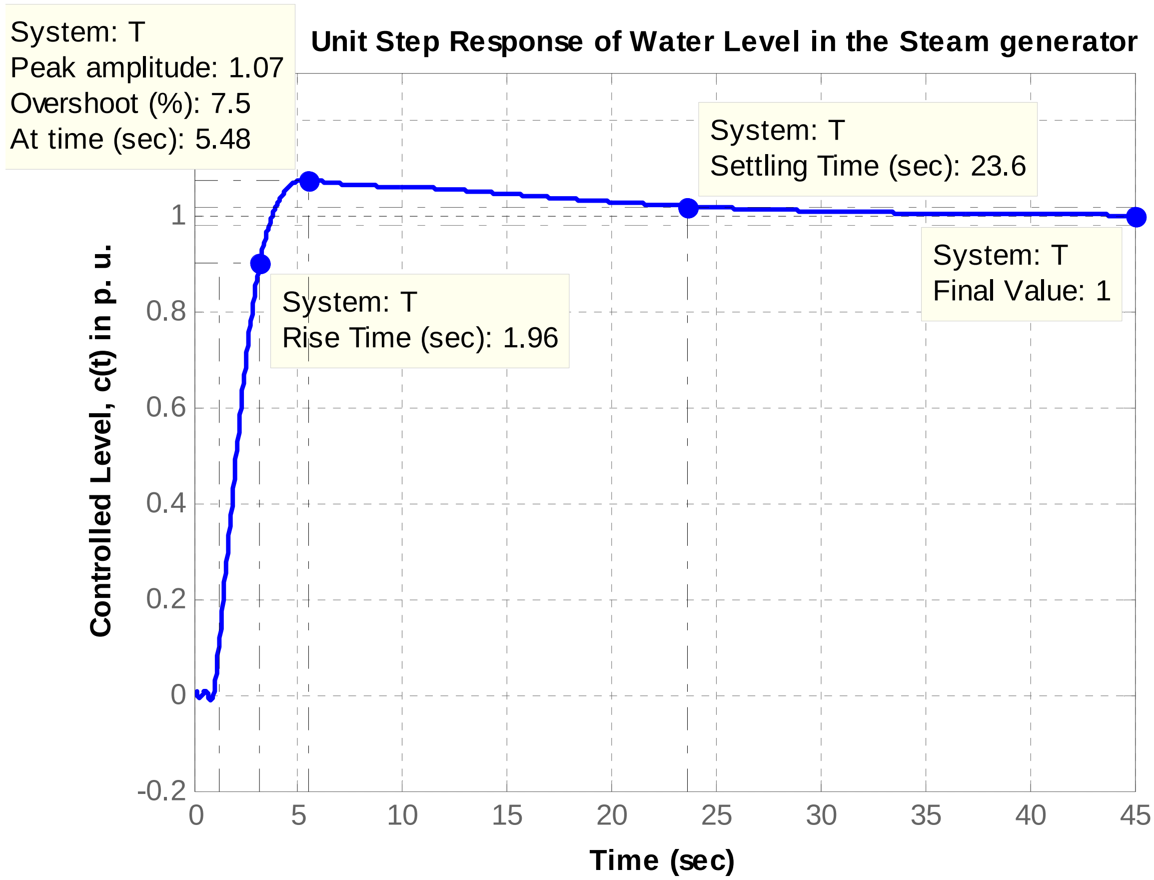
The following MATLAB file was used to plot the Bode magnitude and phase plots for that system and to obtain the response of the system, c(t), at a pure delay of 1 second.

```
numg1 = conv([4 1], [10 1]);     %numerator of G1 without pure delay
deng0 = conv([25 1],[3.333 1]); %the denominator without pure
integrator
deng1 = conv(deng0, [1 0]);      %add pure integrator to denominator
G1 = tf(numg1, deng1);           %G1 is the open-loop TF without pure
delay
[numd,dend]= pade(1,5);          %5th-order Pade approximation of
pure delay
D = (tf (numd, dend));
G = G1*D;
Bode(G);
grid
pause
T = feedback(G,1);       %C.L. TF of the PD_compensated system
T = minreal(T);
step(T);
grid
xlabel ('Time')
ylabel ('Controlled Level, c(t) in p. u.')
title ('Unit Step Response of Water Level in the Steam generator ')
```

a. The Bode magnitude and phase plots for this system, obtained using fifth-order Pade approximation for $\tau = 1$ sec, are shown below with minimum stability margins displayed.



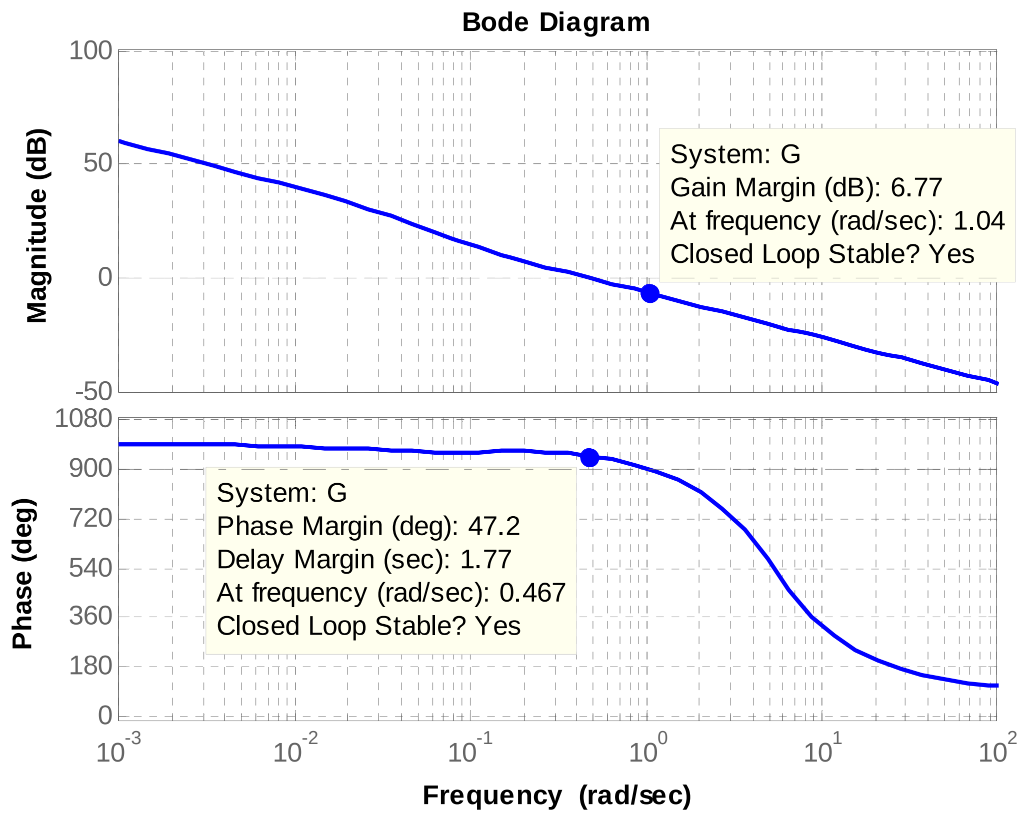
b. The response of the system, c(t), to a unit step input, $r(t) = u(t)$, is shown below for $\tau = 1$ sec. The rise time, T_r, the settling time, T_s, the final value of the output, percent overshoot, $\%OS$, and peak time, T_p, are displayed.

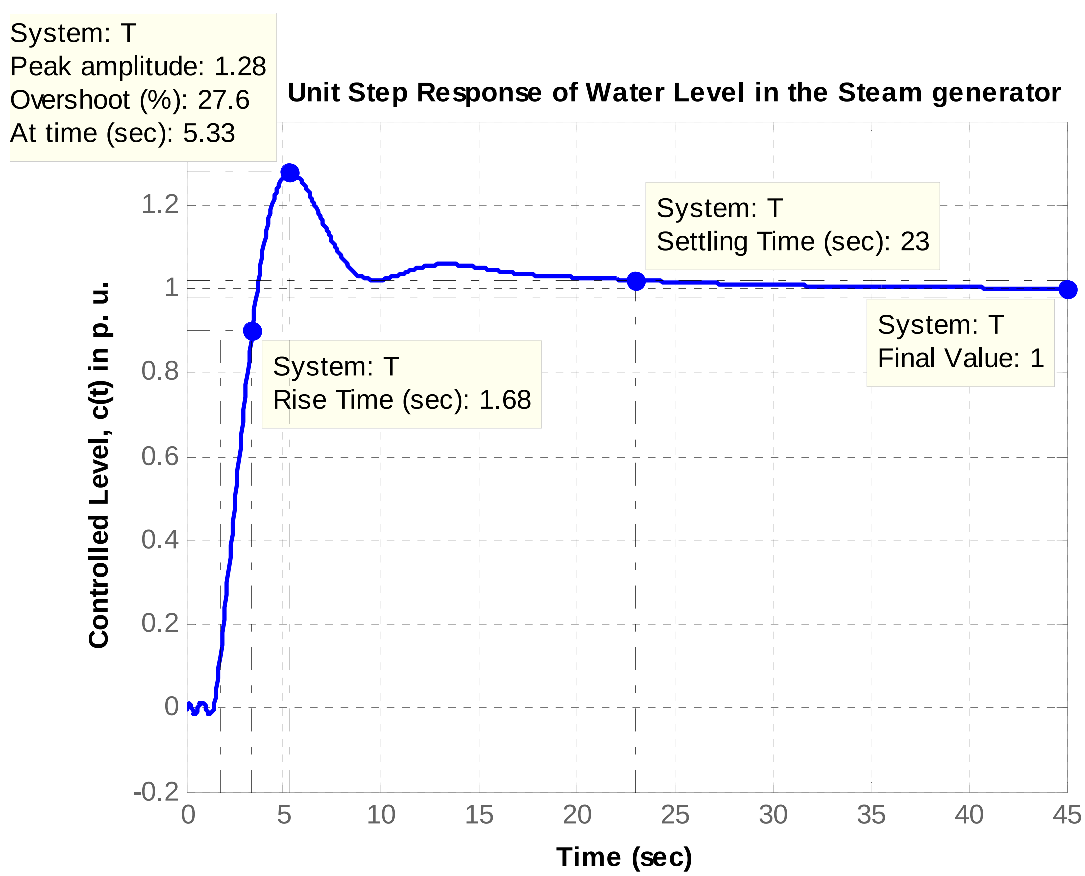


c. The Bode magnitude and phase plots for this system for a pure delay τ = 1.5 seconds, are shown below with minimum stability margins displayed.

Also shown below is the response of the system, c(t), to a unit step input, *r(t)* = *u(t)*, for a pure delay τ = 1.5 seconds. The rise time, T_r, the settling time, T_s, the final value of the output, percent overshoot, *%OS*, and peak time, T_p, are displayed on the graph.

Note that the percent overshoot increased from 7.5% to 27.6% as the delay time increased from 1 to 1.5 seconds.


Bode Diagram
Magnitude (dB)
100
50
0
-50
System: G
Gain Margin (dB): 6.77
At frequency (rad/sec): 1.04
Closed Loop Stable? Yes
Phase (deg)
1080
900
720
540
360
180
0
System: G
Phase Margin (deg): 47.2
Delay Margin (sec): 1.77
At frequency (rad/sec): 0.467
Closed Loop Stable? Yes
10^-3
10^-2
10^-1
10^0
10^1
10^2
Frequency (rad/sec)


System: T
Peak amplitude: 1.28
Overshoot (%): 27.6
At time (sec): 5.33
Unit Step Response of Water Level in the Steam generator
System: T
Settling Time (sec): 23
System: T
Rise Time (sec): 1.68
System: T
Final Value: 1
Controlled Level, c(t) in p. u.
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
0
5
10
15
20
25
30
35
40
45
Time (sec)

53.

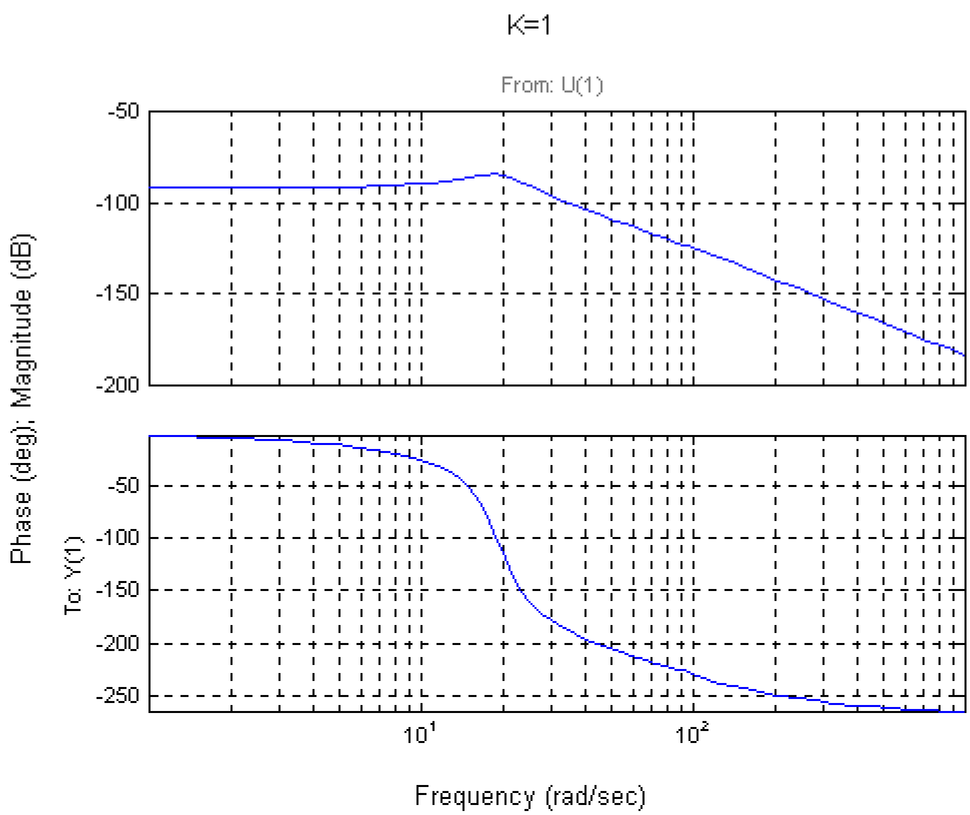
a. From Chapter 8,

$$G_e(s) = \frac{0.6488K\,(s+53.85)}{(s^2 + 8.119s + 376.3)\,(s\char`^2 + 15.47s + 9283)}$$

Cascading the notch filter,

$$G_{et}(s) = \frac{0.6488K\,(s+53.85)(s^2 + 16s + 9200)}{(s^2 + 8.119s + 376.3)\,(s\char`^2 + 15.47s + 9283)(s+60)^2}$$

Plotting the Bode plot,



From the Bode plot: Gain margin = 96.74 dB; phase margin = ∞; 0 dB frequency = N/A; 180^0 frequency = 30.44 rad/s.

b. K = 96.74 dB = 68732

c. In Chapter 6 K = 188444. The difference is due to the notch filter.

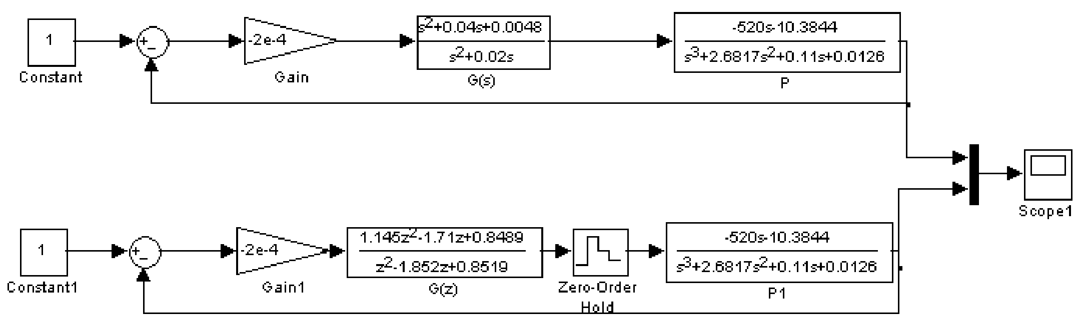
54.

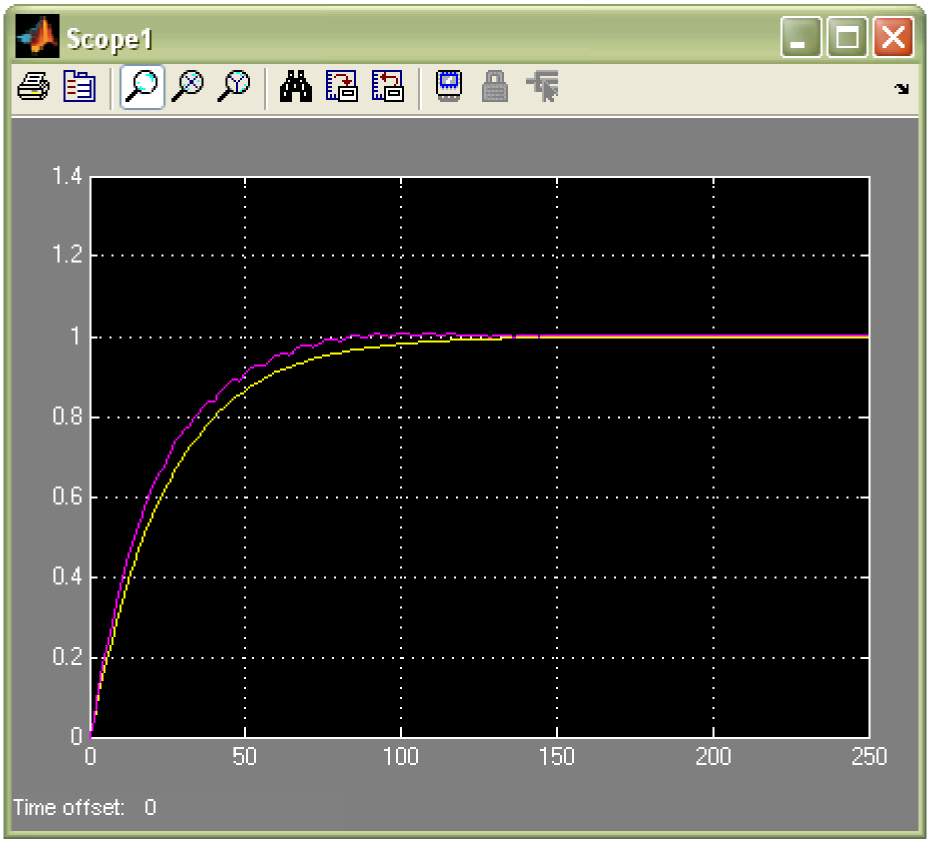
a. A bode plot of the open loop transmission $G_c(s)P(s)$ shows that the open loop transfer function has a crossover frequency of $\omega_c = 0.04\dfrac{rad}{day}$. A convenient range for sampling periods is $3.75day = \dfrac{0.15}{\omega_c} < T < \dfrac{0.5}{\omega_c} = 12.5day$. T=8 days fall within range.

b. We substitute $s = \dfrac{1}{4}\dfrac{z-1}{z+1}$ into $G_c(s)$ we get

$$G_c(z) = \frac{-2\times10^{-4}(1.145z^2 - 1.71z + 0.8489)}{z^2 - 1.852z + 0.8519}$$

c.





55.

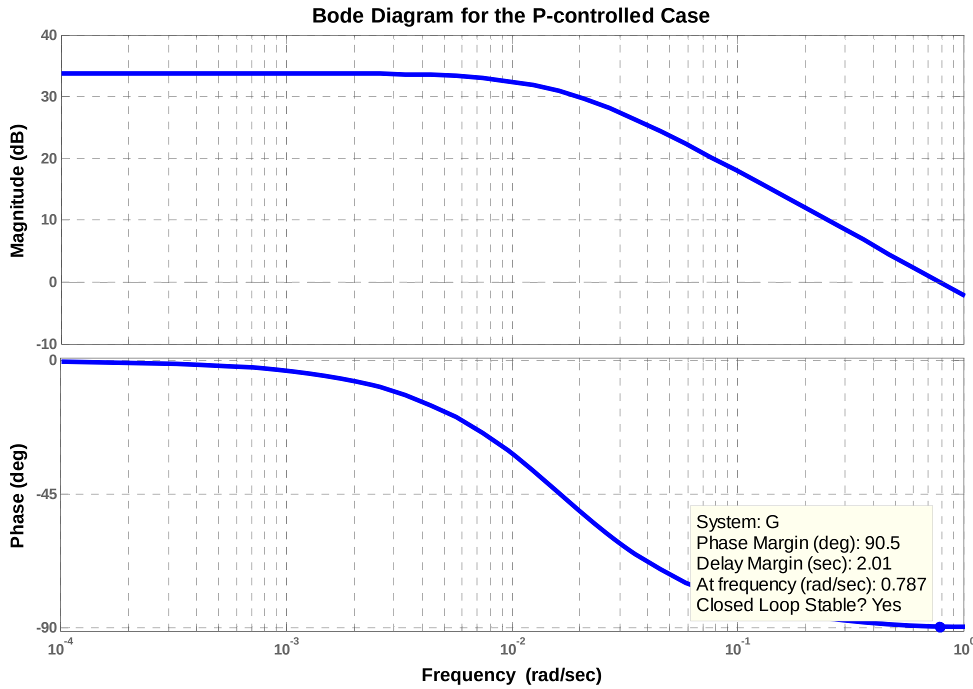
a.

The following MATLAB file was used to plot the Bode magnitude and phase plots for that system and to obtain the response of the system, c(t), to a step input, *r(t)* = 4 *u(t).*

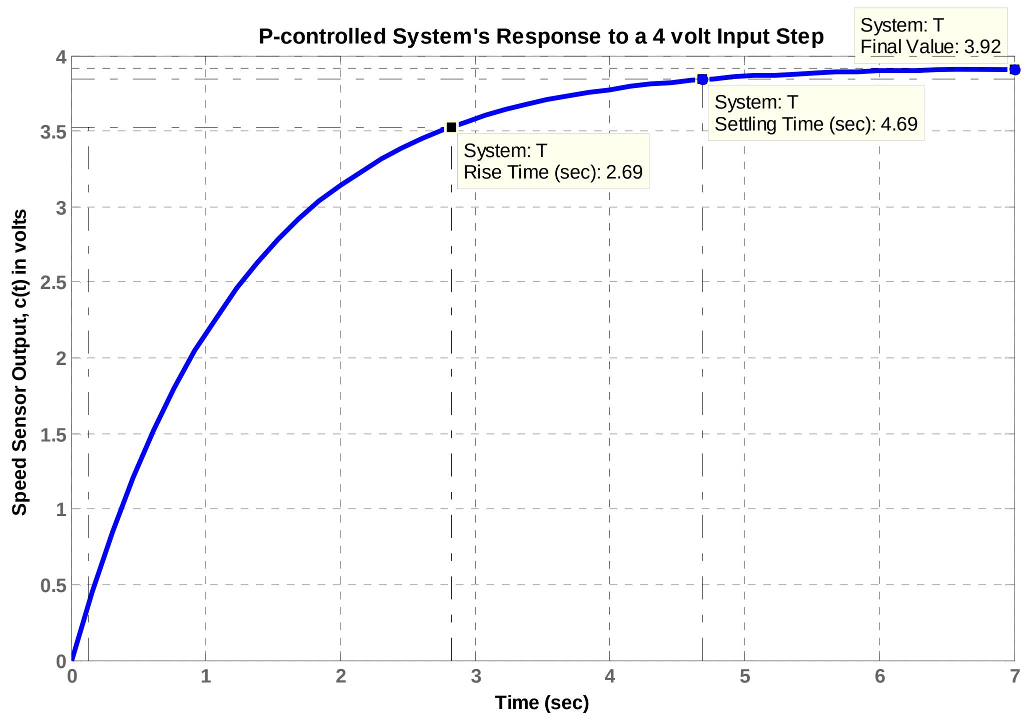
```
K = 0.78;
numg = K*[1 0.6];
deng = poly ([-0.0163 -0.5858]);
G = tf(numg, deng);
bode (G);
grid
pause
T = feedback(G,1);   %T is the closed-loop TF of the P_controlled
system
T = minreal(T);
step(4*T);
grid
```

```
xlabel ('Time')
ylabel ('Speed Sensor Output, c(t) in volts')
title ('P-controlled Systems Response to a 4 volt Input Step')
```

The Bode magnitude and phase plots obtained are shown below with the minimum stability margins displayed on the phase plot.



The response of the system, c(t), to a step input, $r(t) = 4\ u(t)$ is shown below. The rise time, T_r, settling time, T_s, and the final value of the output are noted.



b.

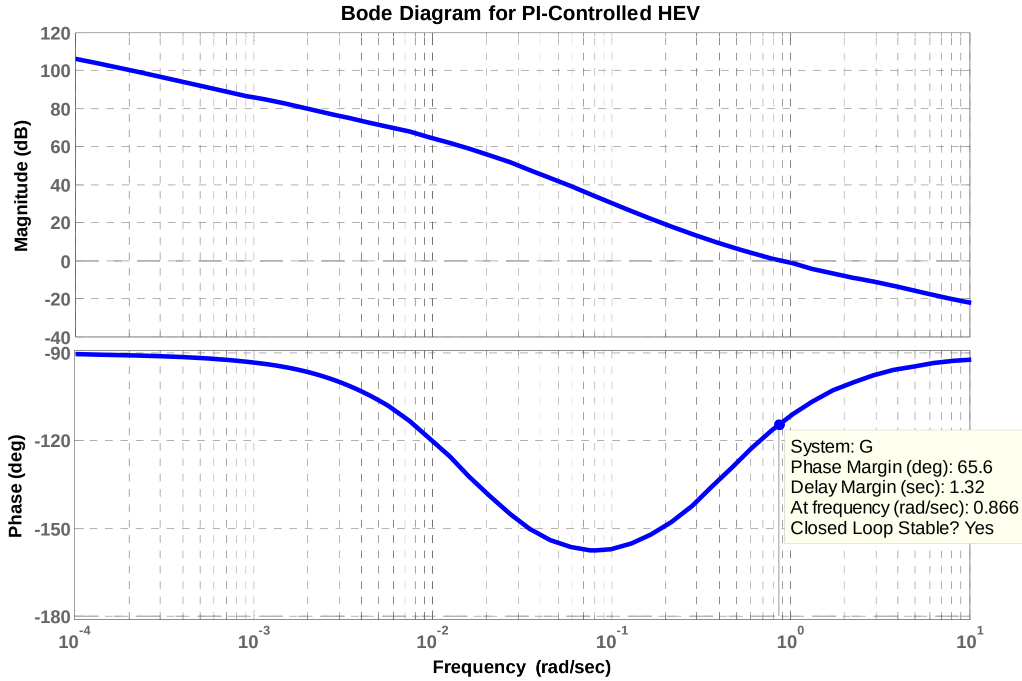
After adding the integral gain to the controller, we substituted the values of $K_1 = 0.78$ and $Z_c = \frac{K_2}{K_1} = 0.4$ into the transfer function of the plant and compensator. Thus we obtained:

$$G(s) = \frac{0.78 \times (s + 0.4)(s + 0.6)}{s(s + 0.5858)(s + 0.0163)} = \frac{0.78s^2 + 0.78s + 0.1872}{s(s + 0.5858)(s + 0.0163)}.$$

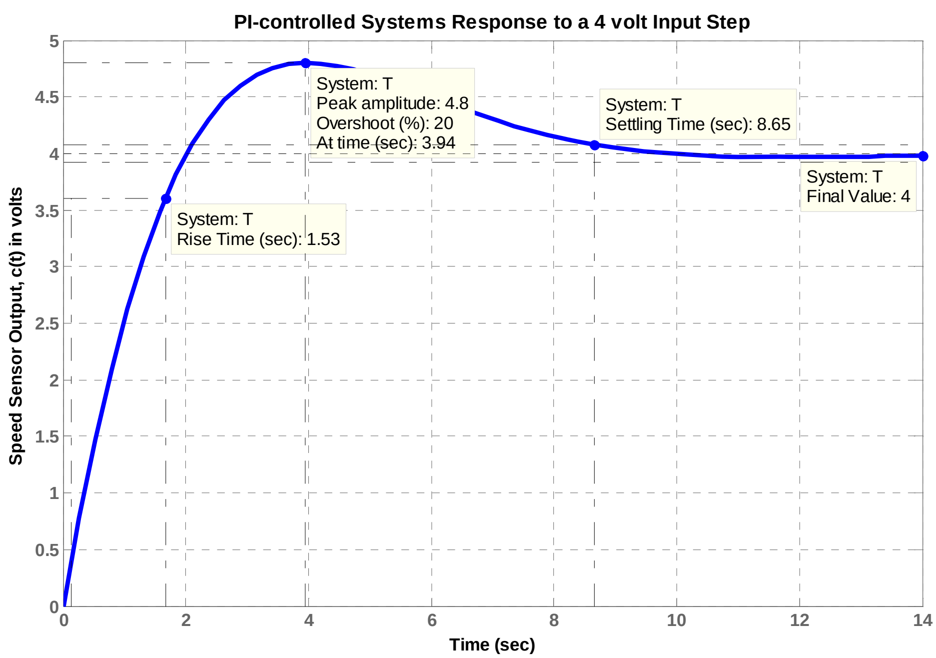
The following MATLAB file was written to plot the Bode magnitude and phase plots for that system and to obtain the response of the system, c(t), to a step input, *r(t)* = 4 *u(t)*.

```
numg = [0.78 0.78 0.1872];
deng = poly ([0 -0.0163 -0.5858]);
G = tf(numg, deng);
bode (G);
grid
pause
T = feedback(G,1);   %T is the closed-loop TF of the PI-controlled
system
T = minreal(T);
step(4*T);
grid
xlabel ('Time')
ylabel ('Speed Sensor Output, c(t) in volts')
title ('PI-controlled Systems Response to a 4 volt Input Step')
```

The Bode magnitude and phase plots obtained are shown below with the minimum stability margins displayed on the phase plot.



The response of the system, c(t), to a step input, $r(t) = 4\,u(t)$ is shown below. The rise time, T_r, settling time, T_s, percent overshoot, $\%OS$, peak time, T_p, settling time, T_s. and the final value of the output are noted.



c.

The response obtained in (a), e.g. in the case of proportionally-controlled HEV, is closer to that of a first-order system rather than a critically-damped second order system. Note also that the Bode magnitude plot has a final slope of – 20 dB/dec and the phase plot has a final value of 90^{o}.

The response obtained in (b), e.g. in the case of PI-controlled HEV, does not resemble well that of a second-order underdamped response. It should be noted, that one of the closed-loop poles (located at – 0.5753) is quite close to one of the closed-loop zeros (located at – 0.6) whereas the second closed-loop zero (located at – 0.4) is almost equal to the real part of the complex-conjugate dominant poles (located at $-0.4034 \pm j\,0.4034$).